

Question Bank — All Items by Benchmark

532 items across 14 test forms, grouped by B.E.S.T. benchmark.

Every item shows its test form, item type, difficulty, correct answer and explanation.

Number Sense and Operations

MA.6.NSO.1.1 — Extend previous understanding of numbers to define rational numbers. Plot, order and compare rational numbers.

Q0019 (Test 1 · Multiple choice · easy)

Camila recorded these four numbers in her math journal: -5.5, $-1\frac{3}{4}$, $\frac{3}{4}$, 1.

Which number is the least?

Choices: $\frac{3}{4}$ | $-1\frac{3}{4}$ | 1 | -5.5

Answer: -5.5

On a number line the least number is the one farthest to the left.

Q0036 (Test 1 · Editing task choice · moderate)

Use the drop-down menus to place these numbers in order from least to greatest.

-4.5, $-\frac{3}{4}$, $\frac{2}{5}$, 3

Least , , , greatest.

Blanks: -4.5 / $-\frac{3}{4}$ / $\frac{2}{5}$ / 3 || -4.5 / $-\frac{3}{4}$ / $\frac{2}{5}$ / 3 || -4.5 / $-\frac{3}{4}$ / $\frac{2}{5}$ / 3 || -4.5 / $-\frac{3}{4}$ / $\frac{2}{5}$ / 3

Answer: -4.5, $-\frac{3}{4}$, $\frac{2}{5}$, 3

Order the numbers by their positions on a number line, from left to right.

Q0057 (Test 2 · Multiple choice · challenging)

Which number is located between 1 and 2 on a number line?

Choices: $1\frac{1}{4}$ | 3 | $2\frac{1}{4}$ | $\frac{1}{2}$

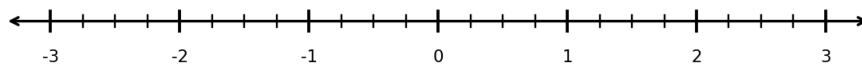
Answer: $1\frac{1}{4}$

$1\frac{1}{4}$ is more than 1 and less than 2, so it sits between them on the number line.

Q0098 (Test 3 · Graphing (GRID) · easy)

Plot these three numbers on the number line.

-2, $-\frac{1}{4}$, $\frac{1}{2}$



Answer: -2.0, -0.25, 0.5

Each small tick mark is one fourth of a unit, so count fourths from 0 to place each number.

Q0121 (Test 4 · Multiple choice · moderate)Elena recorded these four numbers in her math journal: -3.5 , $-\frac{3}{4}$, $\frac{1}{5}$, 3 .

Which number is the least?

Choices: $\frac{1}{5}$ | $-\frac{3}{4}$ | -3.5 | 3 **Answer:** -3.5

On a number line the least number is the one farthest to the left.

Q0153 (Test 5 · Multiselect · moderate)Select **all** the numbers that are less than $-1\frac{1}{2}$.Choices: 0 | 2 | $\frac{3}{4}$ | $-2\frac{1}{2}$ | $-1\frac{3}{4}$ | $-\frac{1}{2}$ **Answer:** $-2\frac{1}{2}$, $-1\frac{3}{4}$ A number is less than $-1\frac{1}{2}$ when it lies to the left of $-1\frac{1}{2}$ on the number line.**Q0205** (Test 6 · Multiple choice · easy)Which number is located between -4 and -3 on a number line?Choices: $-2\frac{3}{4}$ | $-4\frac{1}{2}$ | $-3\frac{1}{4}$ | -2 **Answer:** $-3\frac{1}{4}$ $-3\frac{1}{4}$ is more than -4 and less than -3 , so it sits between them on the number line.**Q0260** (Test 7 · Editing task choice · moderate)

Use the drop-down menus to place these numbers in order from least to greatest.

 -5.5 , $-1\frac{3}{4}$, $\frac{1}{4}$, 3 Least , , , greatest.Blanks: -5.5 / $-1\frac{3}{4}$ / $\frac{1}{4}$ / 3 || -5.5 / $-1\frac{3}{4}$ / $\frac{1}{4}$ / 3 || -5.5 / $-1\frac{3}{4}$ / $\frac{1}{4}$ / 3 || -5.5 / $-1\frac{3}{4}$ / $\frac{1}{4}$ / 3 **Answer:** -5.5 , $-1\frac{3}{4}$, $\frac{1}{4}$, 3

Order the numbers by their positions on a number line, from left to right.

Q0289 (Test 8 · Multiple choice · challenging)Nico recorded these four numbers in her math journal: -2.5 , $-\frac{3}{4}$, $\frac{3}{5}$, 2 .

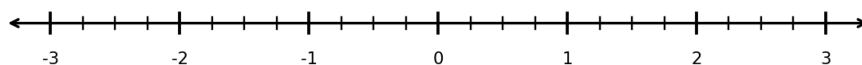
Which number is the least?

Choices: -2.5 | $\frac{3}{5}$ | $-\frac{3}{4}$ | 2 **Answer:** -2.5

On a number line the least number is the one farthest to the left.

Q0322 (Test 9 · Graphing (GRID) · moderate)

Plot these three numbers on the number line.

 $-2\frac{1}{2}$, $-1\frac{3}{4}$, -1 **Answer:** -2.5 , -1.75 , -1.0

Each small tick mark is one fourth of a unit, so count fourths from 0 to place each number.

Q0346 (Test 10 · Multiple choice · easy)

Which number is located between -2 and -1 on a number line?

Choices: $-1\frac{3}{4}$ | 0 | $-\frac{3}{4}$ | $-2\frac{1}{2}$

Answer: $-1\frac{3}{4}$

$-1\frac{3}{4}$ is more than -2 and less than -1, so it sits between them on the number line.

Q0415 (Test 11 · Multiselect · moderate)

Select **all** the numbers that are less than $-1\frac{1}{2}$.

Choices: 0 | 2 | $-1\frac{3}{4}$ | $-2\frac{1}{2}$ | $\frac{3}{4}$ | $-\frac{1}{2}$

Answer: $-2\frac{1}{2}$, $-1\frac{3}{4}$

A number is less than $-1\frac{1}{2}$ when it lies to the left of $-1\frac{1}{2}$ on the number line.

Q0423 (Test 12 · Multiple choice · challenging)

Gabriel recorded these four numbers in her math journal: -3.5, $-1\frac{1}{4}$, $\frac{1}{8}$, 3.

Which number is the least?

Choices: -3.5 | $-1\frac{1}{4}$ | $\frac{1}{8}$ | 3

Answer: -3.5

On a number line the least number is the one farthest to the left.

Q0489 (Test 13 · Editing task choice · easy)

Use the drop-down menus to place these numbers in order from least to greatest.

-4.5, $-1\frac{3}{4}$, $\frac{2}{5}$, 1

Least [[1]], [[2]], [[3]], [[4]] greatest.

Blanks: -4.5 / $-1\frac{3}{4}$ / $\frac{2}{5}$ / 1 || -4.5 / $-1\frac{3}{4}$ / $\frac{2}{5}$ / 1 || -4.5 / $-1\frac{3}{4}$ / $\frac{2}{5}$ / 1 || -4.5 / $-1\frac{3}{4}$ / $\frac{2}{5}$ / 1

Answer: -4.5, $-1\frac{3}{4}$, $\frac{2}{5}$, 1

Order the numbers by their positions on a number line, from left to right.

Q0522 (Test 14 · Multiple choice · moderate)

Which number is located between -1 and 0 on a number line?

Choices: $-\frac{3}{4}$ | $\frac{1}{4}$ | 1 | $-1\frac{1}{2}$

Answer: $-\frac{3}{4}$

$-\frac{3}{4}$ is more than -1 and less than 0, so it sits between them on the number line.

MA.6.NSO.1.2 — Given a mathematical or real-world context, represent quantities that have opposite direction using rational numbers. Compare them on a number line and explain the meaning of zero within its context.

Q0029 (Test 1 · Multiple choice · easy)

A diver off Virginia Key Beach descends 27 feet below the surface, then rises 17 feet. Sea level is represented by 0.

Which integer represents the diver's position now?

Choices: 44 | 10 | -44 | -10

Answer: -10

Below the surface is negative: $-27 + 17 = -10$.

Q0041 (Test 2 · Editing task choice · moderate)

On a January morning the temperature in Homestead was 5 degrees below zero Celsius at a research freezer.

That temperature is °C, and its opposite is °C.

Blanks: -5 / 5 / -10 / 0 || -5 / 5 / -10 / 0

Answer: -5, 5

Below zero is negative, and the opposite of a negative number is positive.

Q0070 (Test 2 · Multiple choice · challenging)

What is the opposite of $-8 \frac{1}{2}$?

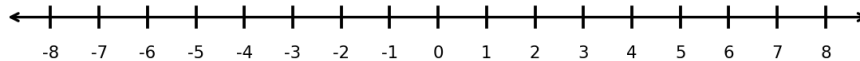
Choices: $8 \frac{1}{2}$ | $4 \frac{1}{4}$ | $-8 \frac{1}{2}$ | -17

Answer: $8 \frac{1}{2}$

The opposite of a number is the same distance from 0 but on the other side.

Q0110 (Test 3 · Graphing (GRID) · easy)

A football team gained 4 yards on one play and lost 2 yards on the next. Plot the two integers that represent these two plays, and plot 0.



Answer: 4.0, -2.0, 0.0

A gain is a positive integer, a loss is a negative integer, and 0 means no gain or loss.

Q0133 (Test 4 · Multiple choice · moderate)

A diver off Crandon Beach descends 25 feet below the surface, then rises 11 feet. Sea level is represented by 0.

Which integer represents the diver's position now?

Choices: -14 | 14 | 36 | -36

Answer: -14

Below the surface is negative: $-25 + 11 = -14$.

Q0155 (Test 5 · Multiple choice · moderate)

Sofia's prepaid transit card shows a balance of 0 dollars.

What does the 0 mean in this situation?

Choices: There is no money left on the card and no amount is owed. | The card has \$28 of credit remaining. | The card has been used 28 times. | The card is worth more than it was yesterday.

Answer: There is no money left on the card and no amount is owed.

In a balance context, 0 is the point between having credit and owing money.

Q0208 (Test 6 · Multiple choice · easy)

What is the opposite of $-5 \frac{3}{4}$?

Choices: $-5 \frac{3}{4}$ | $5 \frac{3}{4}$ | $-11 \frac{1}{2}$ | $2 \frac{7}{8}$

Answer: $5 \frac{3}{4}$

The opposite of a number is the same distance from 0 but on the other side.

Q0234 (Test 7 · Editing task choice · moderate)

On a January morning the temperature in Homestead was 11 degrees below zero Celsius at a research freezer.

That temperature is °C, and its opposite is °C.

Blanks: -11 / 11 / -22 / 0 || -11 / 11 / -22 / 0

Answer: -11, 11

Below zero is negative, and the opposite of a negative number is positive.

Q0294 (Test 8 · Multiple choice · challenging)

A diver off South Beach descends 19 feet below the surface, then rises 12 feet. Sea level is represented by 0.

Which integer represents the diver's position now?

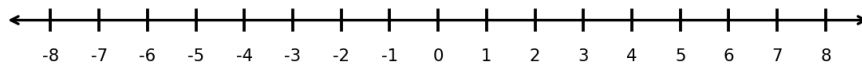
Choices: -7 | -31 | 7 | 31

Answer: -7

Below the surface is negative: $-19 + 12 = -7$.

Q0307 (Test 9 · Graphing (GRID) · moderate)

A football team gained 6 yards on one play and lost 5 yards on the next. Plot the two integers that represent these two plays, and plot 0.



Answer: 6.0, -5.0, 0.0

A gain is a positive integer, a loss is a negative integer, and 0 means no gain or loss.

Q0370 (Test 10 · Multiple choice · easy)

What is the opposite of $-5 \frac{3}{4}$?

Choices: $-11 \frac{1}{2}$ | $5 \frac{3}{4}$ | $-5 \frac{3}{4}$ | $2 \frac{7}{8}$

Answer: $5 \frac{3}{4}$

The opposite of a number is the same distance from 0 but on the other side.

Q0394 (Test 11 · Multiple choice · moderate)

Gabriel's prepaid transit card shows a balance of 0 dollars.

What does the 0 mean in this situation?

Choices: The card has been used 15 times. | The card is worth more than it was yesterday. | The card has \$15 of credit remaining. | There is no money left on the card and no amount is owed.

Answer: There is no money left on the card and no amount is owed.

In a balance context, 0 is the point between having credit and owing money.

Q0444 (Test 12 · Multiple choice · challenging)

A diver off Crandon Beach descends 17 feet below the surface, then rises 11 feet. Sea level is represented by 0.

Which integer represents the diver's position now?

Choices: -6 | 28 | -28 | 6

Answer: -6

Below the surface is negative: $-17 + 11 = -6$.

Q0470 (Test 13 · Editing task choice · easy)

On a January morning the temperature in Homestead was 12 degrees below zero Celsius at a research freezer.

That temperature is $[[1]]$ °C, and its opposite is $[[2]]$ °C.

Blanks: -12 / 12 / -24 / 0 || -12 / 12 / -24 / 0

Answer: -12, 12

Below zero is negative, and the opposite of a negative number is positive.

Q0523 (Test 14 · Multiple choice · moderate)

What is the opposite of $-8 \frac{1}{2}$?

Choices: $-8 \frac{1}{2}$ | $8 \frac{1}{2}$ | $4 \frac{1}{4}$ | -17

Answer: $8 \frac{1}{2}$

The opposite of a number is the same distance from 0 but on the other side.

MA.6.NSO.1.3 — Given a mathematical or real-world context, interpret the absolute value of a number as the distance from zero on a number line. Find the absolute value of rational numbers.

Q0020 (Test 1 · Multiple choice · easy)

What is the value of $|-15 \frac{3}{5}|$?

Choices: 31.2 | -0.06 | -15.6 | 15.6

Answer: 15.6

Absolute value is the distance from 0, so it is never negative.

Q0044 (Test 2 · Selectable hot text · moderate)

Select **all** the numbers whose absolute value is 8.

Choices: 8 | -8 | 16 | -16 | 0 | 9

Answer: 8, -8

Both 8 and -8 are 8 units from 0.

Q0109 (Test 3 · Multiple choice · challenging)

A point is graphed at $-4 \frac{3}{4}$ on a number line.

How many units is the point from 0?

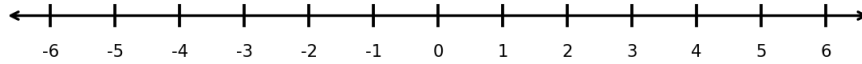
Choices: $-4 \frac{3}{4}$ | $4 \frac{3}{4}$ | $5 \frac{3}{4}$ | $2 \frac{3}{8}$

Answer: $4 \frac{3}{4}$

The distance from 0 is the absolute value of the number.

Q0113 (Test 3 · Graphing (GRID) · easy)

Plot the two numbers on the number line that are exactly 3 units from 0.



Answer: 3.0, -3.0

Two numbers are 3 units from 0: 3 and -3.

Q0116 (Test 4 · Multiple choice · moderate)

What is the value of $|-20 \frac{3}{10}|$?

Choices: -0.05 | -20.3 | 40.6 | 20.3

Answer: 20.3

Absolute value is the distance from 0, so it is never negative.

Q0154 (Test 5 · Multiselect · moderate)

Select **all** the numbers with an absolute value greater than $3 \frac{1}{2}$.

Choices: -1 | 3 | $-6 \frac{1}{2}$ | 7 | $-4 \frac{1}{2}$ | $-2 \frac{1}{2}$

Answer: $-4 \frac{1}{2}$, $-6 \frac{1}{2}$, 7

Compare how far each number is from 0, ignoring its sign.

Q0216 (Test 6 · Multiple choice · easy)

A point is graphed at $-3 \frac{3}{4}$ on a number line.

How many units is the point from 0?

Choices: $4 \frac{3}{4}$ | $3 \frac{3}{4}$ | $-3 \frac{3}{4}$ | $1 \frac{7}{8}$

Answer: $3 \frac{3}{4}$

The distance from 0 is the absolute value of the number.

Q0230 (Test 7 · Selectable hot text · moderate)

Select **all** the numbers whose absolute value is 5.

Choices: 5 | -5 | 10 | -10 | 0 | 6

Answer: 5, -5

Both 5 and -5 are 5 units from 0.

Q0279 (Test 8 · Multiple choice · challenging)

What is the value of $|-8 \frac{7}{10}|$?

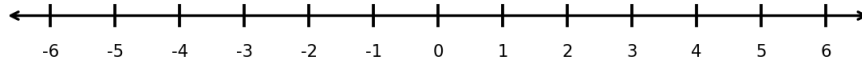
Choices: 8.7 | -0.11 | 17.4 | -8.7

Answer: 8.7

Absolute value is the distance from 0, so it is never negative.

Q0327 (Test 9 · Graphing (GRID) · moderate)

Plot the two numbers on the number line that are exactly 3 units from 0.

**Answer:** 3.0, -3.0

Two numbers are 3 units from 0: 3 and -3.

Q0360 (Test 10 · Multiple choice · easy)A point is graphed at $-3\frac{3}{4}$ on a number line.

How many units is the point from 0?

Choices: $1\frac{7}{8}$ | $-3\frac{3}{4}$ | $4\frac{3}{4}$ | $3\frac{3}{4}$ **Answer:** $3\frac{3}{4}$

The distance from 0 is the absolute value of the number.

Q0409 (Test 11 · Multiselect · moderate)Select **all** the numbers with an absolute value greater than $4\frac{1}{2}$.Choices: -1 | $-4\frac{1}{2}$ | 3 | 7 | $-6\frac{1}{2}$ | $-2\frac{1}{2}$ **Answer:** $-6\frac{1}{2}$, 7

Compare how far each number is from 0, ignoring its sign.

Q0428 (Test 12 · Multiple choice · challenging)What is the value of $|-20\frac{3}{10}|$?

Choices: -0.05 | 20.3 | -20.3 | 40.6

Answer: 20.3

Absolute value is the distance from 0, so it is never negative.

Q0459 (Test 13 · Selectable hot text · easy)Select **all** the numbers whose absolute value is 9.

Choices: 9 | -9 | 18 | -18 | 0 | 10

Answer: 9, -9

Both 9 and -9 are 9 units from 0.

Q0510 (Test 14 · Multiple choice · moderate)A point is graphed at $-2\frac{3}{4}$ on a number line.

How many units is the point from 0?

Choices: $-2\frac{3}{4}$ | $3\frac{3}{4}$ | $1\frac{3}{8}$ | $2\frac{3}{4}$ **Answer:** $2\frac{3}{4}$

The distance from 0 is the absolute value of the number.

MA.6.NSO.1.4 — Solve mathematical and real-world problems involving absolute value, including the comparison of absolute value.

Q0017 (Test 1 · Multiple choice · easy)

Four sensors on a reef near Key Biscayne report these elevations, in feet relative to sea level:

-63, -58, -34, -25

Which elevation is farthest from sea level?

Choices: -63 | -25 | -34 | -58

Answer: -63

Farthest from sea level means the greatest absolute value.

Q0052 (Test 2 · Editing task choice · moderate)

Use the drop-down menu to make the statement true.

 $|-17|$ $|4|$

Blanks: < / > / =

Answer: >

Find both absolute values, then compare those distances.

Q0107 (Test 3 · Multiple choice · challenging)What is the value of $|-17| - |10|$?

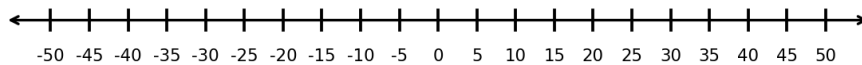
Choices: 7 | 9 | -7 | 27

Answer: 7

Take each absolute value first, then subtract.

Q0130 (Test 4 · Graphing (GRID) · easy)

Diego and a friend swim to depths with the same absolute value. One is 35 feet below sea level. Plot both depths on the number line.

**Answer: -35.0, 35.0**

Depths of -35 and 35 feet are both 35 units from 0.

Q0138 (Test 4 · Multiple choice · moderate)

Four sensors on a reef near Key Biscayne report these elevations, in feet relative to sea level:

-63, -47, -25, -12

Which elevation is farthest from sea level?

Choices: -12 | -63 | -25 | -47

Answer: -63

Farthest from sea level means the greatest absolute value.

Q0179 (Test 5 · Multiselect · moderate)

A change of more than 10 degrees from 0 sets off an alarm.

Select **all** the readings that set off the alarm.

Choices: -12 | 6 | 11 | -10 | -7 | 1

Answer: -12, 11

A reading sets off the alarm when its absolute value is greater than 10.

Q0210 (Test 6 · Multiple choice · easy)What is the value of $|-16| - |8|$?

Choices: -8 | 24 | 10 | 8

Answer: 8

Take each absolute value first, then subtract.

Q0266 (Test 7 · Editing task choice · moderate)

Use the drop-down menu to make the statement true.

 $|-6|$ $|16|$

Blanks: < / > / =

Answer: <

Find both absolute values, then compare those distances.

Q0271 (Test 8 · Multiple choice · challenging)

Four sensors on a reef near Key Biscayne report these elevations, in feet relative to sea level:

-47, -34, -25, -12

Which elevation is farthest from sea level?

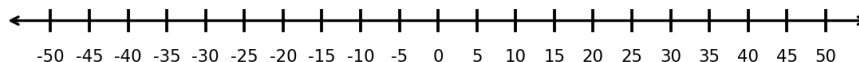
Choices: -12 | -47 | -34 | -25

Answer: -47

Farthest from sea level means the greatest absolute value.

Q0324 (Test 9 · Graphing (GRID) · moderate)

Isabella and a friend swim to depths with the same absolute value. One is 35 feet below sea level. Plot both depths on the number line.

**Answer:** -35.0, 35.0

Depths of -35 and 35 feet are both 35 units from 0.

Q0378 (Test 10 · Multiple choice · easy)What is the value of $|-16| - |10|$?

Choices: 8 | -6 | 26 | 6

Answer: 6

Take each absolute value first, then subtract.

Q0388 (Test 11 · Multiselect · moderate)

A change of more than 8 degrees from 0 sets off an alarm.

Select **all** the readings that set off the alarm.

Choices: -10 | 9 | -8 | 4 | 1 | -5

Answer: -10, 9

A reading sets off the alarm when its absolute value is greater than 8.

Q0452 (Test 12 · Multiple choice · challenging)

Four sensors on a reef near Key Biscayne report these elevations, in feet relative to sea level:

-63, -58, -34, -25

Which elevation is farthest from sea level?

Choices: -58 | -25 | -63 | -34

Answer: -63

Farthest from sea level means the greatest absolute value.

Q0478 (Test 13 · Editing task choice · easy)

Use the drop-down menu to make the statement true.

$|-19|$ $[[1]]$ $|7|$

Blanks: < / > / =

Answer: >

Find both absolute values, then compare those distances.

Q0506 (Test 14 · Multiple choice · moderate)

What is the value of $|-24| - |8|$?

Choices: 16 | 32 | -16 | 18

Answer: 16

Take each absolute value first, then subtract.

MA.6.NSO.2.1 — Multiply and divide positive multi-digit numbers with decimals to the thousandths, including using a standard algorithm with procedural fluency.

Q0038 (Test 1 · Multiple choice · easy)

What is the product of 21.8 and 0.84?

Choices: 1831.2 | 18.312 | 183.12 | 1.8312

Answer: 18.312

Multiply as with whole numbers, then place the decimal point using the total number of decimal places in the factors.

Q0050 (Test 2 · Equation editor · moderate)

A store sells guavas for \$2.40 each.

What is the total cost of 20 of them? Enter your answer in dollars.

Answer: 48

Multiply the unit price by the number of items.

Q0083 (Test 3 · Multiple choice · challenging)

What is $1.25 \div 0.25$?

Choices: 0.5 | 50 | 0.3125 | 5

Answer: 5

Multiply both numbers by the same power of 10 so the divisor is a whole number, then divide.

Q0140 (Test 4 · Editing task choice · easy)

Use the drop-down menus to complete the two statements.

$27 \times 0.25 = \text{[[1]]}$ and $27 \div 0.25 = \text{[[2]]}$

Blanks: 108 / 26.75 / 27.25 / 6.75 || 108 / 26.75 / 27.25 / 6.75

Answer: 6.75, 108

Multiplying by a number less than 1 makes the result smaller; dividing by it makes the result larger.

Q0163 (Test 5 · Multiple choice · moderate)

What is the product of 45.9 and 0.45?

Choices: 2065.5 | 20.655 | 2.0655 | 206.55

Answer: 20.655

Multiply as with whole numbers, then place the decimal point using the total number of decimal places in the factors.

Q0187 (Test 5 · Multiselect · moderate)

Select **all** the expressions with a value of 2.4.

Choices: $24 \div 10$ | 2.5×1 | 1.2×2 | 0.4×6 | $2.4 \div 0.1$ | 2.4×10

Answer: 0.4×6 , $24 \div 10$, 1.2×2

Evaluate each expression and compare it to the target value.

Q0200 (Test 6 · Multiple choice · easy)

What is $0.6 \div 0.05$?

Choices: 0.03 | 1.2 | 120 | 12

Answer: 12

Multiply both numbers by the same power of 10 so the divisor is a whole number, then divide.

Q0254 (Test 7 · Equation editor · moderate)

A store sells avocados for \$2.70 each.

What is the total cost of 6 of them? Enter your answer in dollars.

Answer: 16.2

Multiply the unit price by the number of items.

Q0295 (Test 8 · Multiple choice · challenging)

What is the product of 27.5 and 0.31?

Choices: 8.525 | 0.8525 | 852.5 | 85.25

Answer: 8.525

Multiply as with whole numbers, then place the decimal point using the total number of decimal places in the factors.

Q0320 (Test 9 · Editing task choice · moderate)

Use the drop-down menus to complete the two statements.

$$34.5 \times 0.25 = \text{[[1]]} \text{ and } 34.5 \div 0.25 = \text{[[2]]}$$

Blanks: 138 / 34.25 / 34.75 / 8.625 || 138 / 34.25 / 34.75 / 8.625

Answer: 8.625, 138

Multiplying by a number less than 1 makes the result smaller; dividing by it makes the result larger.

Q0373 (Test 10 · Multiple choice · easy)

What is $0.625 \div 0.125$?

Choices: 0.5 | 5 | 50 | 0.078125

Answer: 5

Multiply both numbers by the same power of 10 so the divisor is a whole number, then divide.

Q0408 (Test 11 · Multiselect · moderate)

Select **all** the expressions with a value of 2.4.

Choices: 2.5×1 | 2.4×10 | $24 \div 10$ | $2.4 \div 0.1$ | 0.4×6 | 1.2×2

Answer: 0.4×6 , $24 \div 10$, 1.2×2

Evaluate each expression and compare it to the target value.

Q0447 (Test 12 · Multiple choice · challenging)

What is the product of 15.2 and 0.69?

Choices: 1.0488 | 1048.8 | 104.88 | 10.488

Answer: 10.488

Multiply as with whole numbers, then place the decimal point using the total number of decimal places in the factors.

Q0487 (Test 13 · Equation editor · easy)

A store sells plantains for \$1.93 each.

What is the total cost of 16 of them? Enter your answer in dollars.

Answer: 30.88

Multiply the unit price by the number of items.

Q0514 (Test 14 · Multiple choice · moderate)

What is $2 \div 0.25$?

Choices: 0.5 | 8 | 0.8 | 80

Answer: 8

Multiply both numbers by the same power of 10 so the divisor is a whole number, then divide.

MA.6.NSO.2.2 — Extend previous understanding of multiplication and division to compute products and quotients of positive fractions by positive fractions, including mixed numbers, with procedural fluency.

Q0012 (Test 1 · Multiple choice · easy)

What is the product of $\frac{3}{5}$ and $\frac{1}{2}$?

Choices: $\frac{3}{10}$ | $\frac{4}{7}$ | $1\frac{1}{10}$ | $1\frac{1}{5}$

Answer: $\frac{3}{10}$

Multiply numerator by numerator and denominator by denominator, then simplify.

Q0046 (Test 2 · Equation editor · moderate)

Camila needs $1\frac{1}{2}$ cups of flour for one batch of pastelitos.

How many cups are needed for 4 batches? Enter your answer as a fraction or a mixed number.

Answer: 6

Multiply the amount for one batch by the number of batches.

Q0100 (Test 3 · Multiple choice · challenging)

What is $1\frac{3}{4} \div \frac{5}{7}$?

Choices: $2\frac{9}{20}$ | $20/49$ | $1\frac{1}{4}$ | $2\frac{13}{28}$

Answer: $2\frac{9}{20}$

Write the mixed number as a fraction, then multiply by the reciprocal of the divisor.

Q0144 (Test 4 · Matching table · easy)

Match each expression with its value.

Answer: $\frac{5}{6} \times \frac{4}{9} \rightarrow \frac{10}{27}$; $\frac{5}{6} \div \frac{4}{9} \rightarrow \frac{15}{8}$; $\frac{4}{9} \div \frac{5}{6} \rightarrow \frac{8}{15}$

Multiply straight across; to divide, multiply by the reciprocal.

Q0161 (Test 5 · Multiple choice · moderate)

What is the product of $\frac{2}{3}$ and $\frac{1}{2}$?

Choices: $\frac{1}{3}$ | $1\frac{1}{6}$ | $1\frac{1}{3}$ | $\frac{3}{5}$

Answer: $\frac{1}{3}$

Multiply numerator by numerator and denominator by denominator, then simplify.

Q0207 (Test 6 · Multiselect · moderate)

Select **all** the expressions equal to $\frac{1}{3}$.

Choices: $\frac{2}{3} \div \frac{1}{2}$ | $\frac{2}{6}$ | $\frac{1}{2} \times \frac{2}{3}$ | $\frac{2}{3} + \frac{1}{2}$ | $\frac{2}{3} \times \frac{1}{2}$ | $\frac{3}{5}$

Answer: $\frac{2}{3} \times \frac{1}{2}$, $\frac{1}{2} \times \frac{2}{3}$, $\frac{2}{6}$

Multiplication of fractions can be done in either order, and the product of the numerators over the product of the denominators is the same value.

Q0224 (Test 6 · Multiple choice · easy)

What is $2\frac{2}{3} \div \frac{2}{5}$?

Choices: $6\frac{2}{3}$ | $\frac{3}{20}$ | $1\frac{1}{15}$ | $3\frac{1}{15}$

Answer: $6\frac{2}{3}$

Write the mixed number as a fraction, then multiply by the reciprocal of the divisor.

Q0256 (Test 7 · Equation editor · moderate)

Andres needs $4\frac{1}{2}$ cups of flour for one batch of pastelitos.

How many cups are needed for 3 batches? Enter your answer as a fraction or a mixed number.

Answer: $13\frac{1}{2}$

Multiply the amount for one batch by the number of batches.

Q0276 (Test 8 · Multiple choice · challenging)

What is the product of $\frac{3}{5}$ and $\frac{3}{8}$?

Choices: $\frac{6}{13}$ | $\frac{9}{40}$ | $\frac{39}{40}$ | $1\frac{3}{5}$

Answer: $\frac{9}{40}$

Multiply numerator by numerator and denominator by denominator, then simplify.

Q0332 (Test 9 · Matching table · moderate)

Match each expression with its value.

Answer: $5/6 \times 4/9 \rightarrow 10/27$; $5/6 \div 4/9 \rightarrow 15/8$; $4/9 \div 5/6 \rightarrow 8/15$

Multiply straight across; to divide, multiply by the reciprocal.

Q0380 (Test 10 · Multiple choice · easy)

What is $1 \frac{3}{5} \div \frac{1}{2}$?

Choices: $4/5$ | $3 \frac{1}{5}$ | $5/16$ | $2 \frac{1}{10}$

Answer: $3 \frac{1}{5}$

Write the mixed number as a fraction, then multiply by the reciprocal of the divisor.

Q0385 (Test 11 · Multiselect · moderate)

Select **all** the expressions equal to $3/8$.

Choices: $4/6$ | $3/4 + 1/2$ | $3/4 \div 1/2$ | $3/8$ | $1/2 \times 3/4$ | $3/4 \times 1/2$

Answer: $3/4 \times 1/2$, $1/2 \times 3/4$, $3/8$

Multiplication of fractions can be done in either order, and the product of the numerators over the product of the denominators is the same value.

Q0438 (Test 12 · Multiple choice · challenging)

What is the product of $3/5$ and $3/8$?

Choices: $6/13$ | $39/40$ | $9/40$ | $1 \frac{3}{5}$

Answer: $9/40$

Multiply numerator by numerator and denominator by denominator, then simplify.

Q0480 (Test 13 · Equation editor · easy)

Marcos needs $4 \frac{1}{2}$ cups of flour for one batch of pastelitos.

How many cups are needed for 4 batches? Enter your answer as a fraction or a mixed number.

Answer: 18

Multiply the amount for one batch by the number of batches.

Q0502 (Test 14 · Multiple choice · moderate)

What is $2 \frac{3}{5} \div \frac{1}{2}$?

Choices: $1 \frac{3}{10}$ | $5/26$ | $3 \frac{1}{10}$ | $5 \frac{1}{5}$

Answer: $5 \frac{1}{5}$

Write the mixed number as a fraction, then multiply by the reciprocal of the divisor.

MA.6.NSO.2.3 — Solve multi-step real-world problems involving any of the four operations with positive multi-digit decimals or positive fractions, including mixed numbers.

Q0006 (Test 1 · Multiple choice · easy)

Andres buys 7 batteries at \$8.51 each and pays a \$2.69 delivery fee.

What is the total cost?

Choices: \$27.34 | \$78.40 | \$59.57 | \$62.26

Answer: \$62.26

Multiply the price by the number of items, then add the one-time fee.

Q0066 (Test 2 · Equation editor · moderate)

A roll holds 2 yards of ribbon. Marcos cuts 14 pieces that are each $\frac{3}{4}$ yard long.

How many yards of ribbon are left? Enter your answer as a fraction or a mixed number.

Answer: $-8 \frac{1}{2}$

Multiply to find the ribbon used, then subtract from the whole roll.

Q0097 (Test 3 · Multiple choice · challenging)

Andres pays with \$100 for 5 items that cost \$6.13 each.

How much change should Andres receive?

Choices: \$93.87 | \$69.35 | \$75.48 | \$30.65

Answer: \$69.35

Find the cost of all the items first, then subtract that from the amount paid.

Q0139 (Test 4 · Editing task choice · easy)

Sofia has \$38.75 and spends \$5.80 on a bus pass. The rest is shared equally among 5 people.

After buying the pass, Sofia has left, and each person receives .

Blanks: \$32.95 / \$44.55 / \$9.75 || \$164.75 / \$6.59 / \$7.75

Answer: \$32.95, \$6.59

Subtract the cost of the pass, then divide what is left by the number of people.

Q0167 (Test 5 · Multiple choice · moderate)

Camila buys 8 rolls of tape at \$4.11 each and pays a \$5.95 delivery fee.

What is the total cost?

Choices: \$80.48 | \$32.88 | \$38.83 | \$51.71

Answer: \$38.83

Multiply the price by the number of items, then add the one-time fee.

Q0221 (Test 6 · Multiselect · moderate)

A ticket costs \$7.31 and 6 tickets are bought.

Select **all** the expressions that give the total cost in dollars.

Choices: 7.31×6 | $7.31 + 6$ | 43.86×6 | $43.86 \div 6$ | $43.86 \div 1$ | 6×7.31

Answer: 7.31×6 , 6×7.31 , $43.86 \div 1$

The total is the price times the number of tickets; order does not matter.

Q0239 (Test 7 · Multiple choice · easy)

Marcos pays with \$20 for 5 items that cost \$8.01 each.

How much change should Marcos receive?

Choices: \$-12.04 | \$11.99 | \$40.05 | \$-20.05

Answer: \$-20.05

Find the cost of all the items first, then subtract that from the amount paid.

Q0253 (Test 7 · Equation editor · moderate)

A roll holds 4 yards of ribbon. Isabella cuts 9 pieces that are each $\frac{3}{4}$ yard long.

How many yards of ribbon are left? Enter your answer as a fraction or a mixed number.

Answer: $-2 \frac{3}{4}$

Multiply to find the ribbon used, then subtract from the whole roll.

Q0280 (Test 8 · Multiple choice · challenging)

Marcos buys 5 cases of water at \$2.63 each and pays a \$4.02 delivery fee.

What is the total cost?

Choices: \$33.25 | \$13.15 | \$17.17 | \$22.73

Answer: \$17.17

Multiply the price by the number of items, then add the one-time fee.

Q0318 (Test 9 · Editing task choice · moderate)

Diego has \$33.27 and spends \$9.11 on a bus pass. The rest is shared equally among 4 people.

After buying the pass, Diego has left, and each person receives .

Blanks: \$-3.17 / \$24.16 / \$42.38 || \$6.04 / \$8.32 / \$96.64

Answer: \$24.16, \$6.04

Subtract the cost of the pass, then divide what is left by the number of people.

Q0365 (Test 10 · Multiple choice · easy)

Diego pays with \$20 for 4 items that cost \$3.38 each.

How much change should Diego receive?

Choices: \$16.62 | \$13.52 | \$6.48 | \$9.86

Answer: \$6.48

Find the cost of all the items first, then subtract that from the amount paid.

Q0386 (Test 11 · Multiselect · moderate)

A ticket costs \$7.12 and 8 tickets are bought.

Select **all** the expressions that give the total cost in dollars.

Choices: 8×7.12 | $56.96 \div 8$ | $7.12 + 8$ | $56.96 \div 1$ | 56.96×8 | 7.12×8

Answer: 7.12×8 , 8×7.12 , $56.96 \div 1$

The total is the price times the number of tickets; order does not matter.

Q0435 (Test 12 · Multiple choice · challenging)

Ana buys 9 cases of water at \$4.45 each and pays a \$3.47 delivery fee.

What is the total cost?

Choices: \$71.28 | \$35.68 | \$43.52 | \$40.05

Answer: \$43.52

Multiply the price by the number of items, then add the one-time fee.

Q0477 (Test 13 · Equation editor · easy)

A roll holds 3 yards of ribbon. Ana cuts 8 pieces that are each $\frac{3}{4}$ yard long.

How many yards of ribbon are left? Enter your answer as a fraction or a mixed number.

Answer: -3

Multiply to find the ribbon used, then subtract from the whole roll.

Q0531 (Test 14 · Multiple choice · moderate)

Ana pays with \$100 for 3 items that cost \$5.68 each.

How much change should Ana receive?

Choices: \$88.64 | \$82.96 | \$17.04 | \$94.32

Answer: \$82.96

Find the cost of all the items first, then subtract that from the amount paid.

MA.6.NSO.3.1 — Given a mathematical or real-world context, find the greatest common factor and least common multiple of two whole numbers.

Q0032 (Test 1 · Multiple choice · easy)

What is the greatest common factor of 8 and 20?

Choices: 6 | 8 | 40 | 4

Answer: 4

List the factors of each number and choose the largest one they share.

Q0047 (Test 2 · Equation editor · moderate)

A teacher has 24 pencils and 36 erasers to put into identical bags with nothing left over.

What is the greatest number of bags she can make?

Answer: 12

The greatest number of identical bags is the greatest common factor.

Q0090 (Test 3 · Multiple choice · challenging)

Buses on one route leave Government Center every 36 minutes and buses on another route leave every 60 minutes. Both leave at 9:00 a.m.

In how many minutes will they next leave at the same time?

Choices: 12 | 2160 | 180 | 96

Answer: 180

The next time they leave together is the least common multiple of the two intervals.

Q0145 (Test 4 · Matching table · easy)

Match each description with its value.

Answer: GCF of 12 and 18 $\rightarrow$ 6; LCM of 12 and 18 $\rightarrow$ 36; GCF of 12 and 36 $\rightarrow$ 12

The GCF is the largest shared factor; the LCM is the smallest shared multiple.

Q0185 (Test 5 · Multiple choice · moderate)

What is the greatest common factor of 18 and 30?

Choices: 12 | 18 | 90 | 6

Answer: 6

List the factors of each number and choose the largest one they share.

Q0214 (Test 6 · Multiselect · moderate)

Select **all** the numbers that are common factors of 24 and 36.

Choices: 4 | 3 | 9 | 6 | 2 | 5

Answer: 2, 3, 4, 6

A common factor divides both numbers with no remainder.

Q0265 (Test 7 · Multiple choice · easy)

Buses on one route leave Government Center every 9 minutes and buses on another route leave every 15 minutes. Both leave at 9:00 a.m.

In how many minutes will they next leave at the same time?

Choices: 135 | 24 | 45 | 3

Answer: 45

The next time they leave together is the least common multiple of the two intervals.

Q0284 (Test 8 · Equation editor · moderate)

A teacher has 18 pencils and 30 erasers to put into identical bags with nothing left over.

What is the greatest number of bags she can make?

Answer: 6

The greatest number of identical bags is the greatest common factor.

Q0292 (Test 8 · Multiple choice · challenging)

What is the greatest common factor of 48 and 72?

Choices: 24 | 26 | 48 | 144

Answer: 24

List the factors of each number and choose the largest one they share.

Q0314 (Test 9 · Matching table · moderate)

Match each description with its value.

Answer: GCF of 24 and 36 → 12; LCM of 24 and 36 → 72; GCF of 24 and 72 → 24

The GCF is the largest shared factor; the LCM is the smallest shared multiple.

Q0358 (Test 10 · Multiple choice · easy)

Buses on one route leave Earlington Heights every 12 minutes and buses on another route leave every 18 minutes. Both leave at 9:00 a.m.

In how many minutes will they next leave at the same time?

Choices: 216 | 6 | 30 | 36

Answer: 36

The next time they leave together is the least common multiple of the two intervals.

Q0390 (Test 11 · Multiselect · moderate)

Select **all** the numbers that are common factors of 24 and 36.

Choices: 2 | 9 | 4 | 3 | 5 | 6

Answer: 2, 3, 4, 6

A common factor divides both numbers with no remainder.

Q0441 (Test 12 · Multiple choice · challenging)

What is the greatest common factor of 45 and 75?

Choices: 225 | 15 | 30 | 45

Answer: 15

List the factors of each number and choose the largest one they share.

Q0461 (Test 13 · Equation editor · easy)

A teacher has 9 pencils and 15 erasers to put into identical bags with nothing left over.

What is the greatest number of bags she can make?

Answer: 3

The greatest number of identical bags is the greatest common factor.

Q0521 (Test 14 · Multiple choice · moderate)

Buses on one route leave Vizcaya every 28 minutes and buses on another route leave every 42 minutes. Both leave at 9:00 a.m.

In how many minutes will they next leave at the same time?

Choices: 84 | 1176 | 70 | 14

Answer: 84

The next time they leave together is the least common multiple of the two intervals.

MA.6.NSO.3.2 — Rewrite the sum of two composite whole numbers having a common factor, as a common factor multiplied by the sum of two whole numbers.

Q0002 (Test 1 · Multiple choice · easy)

Rewrite $4 + 14$ as the greatest common factor of the two numbers multiplied by the sum of two whole numbers.

Which expression is correct?

Choices: $2(2 + 8)$ | $2(2 + 7)$ | $2(9)$ | $7(2 + 2)$

Answer: $2(2 + 7)$

The GCF of 4 and 14 is 2, and $4 = 2 \times 2$, $14 = 2 \times 7$.

Q0068 (Test 2 · Equation editor · moderate)

What is the greatest common factor you would use to rewrite $36 + 84$ as a product? Enter your answer.

Answer: 12

Find the largest number that divides both addends.

Q0082 (Test 3 · Multiple choice · challenging)

What is the value of $8(11 + 7)$?

Choices: 616 | 144 | 26 | 95

Answer: 144

Add inside the parentheses first, then multiply by the factor outside.

Q0134 (Test 4 · Editing task choice · easy)

Use the drop-down menus to rewrite $25 + 35$ using the greatest common factor.

$$25 + 35 = [[1]] \times ([[2]] + [[3]])$$

Blanks: 2 / 5 / 10 || 5 / 6 / 7 / 25 || 5 / 7 / 8 / 35

Answer: 5, 5, 7

Divide each addend by the GCF: $25 \div 5 = 5$ and $35 \div 5 = 7$.

Q0189 (Test 5 · Multiple choice · moderate)

Rewrite $12 + 30$ as the greatest common factor of the two numbers multiplied by the sum of two whole numbers.

Which expression is correct?

Choices: $6(7)$ | $6(2 + 5)$ | $2(6 + 5)$ | $6(2 + 6)$

Answer: $6(2 + 5)$

The GCF of 12 and 30 is 6, and $12 = 6 \times 2$, $30 = 6 \times 5$.

Q0225 (Test 6 · Multiselect · moderate)

Select **all** the expressions equal to $24 + 36$.

Choices: $13(2 + 3)$ | $12(2 + 4)$ | $2(12 + 3)$ | $12(2 + 3)$ | 12×5 | $24 + 36$

Answer: $12(2 + 3)$, $24 + 36$, 12×5

Use the distributive property: the factored form must multiply back to the original sum.

Q0264 (Test 7 · Multiple choice · easy)

What is the value of $9(3 + 1)$?

Choices: 27 | 36 | 13 | 28

Answer: 36

Add inside the parentheses first, then multiply by the factor outside.

Q0275 (Test 8 · Equation editor · moderate)

What is the greatest common factor you would use to rewrite $18 + 54$ as a product? Enter your answer.

Answer: 18

Find the largest number that divides both addends.

Q0306 (Test 9 · Multiple choice · challenging)

Rewrite $12 + 36$ as the greatest common factor of the two numbers multiplied by the sum of two whole numbers.

Which expression is correct?

Choices: $12(1 + 4)$ | $1(12 + 3)$ | $12(4)$ | $12(1 + 3)$

Answer: $12(1 + 3)$

The GCF of 12 and 36 is 12, and $12 = 12 \times 1$, $36 = 12 \times 3$.

Q0312 (Test 9 · Editing task choice · moderate)

Use the drop-down menus to rewrite $84 + 60$ using the greatest common factor.

$$84 + 60 = [[1]] \times ([[2]] + [[3]])$$

Blanks: 2 / 7 / 12 / 24 || 5 / 7 / 8 / 84 || 5 / 6 / 7 / 60

Answer: 12, 7, 5

Divide each addend by the GCF: $84 \div 12 = 7$ and $60 \div 12 = 5$.

Q0359 (Test 10 · Multiple choice · easy)

What is the value of $2(3 + 5)$?

Choices: 11 | 30 | 16 | 10

Answer: 16

Add inside the parentheses first, then multiply by the factor outside.

Q0410 (Test 11 · Multiselect · moderate)

Select **all** the expressions equal to $132 + 84$.

Choices: $13(11 + 7)$ | $12(11 + 7)$ | $11(12 + 7)$ | 12×18 | $12(11 + 8)$ | $132 + 84$

Answer: $12(11 + 7)$, $132 + 84$, 12×18

Use the distributive property: the factored form must multiply back to the original sum.

Q0456 (Test 12 · Multiple choice · challenging)

Rewrite $28 + 44$ as the greatest common factor of the two numbers multiplied by the sum of two whole numbers.

Which expression is correct?

Choices: $4(7 + 12)$ | $4(18)$ | $7(4 + 11)$ | $4(7 + 11)$

Answer: $4(7 + 11)$

The GCF of 28 and 44 is 4, and $28 = 4 \times 7$, $44 = 4 \times 11$.

Q0474 (Test 13 · Equation editor · easy)

What is the greatest common factor you would use to rewrite $20 + 8$ as a product? Enter your answer.

Answer: 4

Find the largest number that divides both addends.

Q0501 (Test 14 · Multiple choice · moderate)

What is the value of $4(9 + 7)$?

Choices: 64 | 252 | 20 | 43

Answer: 64

Add inside the parentheses first, then multiply by the factor outside.

MA.6.NSO.3.3 — Evaluate positive rational numbers and integers with natural number exponents.

Q0030 (Test 1 · Multiple choice · easy)

What is the value of 10^2 ?

Choices: 100 | 10 | 20 | 30

Answer: 100

10^2 means 10 multiplied by itself 2 times, not 10×2 .

Q0039 (Test 2 · Equation editor · moderate)

What is the value of 1.2^2 ? Enter your answer.

Answer: 1.44

Squaring means multiplying the number by itself.

Q0104 (Test 3 · Multiple choice · challenging)

What is the value of $(\frac{2}{5})^2$?

Choices: $\frac{4}{25}$ | $\frac{2}{5}$ | $\frac{2}{25}$ | $\frac{4}{5}$

Answer: $\frac{4}{25}$

Square the numerator and square the denominator.

Q0135 (Test 4 · Matching table · easy)

Match each power with its value.

Answer: $5^2 \rightarrow 25$; $5^3 \rightarrow 125$; $2^2 \rightarrow 4$

The exponent tells how many times the base is used as a factor.

Q0170 (Test 5 · Multiple choice · moderate)

What is the value of 6^4 ?

Choices: 30 | 24 | 216 | 1296

Answer: 1296

6^4 means 6 multiplied by itself 4 times, not 6×4 .

Q0191 (Test 6 · Multiselect · moderate)

Select **all** the expressions with a value of 25.

Choices: 5×5 | 2^5 | 5^3 | 25×1 | 5^2 | 5×2

Answer: 5^2 , 5×5 , 25×1

A power repeats the base as a factor; it is not the base times the exponent.

Q0262 (Test 7 · Multiple choice · easy)

What is the value of $(\frac{3}{4})^2$?

Choices: $\frac{9}{16}$ | $\frac{3}{16}$ | $\frac{9}{4}$ | $\frac{3}{4}$

Answer: $\frac{9}{16}$

Square the numerator and square the denominator.

Q0293 (Test 8 · Equation editor · moderate)

What is the value of 0.3^2 ? Enter your answer.

Answer: 0.09

Squaring means multiplying the number by itself.

Q0309 (Test 9 · Multiple choice · challenging)

What is the value of 2^4 ?

Choices: 10 | 19 | 8 | 16

Answer: 16

2^4 means 2 multiplied by itself 4 times, not 2×4 .

Q0343 (Test 10 · Matching table · moderate)

Match each power with its value.

Answer: $10^2 \rightarrow 100$; $10^3 \rightarrow 1000$; $2^3 \rightarrow 8$

The exponent tells how many times the base is used as a factor.

Q0366 (Test 10 · Multiple choice · easy)

What is the value of $(\frac{2}{5})^2$?

Choices: $\frac{2}{5}$ | $\frac{4}{5}$ | $\frac{4}{25}$ | $\frac{2}{25}$

Answer: $\frac{4}{25}$

Square the numerator and square the denominator.

Q0411 (Test 11 · Multiselect · moderate)

Select **all** the expressions with a value of 36.

Choices: 6^2 | 6^3 | 36×1 | 6×6 | 6×2 | 2^6

Answer: 6^2 , 6×6 , 36×1

A power repeats the base as a factor; it is not the base times the exponent.

Q0431 (Test 12 · Multiple choice · challenging)

What is the value of 2^3 ?

Choices: 4 | 11 | 6 | 8

Answer: 8

2^3 means 2 multiplied by itself 3 times, not 2×3 .

Q0484 (Test 13 · Equation editor · easy)

What is the value of 0.3^2 ? Enter your answer.

Answer: 0.09

Squaring means multiplying the number by itself.

Q0518 (Test 14 · Multiple choice · moderate)

What is the value of $(\frac{1}{2})^2$?

Choices: $\frac{1}{4}$ | $\frac{3}{8}$ | $\frac{1}{8}$ | $\frac{1}{2}$

Answer: $\frac{1}{4}$

Square the numerator and square the denominator.

MA.6.NSO.3.4 — Express composite whole numbers as a product of prime factors with natural number exponents.

Q0004 (Test 1 · Multiple choice · easy)

Which expression shows the prime factorization of 45?

Choices: $3^2 \times 5$ | 2×11 | $2 \times 3^2 \times 5$ | 47

Answer: $3^2 \times 5$

Break the number into factors until every factor is prime, then use exponents.

Q0073 (Test 2 · Equation editor · moderate)

Enter the prime factorization of 84. Use ^ for exponents and $\times$ between factors, for example $2^2 \times 3$.

Answer: $2^2 \times 3 \times 7$

Divide by the smallest prime repeatedly until only primes remain.

Q0106 (Test 3 · Multiple choice · challenging)

A number has the prime factorization $2^2 \times 3^2 \times 11$.

What is the number?

Choices: 396 | 792 | 21 | 198

Answer: 396

Multiply all of the prime factors together.

Q0149 (Test 4 · Selectable hot text · easy)

Select **all** the prime factors of 50.

Choices: 1 | 2 | 4 | 5 | 50

Answer: 2, 5

A prime factor divides the number and has exactly two factors.

Q0169 (Test 5 · Multiple choice · moderate)

Which expression shows the prime factorization of 126?

Choices: 2^7 | $2^2 \times 3^2 \times 7$ | $2 \times 3^2 \times 7$ | $3^2 \times 7$

Answer: $2 \times 3^2 \times 7$

Break the number into factors until every factor is prime, then use exponents.

Q0211 (Test 6 · Multiselect · moderate)

Select **all** the expressions with a value of 84.

Choices: $2^2 \times 3 \times 7$ | $84 + 1$ | 84×2 | 84×1 | $2 \times 2 \times 3 \times 7$ | $2^2 \times 3^2 \times 7$

Answer: $2^2 \times 3 \times 7$, $2 \times 2 \times 3 \times 7$, 84×1

Each correct expression multiplies back to the original number.

Q0255 (Test 7 · Multiple choice · easy)

A number has the prime factorization 2×5^2 .

What is the number?

Choices: 50 | 100 | 25 | 12

Answer: 50

Multiply all of the prime factors together.

Q0273 (Test 8 · Equation editor · moderate)

Enter the prime factorization of 84. Use ^ for exponents and $\times$ between factors, for example $2^2 \times 3$.

Answer: $2^2 \times 3 \times 7$

Divide by the smallest prime repeatedly until only primes remain.

Q0328 (Test 9 · Multiple choice · challenging)

Which expression shows the prime factorization of 396?

Choices: 2×199 | $2^3 \times 3^2 \times 11$ | $2^2 \times 3^2 \times 11$ | $2 \times 3^2 \times 11$

Answer: $2^2 \times 3^2 \times 11$

Break the number into factors until every factor is prime, then use exponents.

Q0377 (Test 10 · Selectable hot text · moderate)

Select **all** the prime factors of 126.

Choices: 1 | 2 | 3 | 4 | 7 | 126

Answer: 2, 3, 7

A prime factor divides the number and has exactly two factors.

Q0406 (Test 11 · Multiple choice · easy)

A number has the prime factorization $3^2 \times 5$.

What is the number?

Choices: 90 | 45 | 22 | 11

Answer: 45

Multiply all of the prime factors together.

Q0412 (Test 11 · Multiselect · moderate)

Select **all** the expressions with a value of 132.

Choices: 132×1 | $2 \times 2 \times 3 \times 11$ | 132×2 | $2^2 \times 3 \times 11$ | $2^2 \times 3^2 \times 11$ | $132 + 1$

Answer: $2^2 \times 3 \times 11$, $2 \times 2 \times 3 \times 11$, 132×1

Each correct expression multiplies back to the original number.

Q0433 (Test 12 · Multiple choice · challenging)

Which expression shows the prime factorization of 252?

Choices: 2×127 | $2^2 \times 3^2 \times 7$ | $2 \times 3^2 \times 7$ | $2^3 \times 3^2 \times 7$

Answer: $2^2 \times 3^2 \times 7$

Break the number into factors until every factor is prime, then use exponents.

Q0476 (Test 13 · Equation editor · easy)

Enter the prime factorization of 50. Use ^ for exponents and × between factors, for example $2^2 \times 3$.

Answer: 2×5^2

Divide by the smallest prime repeatedly until only primes remain.

Q0530 (Test 14 · Multiple choice · moderate)

A number has the prime factorization $2 \times 3^2 \times 5$.

What is the number?

Choices: 180 | 45 | 90 | 13

Answer: 90

Multiply all of the prime factors together.

MA.6.NSO.3.5 — Rewrite positive rational numbers in different but equivalent forms including fractions, terminating decimals and percentages.

Q0014 (Test 1 · Multiple choice · easy)

Which percent is equivalent to $\frac{1}{4}$?

Choices: 400% | 25% | 35% | 2.5%

Answer: 25%

Write the fraction as a decimal, then multiply by 100.

Q0074 (Test 2 · Equation editor · moderate)

Write 0.6 as a percent. Enter your answer without the % symbol.

Answer: 60

Multiply the decimal by 100 to write it as a percent.

Q0105 (Test 3 · Multiple choice · challenging)

Which fraction is equivalent to 80%, written in simplest form?

Choices: 8 | $\frac{8}{5}$ | $\frac{5}{4}$ | $\frac{4}{5}$

Answer: $\frac{4}{5}$

A percent means hundredths, so write it over 100 and simplify.

Q0131 (Test 4 · Matching table · easy)

Match each number with the equivalent percent.

Answer: $\frac{1}{4} \rightarrow 25\%$; $0.5 \rightarrow 50\%$; $\frac{7}{10} \rightarrow 70\%$

Write each number as a decimal and multiply by 100 to get its percent.

Q0171 (Test 5 · Multiple choice · moderate)

Which percent is equivalent to $\frac{3}{20}$?

Choices: 15% | 666.6667% | 25% | 1.5%

Answer: 15%

Write the fraction as a decimal, then multiply by 100.

Q0215 (Test 6 · Multiselect · moderate)

Select **all** the forms that are equivalent to $\frac{3}{20}$.

Choices: 0.15 | 15% | 1.5 | $\frac{20}{3}$ | 150% | $\frac{3}{20}$

Answer: $\frac{3}{20}$, 0.15, 15%

Equivalent forms name the same value as a fraction, a decimal and a percent.

Q0233 (Test 7 · Multiple choice · easy)

Which fraction is equivalent to 60%, written in simplest form?

Choices: $\frac{3}{5}$ | $\frac{6}{5}$ | $\frac{5}{3}$ | 6

Answer: $\frac{3}{5}$

A percent means hundredths, so write it over 100 and simplify.

Q0283 (Test 8 · Equation editor · moderate)

Write 0.75 as a percent. Enter your answer without the % symbol.

Answer: 75

Multiply the decimal by 100 to write it as a percent.

Q0310 (Test 9 · Multiple choice · challenging)

Which percent is equivalent to $\frac{1}{4}$?

Choices: 35% | 2.5% | 400% | 25%

Answer: 25%

Write the fraction as a decimal, then multiply by 100.

Q0345 (Test 10 · Matching table · moderate)

Match each number with the equivalent percent.

Answer: $\frac{1}{2} \rightarrow 50\%$; $0.6 \rightarrow 60\%$; $\frac{1}{4} \rightarrow 25\%$

Write each number as a decimal and multiply by 100 to get its percent.

Q0383 (Test 11 · Multiple choice · easy)

Which fraction is equivalent to 45%, written in simplest form?

Choices: $\frac{9}{2}$ | $\frac{9}{20}$ | $\frac{20}{9}$ | $\frac{9}{10}$

Answer: $\frac{9}{20}$

A percent means hundredths, so write it over 100 and simplify.

Q0445 (Test 12 · Multiselect · moderate)

Select **all** the forms that are equivalent to $\frac{5}{8}$.

Choices: 6.25 | $\frac{8}{5}$ | $\frac{5}{8}$ | 0.625 | 625% | 62.5%

Answer: $\frac{5}{8}$, 0.625, 62.5%

Equivalent forms name the same value as a fraction, a decimal and a percent.

Q0448 (Test 12 · Multiple choice · challenging)

Which percent is equivalent to $\frac{3}{20}$?

Choices: 15% | 666.6667% | 1.5% | 25%

Answer: 15%

Write the fraction as a decimal, then multiply by 100.

Q0491 (Test 13 · Equation editor · easy)

Write 0.75 as a percent. Enter your answer without the % symbol.

Answer: 75

Multiply the decimal by 100 to write it as a percent.

Q0499 (Test 14 · Multiple choice · moderate)

Which fraction is equivalent to 60%, written in simplest form?

Choices: $\frac{5}{3}$ | 6 | $\frac{3}{5}$ | $\frac{6}{5}$

Answer: $\frac{3}{5}$

A percent means hundredths, so write it over 100 and simplify.

MA.6.NSO.4.1 — Apply and extend previous understandings of operations with whole numbers to add and subtract integers with procedural fluency.

Q0025 (Test 1 · Multiple choice · easy)

What is the value of $13 + (-6)$?

Choices: 9 | 7 | -7 | 19

Answer: 7

Adding a negative number moves left on the number line; adding a positive number moves right.

Q0055 (Test 2 · Equation editor · moderate)

At 6 a.m. the temperature was -12°F in a walk-in freezer. It rose 16°F and then fell 10°F .

What is the temperature now, in $^{\circ}\text{F}$? Enter your answer.

Answer: -6

Add the rise and subtract the fall from the starting temperature.

Q0088 (Test 3 · Multiple choice · challenging)

What is the value of $-29 - (-22)$?

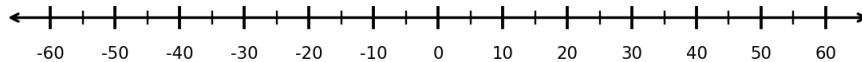
Choices: -4 | -51 | 7 | -7

Answer: -7

Subtracting a number is the same as adding its opposite.

Q0115 (Test 4 · Graphing (GRID) · easy)

Plot the value of $-35 + 30$ on the number line.



Answer: -5.0

Combine the two integers, then find that value on the number line.

Q0158 (Test 5 · Multiple choice · moderate)

What is the value of $-7 + (25)$?

Choices: -18 | 32 | 18 | -32

Answer: 18

Adding a negative number moves left on the number line; adding a positive number moves right.

Q0202 (Test 6 · Multiselect · moderate)

Select **all** the expressions with a value of 6.

Choices: $21 - (15)$ | $21 + (-15)$ | $21 - (-15)$ | $21 \times (-15)$ | $-21 + (-15)$ | $(-15) + 21$

Answer: $21 + (-15)$, $(-15) + 21$, $21 - (15)$

Addition can be done in either order, and subtracting a number equals adding its opposite.

Q0252 (Test 7 · Multiple choice · easy)

What is the value of $-3 - (-14)$?

Choices: -17 | -11 | 11 | 14

Answer: 11

Subtracting a number is the same as adding its opposite.

Q0277 (Test 8 · Equation editor · moderate)

At 6 a.m. the temperature was -5°F in a walk-in freezer. It rose 20°F and then fell 15°F .

What is the temperature now, in $^{\circ}\text{F}$? Enter your answer.

Answer: 0

Add the rise and subtract the fall from the starting temperature.

Q0338 (Test 9 · Multiple choice · challenging)

What is the value of $12 + (16)$?

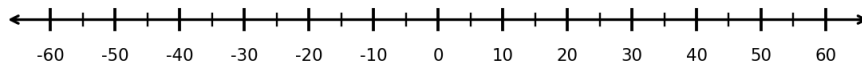
Choices: -4 | -28 | 28 | 30

Answer: 28

Adding a negative number moves left on the number line; adding a positive number moves right.

Q0350 (Test 10 · Graphing (GRID) · moderate)

Plot the value of $-40 + 30$ on the number line.



Answer: -10.0

Combine the two integers, then find that value on the number line.

Q0414 (Test 11 · Multiple choice · easy)

What is the value of $14 - (-3)$?

Choices: -17 | 11 | 20 | 17

Answer: 17

Subtracting a number is the same as adding its opposite.

Q0436 (Test 12 · Multiselect · moderate)

Select **all** the expressions with a value of -29.

Choices: $-17 - (-12)$ | $(-12) + -17$ | $17 + (-12)$ | $-17 \times (-12)$ | $-17 - (12)$ | $-17 + (-12)$

Answer: $-17 + (-12)$, $(-12) + -17$, $-17 - (12)$

Addition can be done in either order, and subtracting a number equals adding its opposite.

Q0479 (Test 13 · Multiple choice · challenging)

What is the value of $30 + (-2)$?

Choices: 28 | -28 | 30 | 32

Answer: 28

Adding a negative number moves left on the number line; adding a positive number moves right.

Q0490 (Test 13 · Equation editor · easy)

At 6 a.m. the temperature was -6°F in a walk-in freezer. It rose 20°F and then fell 6°F .

What is the temperature now, in $^{\circ}\text{F}$? Enter your answer.

Answer: 8

Add the rise and subtract the fall from the starting temperature.

Q0519 (Test 14 · Multiple choice · moderate)

What is the value of $-21 - (15)$?

Choices: -6 | -36 | 36 | -33

Answer: -36

Subtracting a number is the same as adding its opposite.

MA.6.NSO.4.2 — Apply and extend previous understandings of operations with whole numbers to multiply and divide integers with procedural fluency.

Q0015 (Test 1 · Multiple choice · easy)

What is the value of $(8)(-4)$?

Choices: -32 | -28 | 4 | 32

Answer: -32

The product of two numbers with the same sign is positive; with different signs it is negative.

Q0059 (Test 2 · Equation editor · moderate)

A pond loses 8 centimeters of water each week during a dry spell.

Which integer represents the total change in water level after 5 weeks? Enter your answer.

Answer: -40

A loss each week is a negative change repeated, so multiply.

Q0095 (Test 3 · Multiple choice · challenging)

What is the value of $-45 \div (9)$?

Choices: 5 | -54 | -5 | 9

Answer: -5

Divide the absolute values, then decide the sign from the two signs.

Q0143 (Test 4 · Editing task choice · easy)

Use the drop-down menus to complete the equations.

$(-7)(-8) = \boxed{1}$ and $(56) \div (-8) = \boxed{2}$

Blanks: -56 / -15 / 1 / 56 || -8 / -7 / 7 / 56

Answer: 56, -7

Same signs give a positive result; different signs give a negative result.

Q0180 (Test 5 · Multiple choice · moderate)

What is the value of $(8)(8)$?

Choices: 16 | -64 | 64 | 68

Answer: 64

The product of two numbers with the same sign is positive; with different signs it is negative.

Q0193 (Test 6 · Multiselect · moderate)

Select **all** the expressions with a value of 45.

Choices: $(-5) \times (-9)$ | $(-9) + (-5)$ | $(-9) \div (-5)$ | $(9) \times (-5)$ | $(-9) \times (-5)$ | $(9) \times (5)$

Answer: $(-9) \times (-5)$, $(-5) \times (-9)$, $(9) \times (5)$

Multiplying two negatives gives the same product as multiplying the two positives.

Q0244 (Test 7 · Multiple choice · easy)What is the value of $24 \div (-6)$?

Choices: 4 | 30 | -4 | -6

Answer: -4

Divide the absolute values, then decide the sign from the two signs.

Q0268 (Test 8 · Equation editor · moderate)

A pond loses 4 centimeters of water each week during a dry spell.

Which integer represents the total change in water level after 4 weeks? Enter your answer.

Answer: -16

A loss each week is a negative change repeated, so multiply.

Q0323 (Test 9 · Multiple choice · challenging)What is the value of $(12)(-9)$?

Choices: -108 | 3 | -104 | 108

Answer: -108

The product of two numbers with the same sign is positive; with different signs it is negative.

Q0372 (Test 10 · Editing task choice · moderate)

Use the drop-down menus to complete the equations.

 $(-6)(-8) = \boxed{1}$ and $(48) \div (-8) = \boxed{2}$

Blanks: -48 / -14 / 2 / 48 || -8 / -6 / 6 / 48

Answer: 48, -6

Same signs give a positive result; different signs give a negative result.

Q0395 (Test 11 · Multiple choice · easy)What is the value of $-81 \div (9)$?

Choices: 9 | -90 | -7 | -9

Answer: -9

Divide the absolute values, then decide the sign from the two signs.

Q0443 (Test 12 · Multiselect · moderate)Select **all** the expressions with a value of 48.Choices: $(6) \times (8)$ | $(-6) \div (-8)$ | $(-6) \times (-8)$ | $(-6) + (-8)$ | $(-8) \times (-6)$ | $(6) \times (-8)$ **Answer: $(-6) \times (-8)$, $(-8) \times (-6)$, $(6) \times (8)$**

Multiplying two negatives gives the same product as multiplying the two positives.

Q0485 (Test 13 · Multiple choice · challenging)What is the value of $(-9)(9)$?

Choices: -77 | 0 | 81 | -81

Answer: -81

The product of two numbers with the same sign is positive; with different signs it is negative.

Q0505 (Test 14 · Equation editor · easy)

A pond loses 4 centimeters of water each week during a dry spell.

Which integer represents the total change in water level after 6 weeks? Enter your answer.

Answer: -24

A loss each week is a negative change repeated, so multiply.

Q0527 (Test 14 · Multiple choice · moderate)

What is the value of $-48 \div (-12)$?

Choices: 4 | -36 | -4 | -12

Answer: 4

Divide the absolute values, then decide the sign from the two signs.

Algebraic Reasoning

MA.6.AR.1.1 — Given a mathematical or real-world context, translate written descriptions into algebraic expressions and translate algebraic expressions into written descriptions.

Q0069 (Test 2 · Multiple choice · easy)

Which expression represents the sum of a number x and 10, divided by 2?

Choices: $x/2 + 210$ | $2(x + 10)$ | $x + 10/2$ | $(x + 10)/2$

Answer: $(x + 10)/2$

Translate each word phrase in order, keeping subtraction and division in the order they are written.

Q0093 (Test 3 · Editing task choice · moderate)

Use the drop-down menu to complete the sentence.

The expression $10x$ represents the product of 10 and a number x .

Blanks: $10x$ / $x + 10$ / $x - 10$

Answer: $10x$

Match each word in the description to an operation in the expression.

Q0120 (Test 4 · Multiple choice · challenging)

Nico wrote the expression $x - 8$.

Which written description matches the expression?

Choices: a number x divided by 8 | 8 less than a number x | the product of a number x and 8 | a number x decreased by 8

Answer: a number x decreased by 8

Read the expression one operation at a time and match it to the words.

Q0177 (Test 5 · Matching table · easy)

Match each written description with the expression it represents.

Answer: a number x decreased by 11 $\rightarrow x - 11$; the sum of a number x and 11, divided by 2 $\rightarrow (x + 11)/2$; the product of 11 and a number $x \rightarrow 11x$

Translate each phrase one operation at a time.

Q0217 (Test 6 · Multiple choice · moderate)

Which expression represents a number x decreased by 4?

Choices: $x/4$ | $4 - x$ | $x + 4$ | $x - 4$

Answer: $x - 4$

Translate each word phrase in order, keeping subtraction and division in the order they are written.

Q0232 (Test 7 · Multiselect · moderate)

Select **all** the expressions equivalent to 2 more than 5 times a number x .

Choices: $7x$ | $5x + 2$ | $5(x + 2)$ | $2 + 5x$ | $x*5 + 2$ | $5x - 2$

Answer: $5x + 2$, $2 + 5x$, $x*5 + 2$

Addition can be written in either order, but the coefficient and the constant cannot be combined.

Q0302 (Test 8 · Multiple choice · easy)

Ximena wrote the expression $x/6$.

Which written description matches the expression?

Choices: the quotient of a number x and 6 | 6 less than a number x | a number x divided by 6 | the product of a number x and 6

Answer: the quotient of a number x and 6

Read the expression one operation at a time and match it to the words.

Q0329 (Test 9 · Editing task choice · moderate)

Use the drop-down menu to complete the sentence.

The expression $[[1]]$ represents 6 more than a number x .

Blanks: $6 - x$ / $x + 6$ / $x - 6$

Answer: $x + 6$

Match each word in the description to an operation in the expression.

Q0356 (Test 10 · Multiple choice · challenging)

Which expression represents a number x decreased by 9?

Choices: $x + 9$ | $9 - x$ | $x - 9$ | $x/9$

Answer: $x - 9$

Translate each word phrase in order, keeping subtraction and division in the order they are written.

Q0381 (Test 11 · Matching table · moderate)

Match each written description with the expression it represents.

Answer: the quotient of a number x and 6 $\rightarrow x/6$; a number x decreased by 6 $\rightarrow x - 6$; the sum of a number x and 6, divided by 2 $\rightarrow (x + 6)/2$

Translate each phrase one operation at a time.

Q0446 (Test 12 · Multiple choice · easy)

Lucia wrote the expression $x - 2$.

Which written description matches the expression?

Choices: the product of a number x and 2 | 2 less than a number x | a number x decreased by 2 | a number x divided by 2

Answer: a number x decreased by 2

Read the expression one operation at a time and match it to the words.

Q0483 (Test 13 · Multiselect · moderate)

Select **all** the expressions equivalent to 6 more than 4 times a number x .

Choices: $4x - 6$ | $4x + 6$ | $10x$ | $x*4 + 6$ | $6 + 4x$ | $4(x + 6)$

Answer: $4x + 6$, $6 + 4x$, $x*4 + 6$

Addition can be written in either order, but the coefficient and the constant cannot be combined.

MA.6.AR.1.2 — Translate a real-world written description into an algebraic inequality in the form of $x > a$, $x < a$, $x \geq a$ or $x \leq a$. Represent the inequality on a number line.

Q0018 (Test 1 · Multiple choice · easy)

A parking garage charges more than 20 dollars for the whole day.

Let x represent the cost in dollars. Which inequality represents this situation?

Choices: $x < 20$ | $x \geq 20$ | $x > 20$ | $x \leq 20$

Answer: $x > 20$

"At least" and "no more than" include the boundary value; "more than" and "under" do not.

Q0102 (Test 3 · Selectable hot text · moderate)

Select **all** the values of x that make $x \leq 48$ true.

Choices: 44 | 47 | 48 | 49 | 53 | 57

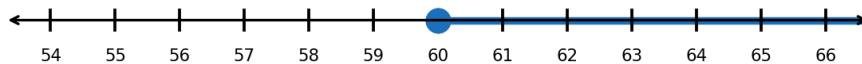
Answer: 44, 47, 48

Test each value in the inequality and keep the ones that make it true.

Q0119 (Test 4 · Multiple choice · challenging)

The number line shows the solution to an inequality.

Which inequality does the graph represent?



Choices: $x \geq 60$ | $x < 60$ | $x \leq 60$ | $x > 60$

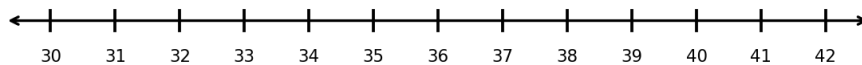
Answer: $x \geq 60$

A closed circle includes the endpoint; an open circle does not. The shading shows which direction makes the inequality true.

Q0160 (Test 5 · Graphing (GRID) · easy)

To ride the roller coaster at Kendall Indian Hammocks, a rider must be at least 36 inches tall.

Plot the boundary value of the solution on the number line.



Answer: 36.0

The boundary value is 36, the number the inequality compares x to.

Q0223 (Test 6 · Multiple choice · moderate)

To ride the roller coaster at Tropical Park, a rider must be at least 36 inches tall.

Let x represent the height in inches. Which inequality represents this situation?

Choices: $x > 36$ | $x \leq 36$ | $x < 36$ | $x \geq 36$

Answer: $x \geq 36$

"At least" and "no more than" include the boundary value; "more than" and "under" do not.

Q0247 (Test 7 · Multiselect · moderate)

A carry-on bag must weigh no more than 60 pounds.

Select **all** the values of x that make the situation true.

Choices: 66 | 60 | 58 | 53 | 63 | 71

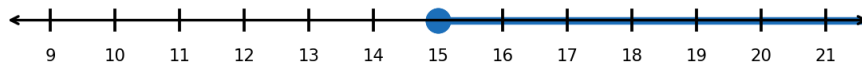
Answer: 53, 58, 60

Substitute each value and check whether the inequality is true.

Q0287 (Test 8 · Multiple choice · easy)

The number line shows the solution to an inequality.

Which inequality does the graph represent?



Choices: $x \geq 15$ | $x \leq 15$ | $x > 15$ | $x < 15$

Answer: $x \geq 15$

A closed circle includes the endpoint; an open circle does not. The shading shows which direction makes the inequality true.

Q0339 (Test 9 · Selectable hot text · moderate)

Select **all** the values of x that make $x > 60$ true.

Choices: 56 | 59 | 60 | 61 | 65 | 69

Answer: 61, 65, 69

Test each value in the inequality and keep the ones that make it true.

Q0354 (Test 10 · Multiple choice · challenging)

A carry-on bag must weigh no more than 48 pounds.

Let x represent the weight in pounds. Which inequality represents this situation?

Choices: $x < 48$ | $x > 48$ | $x \leq 48$ | $x \geq 48$

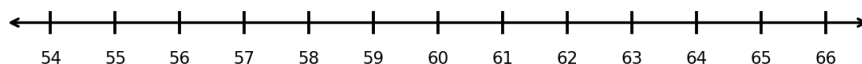
Answer: $x \leq 48$

"At least" and "no more than" include the boundary value; "more than" and "under" do not.

Q0402 (Test 11 · Graphing (GRID) · moderate)

A carry-on bag must weigh no more than 60 pounds.

Plot the boundary value of the solution on the number line.



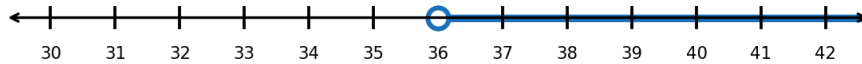
Answer: 60.0

The boundary value is 60, the number the inequality compares x to.

Q0453 (Test 12 · Multiple choice · easy)

The number line shows the solution to an inequality.

Which inequality does the graph represent?



Choices: $x \leq 36$ | $x < 36$ | $x \geq 36$ | $x > 36$

Answer: $x > 36$

A closed circle includes the endpoint; an open circle does not. The shading shows which direction makes the inequality true.

Q0492 (Test 13 · Multiselect · moderate)

A carry-on bag must weigh no more than 36 pounds.

Select **all** the values of x that make the situation true.

Choices: 47 | 36 | 34 | 42 | 39 | 29

Answer: 29, 34, 36

Substitute each value and check whether the inequality is true.

Q0524 (Test 14 · Multiple choice · challenging)

A carry-on bag must weigh no more than 15 pounds.

Let x represent the weight in pounds. Which inequality represents this situation?

Choices: $x > 15$ | $x \leq 15$ | $x < 15$ | $x \geq 15$

Answer: $x \leq 15$

"At least" and "no more than" include the boundary value; "more than" and "under" do not.

MA.6.AR.1.3 — Evaluate algebraic expressions using substitution and order of operations.

Q0037 (Test 1 · Multiple choice · easy)

What is the value of $5x^2 - 6$ when $x = 8$?

Choices: 74 | 314 | 290 | 1594

Answer: 314

Evaluate the exponent before multiplying, then subtract.

Q0051 (Test 2 · Equation editor · moderate)

A rideshare charges a base fare of \$9.00 plus \$4.00 per mile.

What is the cost, in dollars, of a ride of $x = 2$ miles? Enter your answer.

Answer: 17

Multiply the rate by the miles, then add the base fare.

Q0136 (Test 4 · Multiple choice · challenging)

What is the value of $(2x + 2) \div 4$ when $x = 7$?

Choices: 16 | 4 | 5 | 14

Answer: 4

Work inside the parentheses first, then divide.

Q0186 (Test 5 · Matching table · easy)

Each expression is evaluated for $x = 8$. Match each expression with its value.

Answer: $6x + 4 \rightarrow 52$; $6x^2 \rightarrow 384$; $4 - x \rightarrow -4$

Substitute the value of x , then follow the order of operations.

Q0209 (Test 6 · Multiple choice · moderate)

What is the value of $3x^2 - 5$ when $x = 6$?

Choices: 319 | 93 | 31 | 103

Answer: 103

Evaluate the exponent before multiplying, then subtract.

Q0259 (Test 7 · Multiselect · moderate)

Select **all** the expressions with a value of 7 when $x = 2$.

Choices: $2(x) + 3$ | $2(x + 3)$ | $2x + 3$ | $2x - 3$ | $3 + 2x$ | $5x$

Answer: $2x + 3$, $3 + 2x$, $2(x) + 3$

Substitute x and evaluate each expression using the order of operations.

Q0300 (Test 8 · Multiple choice · easy)

What is the value of $(2x + 8) \div 2$ when $x = 8$?

Choices: 24 | 12 | 16 | 20

Answer: 12

Work inside the parentheses first, then divide.

Q0340 (Test 9 · Equation editor · moderate)

A rideshare charges a base fare of \$9.00 plus \$2.00 per mile.

What is the cost, in dollars, of a ride of $x = 3$ miles? Enter your answer.

Answer: 15

Multiply the rate by the miles, then add the base fare.

Q0353 (Test 10 · Multiple choice · challenging)

What is the value of $3x^2 - 12$ when $x = 3$?

Choices: 6 | 69 | -9 | 15

Answer: 15

Evaluate the exponent before multiplying, then subtract.

Q0384 (Test 11 · Matching table · moderate)

Each expression is evaluated for $x = 3$. Match each expression with its value.

Answer: $6x + 3 \rightarrow 21$; $6x^2 \rightarrow 54$; $3 - x \rightarrow 0$

Substitute the value of x , then follow the order of operations.

Q0442 (Test 12 · Multiple choice · easy)

What is the value of $(3x + 49) \div 5$ when $x = 2$?

Choices: 50 | 11 | 15 | 55

Answer: 11

Work inside the parentheses first, then divide.

Q0464 (Test 13 · Multiselect · moderate)Select **all** the expressions with a value of 22 when $x = 6$.Choices: $10 + 2x$ | $2x + 10$ | $2(x + 10)$ | $2x - 10$ | $12x$ | $2(x) + 10$ **Answer:** $2x + 10$, $10 + 2x$, $2(x) + 10$ Substitute x and evaluate each expression using the order of operations.**Q0513** (Test 14 · Multiple choice · challenging)What is the value of $5x^2 - 7$ when $x = 8$?

Choices: 285 | 313 | 1593 | 73

Answer: 313

Evaluate the exponent before multiplying, then subtract.

MA.6.AR.1.4 — Apply the properties of operations to generate equivalent algebraic expressions with integer coefficients.**Q0027** (Test 1 · Multiple choice · easy)Which expression is equivalent to $2(x + 5)$?Choices: $2x + 5$ | $x + 10$ | $2x + 10$ | $2x + 7$ **Answer:** $2x + 10$

Multiply the number outside the parentheses by each term inside.

Q0054 (Test 2 · Editing task choice · moderate)

Use the drop-down menus to complete the equivalent expression.

$$8(6x + 6) = \boxed{[1]}x + \boxed{[2]}$$

Blanks: 6 / 14 / 48 / 54 || 6 / 14 / 48 / 54

Answer: 48, 48

Distribute the factor to both terms inside the parentheses.

Q0085 (Test 3 · Multiple choice · challenging)Which expression is equivalent to $2x + 12 + 4x$?Choices: $8x + 12$ | $6x + 12$ | $6x + 12x$ | $18x$ **Answer:** $6x + 12$ Only like terms can be combined, so add the x -terms and leave the constant.**Q0162** (Test 5 · Matching table · easy)

Match each expression with an equivalent expression.

Answer: $2(x + 2) \rightarrow 2x + 4$; $2x + 7x \rightarrow 9x$; $2x + 2 - 2 \rightarrow 2x$

Use the distributive property and combine like terms.

Q0203 (Test 6 · Multiple choice · moderate)Which expression is equivalent to $5(x + 6)$?Choices: $x + 30$ | $5x + 11$ | $5x + 30$ | $5x + 6$ **Answer:** $5x + 30$

Multiply the number outside the parentheses by each term inside.

Q0240 (Test 7 · Multiselect · moderate)Select **all** the expressions equivalent to $8(x + 9)$.Choices: $8(x + 10)$ | $72 + 8x$ | $8x + 9$ | $8x + 72$ | $17(x)$ | $8(x + 9)$ **Answer:** $8(x + 9)$, $8x + 72$, $72 + 8x$ Equivalent expressions have the same value for every value of x .

Q0281 (Test 8 · Multiple choice · easy)Which expression is equivalent to $7x + 12 + 8x$?Choices: $15x + 12x$ | $15x + 12$ | $56x + 12$ | $27x$ **Answer:** $15x + 12$

Only like terms can be combined, so add the x-terms and leave the constant.

Q0313 (Test 9 · Editing task choice · moderate)

Use the drop-down menus to complete the equivalent expression.

$$8(5x + 9) = \boxed{[1]}x + \boxed{[2]}$$

Blanks: 5 / 13 / 40 / 49 || 9 / 17 / 72 / 77

Answer: 40, 72

Distribute the factor to both terms inside the parentheses.

Q0368 (Test 10 · Multiple choice · challenging)Which expression is equivalent to $4(x + 10)$?Choices: $x + 40$ | $4x + 10$ | $4x + 40$ | $4x + 14$ **Answer:** $4x + 40$

Multiply the number outside the parentheses by each term inside.

Q0413 (Test 11 · Matching table · moderate)

Match each expression with an equivalent expression.

Answer: $7(x + 3) \rightarrow 7x + 21$; $7x + 6x \rightarrow 13x$; $7x + 3 - 3 \rightarrow 7x$

Use the distributive property and combine like terms.

Q0421 (Test 12 · Multiple choice · easy)Which expression is equivalent to $9x + 4 + 8x$?Choices: $72x + 4$ | $17x + 4$ | $17x + 4x$ | $21x$ **Answer:** $17x + 4$

Only like terms can be combined, so add the x-terms and leave the constant.

Q0458 (Test 13 · Multiselect · moderate)Select **all** the expressions equivalent to $6(x + 8)$.Choices: $6x + 8$ | $48 + 6x$ | $14(x)$ | $6(x + 8)$ | $6x + 48$ | $6(x + 9)$ **Answer:** $6(x + 8)$, $6x + 48$, $48 + 6x$

Equivalent expressions have the same value for every value of x.

Q0520 (Test 14 · Multiple choice · challenging)Which expression is equivalent to $9(x + 6)$?Choices: $9x + 15$ | $9x + 6$ | $9x + 54$ | $x + 54$ **Answer:** $9x + 54$

Multiply the number outside the parentheses by each term inside.

MA.6.AR.2.1 — Given an equation or inequality and a specified set of integer values, determine which values make the equation or inequality true or false.**Q0005** (Test 1 · Multiple choice · easy)Which value of x makes the equation $2x + 3 = 23$ true?

Choices: 10 | 20 | 11 | 9

Answer: 10

Substitute each value for x and check which one makes both sides equal.

Q0067 (Test 2 · Selectable hot text · moderate)

Select **all** the values that make $6x + 9 = 75$ true.

Choices: 9 | 11 | 14 | 18 | 75

Answer: 11

An equation is true only for the value that balances both sides.

Q0089 (Test 3 · Multiple choice · challenging)

Which value of x makes the inequality $x > 19$ true?

Choices: 13 | 17 | 19 | 22

Answer: 22

Only numbers greater than 19 make the inequality true, and 19 itself does not.

Q0142 (Test 4 · Matching table · easy)

For each value of x , decide whether $x \geq 20$ is true or false.

Answer: 15 → False; **18** → False; **20** → True; **29** → True

Values of 20 or greater make the inequality true.

Q0228 (Test 6 · Multiple choice · moderate)

Which value of x makes the equation $4x + 10 = 22$ true?

Choices: 3 | 12 | 4 | 2

Answer: 3

Substitute each value for x and check which one makes both sides equal.

Q0257 (Test 7 · Multiselect · moderate)

Select **all** the values of x that make $x \leq 6$ true.

Choices: 11 | 3 | 5 | 8 | 14 | 6

Answer: 3, 5, 6

The symbol $\leq$ includes 6 itself.

Q0274 (Test 8 · Multiple choice · easy)

Which value of x makes the inequality $x > 14$ true?

Choices: 15 | 14 | 12 | 8

Answer: 15

Only numbers greater than 14 make the inequality true, and 14 itself does not.

Q0335 (Test 9 · Selectable hot text · moderate)

Select **all** the values that make $2x + 11 = 35$ true.

Choices: 10 | 11 | 12 | 15 | 19 | 35

Answer: 12

An equation is true only for the value that balances both sides.

Q0374 (Test 10 · Multiple choice · challenging)

Which value of x makes the equation $8x + 20 = 44$ true?

Choices: 3 | 4 | 2 | 24

Answer: 3

Substitute each value for x and check which one makes both sides equal.

Q0403 (Test 11 · Matching table · moderate)

For each value of x , decide whether $x \geq 10$ is true or false.

Answer: 8 → False; **10** → True; **11** → True; **19** → True

Values of 10 or greater make the inequality true.

Q0427 (Test 12 · Multiple choice · easy)

Which value of x makes the inequality $x > 27$ true?

Choices: 28 | 21 | 25 | 27

Answer: 28

Only numbers greater than 27 make the inequality true, and 27 itself does not.

Q0471 (Test 13 · Multiselect · moderate)

Select **all** the values of x that make $x \leq 9$ true.

Choices: 11 | 6 | 14 | 9 | 8 | 17

Answer: 6, 8, 9

The symbol $\leq$ includes 9 itself.

Q0497 (Test 14 · Multiple choice · challenging)

Which value of x makes the equation $7x + 16 = 30$ true?

Choices: 2 | 3 | 1 | 14

Answer: 2

Substitute each value for x and check which one makes both sides equal.

MA.6.AR.2.2 — Write and solve one-step equations in one variable within a mathematical or real-world context using addition and subtraction, where all terms and solutions are integers.

Q0022 (Test 1 · Multiple choice · easy)

What is the solution to the equation $x + 16 = 38$?

Choices: 22 | 38 | 54 | 16

Answer: 22

Subtract the same number from both sides to undo the addition.

Q0071 (Test 2 · Equation editor · moderate)

Solve the equation $x - 25 = 31$ for x . Enter your answer.

Answer: 56

Add the same number to both sides to undo the subtraction.

Q0078 (Test 3 · Multiple choice · challenging)

Sofia had some stickers and gave away 19 of them. Now Sofia has 34 stickers.

Which equation can be used to find x , the number of stickers Sofia started with?

Choices: $x - 19 = 34$ | $19 - x = 34$ | $x + 19 = 34$ | $x + 34 = 19$

Answer: $x - 19 = 34$

Starting amount minus the stickers given away equals what is left.

Q0125 (Test 4 · Editing task choice · easy)

Use the drop-down menus to give each solution.

If $x + 11 = 22$, then $x =$

If $x - 27 = 11$, then $x =$

Blanks: 11 / 22 / 33 || -16 / 11 / 27 / 38

Answer: 11, 38

Undo addition with subtraction, and undo subtraction with addition.

Q0164 (Test 5 · Multiple choice · moderate)What is the solution to the equation $x + 8 = 41$?

Choices: 8 | 49 | 41 | 33

Answer: 33

Subtract the same number from both sides to undo the addition.

Q0237 (Test 7 · Multiselect · moderate)Select **all** the equations that have the solution $x = 37$.Choices: $x + 9 = 37$ | $x - 9 = 37$ | $x + 37 = 9$ | $9 + x = 46$ | $x + 9 = 46$ | $x - 9 = 28$ **Answer: $x + 9 = 46$, $x - 9 = 28$, $9 + x = 46$** Substitute 37 for x and keep the equations that stay balanced.**Q0298** (Test 8 · Multiple choice · easy)

Isabella had some stickers and gave away 3 of them. Now Isabella has 26 stickers.

Which equation can be used to find x , the number of stickers Isabella started with?Choices: $x + 3 = 26$ | $x - 3 = 26$ | $3 - x = 26$ | $x + 26 = 3$ **Answer: $x - 3 = 26$**

Starting amount minus the stickers given away equals what is left.

Q0319 (Test 9 · Equation editor · moderate)Solve the equation $x - 15 = 37$ for x . Enter your answer.**Answer: 52**

Add the same number to both sides to undo the subtraction.

Q0375 (Test 10 · Multiple choice · challenging)What is the solution to the equation $x + 11 = 38$?

Choices: 38 | 27 | 49 | 11

Answer: 27

Subtract the same number from both sides to undo the addition.

Q0389 (Test 11 · Editing task choice · moderate)

Use the drop-down menus to give each solution.

If $x + 16 = 31$, then $x = [[1]]$.If $x - 18 = 15$, then $x = [[2]]$.

Blanks: 15 / 16 / 31 / 47 || -3 / 15 / 18 / 33

Answer: 15, 33

Undo addition with subtraction, and undo subtraction with addition.

Q0419 (Test 12 · Multiple choice · easy)

Elena had some stickers and gave away 9 of them. Now Elena has 11 stickers.

Which equation can be used to find x , the number of stickers Elena started with?Choices: $9 - x = 11$ | $x + 11 = 9$ | $x - 9 = 11$ | $x + 9 = 11$ **Answer: $x - 9 = 11$**

Starting amount minus the stickers given away equals what is left.

Q0468 (Test 13 · Multiselect · moderate)

Select **all** the equations that have the solution $x = 15$.

Choices: $x + 8 = 23$ | $x + 15 = 8$ | $x + 8 = 15$ | $8 + x = 23$ | $x - 8 = 7$ | $x - 8 = 15$

Answer: $x + 8 = 23$, $x - 8 = 7$, $8 + x = 23$

Substitute 15 for x and keep the equations that stay balanced.

Q0532 (Test 14 · Multiple choice · challenging)

What is the solution to the equation $x + 9 = 46$?

Choices: 46 | 37 | 55 | 9

Answer: 37

Subtract the same number from both sides to undo the addition.

MA.6.AR.2.3 — Write and solve one-step equations in one variable within a mathematical or real-world context using multiplication and division, where all terms and solutions are integers.

Q0024 (Test 1 · Multiple choice · easy)

What is the solution to the equation $9x = 72$?

Choices: 72 | 63 | 8 | 648

Answer: 8

Divide both sides by the coefficient to undo the multiplication.

Q0065 (Test 2 · Equation editor · moderate)

Solve the equation $x \div 6 = 5$ for x . Enter your answer.

Answer: 30

Multiply both sides by the divisor to undo the division.

Q0099 (Test 3 · Multiple choice · challenging)

Tomas shared 99 guavas equally among 11 baskets.

Which equation can be used to find x , the number of guavas in each basket?

Choices: $x + 11 = 99$ | $x \div 11 = 99$ | $11x = 99$ | $99x = 11$

Answer: $11x = 99$

The number of baskets times the guavas in each basket is the total.

Q0152 (Test 4 · Editing task choice · easy)

Use the drop-down menus to give each solution.

If $4x = 36$, then $x = \text{[[1]]}$.

If $x \div 2 = 9$, then $x = \text{[[2]]}$.

Blanks: 4 / 9 / 36 / 144 || 2 / 4 / 9 / 18

Answer: 9, 18

Divide to undo multiplication; multiply to undo division.

Q0156 (Test 5 · Multiple choice · moderate)

What is the solution to the equation $2x = 30$?

Choices: 28 | 60 | 30 | 15

Answer: 15

Divide both sides by the coefficient to undo the multiplication.

Q0204 (Test 6 · Multiselect · moderate)Select **all** the equations that have the solution $x = 6$.Choices: $11x = 66$ | $11x = 6$ | $x \div 11 = 6/11$ | $66 = 11x$ | $x + 11 = 6$ | $x \div 11 = 66$ **Answer:** $11x = 66$, $x \div 11 = 6/11$, $66 = 11x$ Substitute 6 for x and keep the equations that are true.**Q0282** (Test 8 · Multiple choice · easy)

Elena shared 30 guavas equally among 5 baskets.

Which equation can be used to find x , the number of guavas in each basket?Choices: $30x = 5$ | $x + 5 = 30$ | $x \div 5 = 30$ | $5x = 30$ **Answer:** $5x = 30$

The number of baskets times the guavas in each basket is the total.

Q0330 (Test 9 · Equation editor · moderate)Solve the equation $x \div 4 = 11$ for x . Enter your answer.**Answer:** 44

Multiply both sides by the divisor to undo the division.

Q0355 (Test 10 · Multiple choice · challenging)What is the solution to the equation $2x = 10$?

Choices: 5 | 8 | 20 | 10

Answer: 5

Divide both sides by the coefficient to undo the multiplication.

Q0396 (Test 11 · Editing task choice · moderate)

Use the drop-down menus to give each solution.

If $4x = 44$, then $x = [[1]]$.If $x \div 5 = 11$, then $x = [[2]]$.

Blanks: 4 / 11 / 44 / 176 || 2 / 5 / 11 / 55

Answer: 11, 55

Divide to undo multiplication; multiply to undo division.

Q0451 (Test 12 · Multiple choice · easy)

Paola shared 48 guavas equally among 4 baskets.

Which equation can be used to find x , the number of guavas in each basket?Choices: $x + 4 = 48$ | $48x = 4$ | $4x = 48$ | $x \div 4 = 48$ **Answer:** $4x = 48$

The number of baskets times the guavas in each basket is the total.

Q0472 (Test 13 · Multiselect · moderate)Select **all** the equations that have the solution $x = 6$.Choices: $7x = 42$ | $x + 7 = 6$ | $7x = 6$ | $42 = 7x$ | $x \div 7 = 42$ | $x \div 7 = 6/7$ **Answer:** $7x = 42$, $x \div 7 = 6/7$, $42 = 7x$ Substitute 6 for x and keep the equations that are true.

Q0526 (Test 14 · Multiple choice · challenging)

What is the solution to the equation $4x = 12$?

Choices: 8 | 3 | 48 | 12

Answer: 3

Divide both sides by the coefficient to undo the multiplication.

MA.6.AR.2.4 — Determine the unknown decimal or fraction in an equation involving any of the four operations, relating three numbers, with the unknown in any position.

Q0009 (Test 1 · Multiple choice · easy)

What value of x makes the equation $x + 2.26 = 5.46$ true?

Choices: 3.2 | 5.46 | 2.26 | 7.72

Answer: 3.2

Subtract the known addend from the total.

Q0064 (Test 2 · Equation editor · moderate)

What value of x makes the equation $x \div 4 = 9.74$ true? Enter your answer.

Answer: 38.96

Multiply both sides by the divisor to find the unknown.

Q0084 (Test 3 · Multiple choice · challenging)

What value of x makes the equation $\frac{2}{3} \times x = \frac{2}{15}$ true?

Choices: $\frac{4}{5}$ | $\frac{1}{5}$ | $\frac{4}{45}$ | $\frac{10}{3}$

Answer: $\frac{1}{5}$

Divide the product by the known factor.

Q0123 (Test 4 · Editing task choice · easy)

Use the drop-down menus to complete the solutions.

If $x + 3.54 = 8.77$, then $x = \text{[[1]]}$.

If $x - 3.54 = 5.23$, then $x = \text{[[2]]}$.

Blanks: 12.31 / 3.54 / 5.23 || 1.69 / 5.23 / 8.77

Answer: 5.23, 8.77

Undo the operation shown in each equation.

Q0172 (Test 5 · Multiple choice · moderate)

What value of x makes the equation $x + 5.98 = 9.63$ true?

Choices: 15.61 | 3.65 | 5.98 | 9.63

Answer: 3.65

Subtract the known addend from the total.

Q0213 (Test 6 · Multiselect · moderate)

Select **all** the equations that have the solution $x = 7.5$.

Choices: $x + 1.5 = 9$ | $x - 0.5 = 7$ | $x - 2 = 10.5$ | $2x = 15$ | $x + 1.5 = 7.5$ | $x \div 2 = 7.5$

Answer: $x + 1.5 = 9$, $x - 0.5 = 7$, $2x = 15$

Substitute 7.5 for x and keep the equations that are true.

Q0251 (Test 7 · Multiple choice · easy)

What value of x makes the equation $\frac{2}{3} \times x = \frac{1}{3}$ true?

Choices: $\frac{4}{3}$ | $\frac{2}{9}$ | 1 | $\frac{1}{2}$

Answer: $\frac{1}{2}$

Divide the product by the known factor.

Q0305 (Test 9 · Equation editor · moderate)What value of x makes the equation $x \div 2.5 = 6.8$ true? Enter your answer.**Answer:** 17

Multiply both sides by the divisor to find the unknown.

Q0361 (Test 10 · Multiple choice · challenging)What value of x makes the equation $x + 1.04 = 8.82$ true?

Choices: 7.78 | 8.82 | 1.04 | 9.86

Answer: 7.78

Subtract the known addend from the total.

Q0393 (Test 11 · Editing task choice · moderate)

Use the drop-down menus to complete the solutions.

If $x + 2.54 = 5.36$, then $x = \text{[[1]]}$.If $x - 2.54 = 2.82$, then $x = \text{[[2]]}$.

Blanks: 2.54 / 2.82 / 7.9 || 0.28 / 2.82 / 5.36

Answer: 2.82, 5.36

Undo the operation shown in each equation.

Q0424 (Test 12 · Multiple choice · easy)What value of x makes the equation $5/6 \times x = 5/4$ true?Choices: $3/2$ | $25/12$ | $5/9$ | $25/24$ **Answer:** $3/2$

Divide the product by the known factor.

Q0494 (Test 13 · Multiselect · moderate)Select **all** the equations that have the solution $x = 6.1$.Choices: $x - 2 = 9.1$ | $x + 1.5 = 7.6$ | $x \div 2 = 6.1$ | $x + 1.5 = 6.1$ | $2x = 12.2$ | $x - 0.5 = 5.6$ **Answer:** $x + 1.5 = 7.6$, $x - 0.5 = 5.6$, $2x = 12.2$ Substitute 6.1 for x and keep the equations that are true.**Q0504** (Test 14 · Multiple choice · challenging)What value of x makes the equation $x + 5.04 = 6.74$ true?

Choices: 5.04 | 11.78 | 6.74 | 1.7

Answer: 1.7

Subtract the known addend from the total.

MA.6.AR.3.1 — Given a real-world context, write and interpret ratios to show the relative sizes of two quantities using appropriate notation: a/b , a to b , or $a:b$ where $b \neq 0$.**Q0016** (Test 1 · Multiple choice · easy)

A sixth-grade class at Riviera Middle has 14 students who ride the bus and 18 students who walk.

What is the ratio of students who ride the bus to students who walk, in simplest form?

Choices: 16:9 | 9:7 | 7:9 | 7:16

Answer: 7:9

Write bus students first, then divide both numbers by their greatest common factor.

Q0042 (Test 2 · Equation editor · moderate)

A recipe uses 16 cups of rice and 10 cups of water.

Write the ratio of rice to water in simplest form. Use the form a:b.

Answer: 8:5

Divide both numbers by their greatest common factor.

Q0101 (Test 3 · Multiple choice · challenging)

A bag holds 25 mangoes and 30 guavas.

What is the ratio of mangoes to the total number of pieces of fruit, in simplest form?

Choices: 5:11 | 11:5 | 6:5 | 5:6

Answer: 5:11

A part-to-whole ratio compares one part to the total of all the parts.

Q0128 (Test 4 · Editing task choice · easy)

A parking lot has 24 cars and 15 trucks.

Use the drop-down menus to write the ratio of cars to trucks in simplest form.

[[1]] : [[2]]

Blanks: 5 / 8 / 9 / 24 || 5 / 6 / 8 / 15

Answer: 8, 5

Divide both quantities by their greatest common factor.

Q0188 (Test 5 · Multiple choice · moderate)

A sixth-grade class at Glades Middle has 36 students who ride the bus and 28 students who walk.

What is the ratio of students who ride the bus to students who walk, in simplest form?

Choices: 16:7 | 9:7 | 7:9 | 9:16

Answer: 9:7

Write bus students first, then divide both numbers by their greatest common factor.

Q0198 (Test 6 · Multiselect · moderate)

Select **all** the ratios equivalent to 28:36.

Choices: 9:7 | 7:11 | 35:45 | 8:9 | 7:9 | 42:54

Answer: 7:9, 35:45, 42:54

Equivalent ratios simplify to the same pair of numbers in the same order.

Q0229 (Test 7 · Multiple choice · easy)

A bag holds 12 mangoes and 20 guavas.

What is the ratio of mangoes to the total number of pieces of fruit, in simplest form?

Choices: 3:8 | 3:5 | 5:3 | 8:3

Answer: 3:8

A part-to-whole ratio compares one part to the total of all the parts.

Q0278 (Test 8 · Equation editor · moderate)

A recipe uses 32 cups of rice and 20 cups of water.

Write the ratio of rice to water in simplest form. Use the form a:b.

Answer: 8:5

Divide both numbers by their greatest common factor.

Q0347 (Test 10 · Multiple choice · challenging)

A sixth-grade class at Palmetto Middle has 25 students who ride the bus and 15 students who walk.

What is the ratio of students who ride the bus to students who walk, in simplest form?

Choices: 3:5 | 8:3 | 5:8 | 5:3

Answer: 5:3

Write bus students first, then divide both numbers by their greatest common factor.

Q0401 (Test 11 · Editing task choice · moderate)

A parking lot has 6 cars and 4 trucks.

Use the drop-down menus to write the ratio of cars to trucks in simplest form.

[[1]] : [[2]]

Blanks: 2 / 3 / 4 / 6 || 2 / 3 / 4

Answer: 3, 2

Divide both quantities by their greatest common factor.

Q0429 (Test 12 · Multiple choice · easy)

A bag holds 21 mangoes and 12 guavas.

What is the ratio of mangoes to the total number of pieces of fruit, in simplest form?

Choices: 7:11 | 7:4 | 4:7 | 11:7

Answer: 7:11

A part-to-whole ratio compares one part to the total of all the parts.

Q0462 (Test 13 · Multiselect · moderate)

Select **all** the ratios equivalent to 8:14.

Choices: 4:7 | 5:7 | 7:4 | 16:28 | 12:21 | 4:9

Answer: 4:7, 12:21, 16:28

Equivalent ratios simplify to the same pair of numbers in the same order.

Q0498 (Test 14 · Multiple choice · challenging)

A sixth-grade class at Riviera Middle has 36 students who ride the bus and 8 students who walk.

What is the ratio of students who ride the bus to students who walk, in simplest form?

Choices: 11:2 | 2:9 | 9:2 | 9:11

Answer: 9:2

Write bus students first, then divide both numbers by their greatest common factor.

MA.6.AR.3.2 — Given a real-world context, determine a rate for a ratio of quantities with different units. Calculate and interpret the corresponding unit rate.

Q0013 (Test 1 · Multiple choice · easy)

A store sells 4 pounds of avocados for \$32.00.

What is the unit price, in dollars per pound?

Choices: \$0.50 | \$32.00 | \$36.00 | \$8.00

Answer: \$8.00

Divide the total cost by the number of pounds to get the cost for one pound.

Q0058 (Test 2 · Equation editor · moderate)

A pack of 12 bottles of water costs \$58.68.

What is the cost per bottle, in dollars? Enter your answer.

Answer: 4.89

Divide the total cost by the number of bottles.

Q0079 (Test 3 · Multiple choice · challenging)

A bus travels 240 miles in 4 hours at a constant speed.

At this rate, how far does the bus travel in 1 hour?

Choices: 960 miles | 1/60 miles | 236 miles | 60 miles

Answer: 60 miles

A unit rate tells how much for exactly one unit, so divide miles by hours.

Q0151 (Test 4 · Matching table · easy)

Match each rate with its unit rate (the amount for exactly 1 unit).

Answer: 16 miles in 4 hours → 4; 24 dollars for 2 pounds → 12; 54 pages in 6 minutes → 9

Divide the first quantity by the second in each rate.

Q0175 (Test 5 · Multiple choice · moderate)

A store sells 8 pounds of avocados for \$48.00.

What is the unit price, in dollars per pound?

Choices: \$1.33 | \$56.00 | \$48.00 | \$6.00

Answer: \$6.00

Divide the total cost by the number of pounds to get the cost for one pound.

Q0194 (Test 6 · Multiselect · moderate)

A 6-pound box costs \$18.00.

Select **all** the expressions that give the cost per pound, in dollars.

Choices: $18 - 6$ | 18×6 | $6 \div 18$ | $18/6$ | $36 \div 12$ | $18 \div 6$

Answer: $18 \div 6$, $18/6$, $36 \div 12$

The unit rate is the total cost divided by the number of pounds.

Q0238 (Test 7 · Multiple choice · easy)

A bus travels 180 miles in 3 hours at a constant speed.

At this rate, how far does the bus travel in 1 hour?

Choices: 1/60 miles | 60 miles | 177 miles | 540 miles

Answer: 60 miles

A unit rate tells how much for exactly one unit, so divide miles by hours.

Q0299 (Test 8 · Equation editor · moderate)

A pack of 8 bottles of water costs \$38.08.

What is the cost per bottle, in dollars? Enter your answer.

Answer: 4.76

Divide the total cost by the number of bottles.

Q0336 (Test 9 · Multiple choice · challenging)

A store sells 6 pounds of guavas for \$36.00.

What is the unit price, in dollars per pound?

Choices: \$1.00 | \$36.00 | \$42.00 | \$6.00

Answer: \$6.00

Divide the total cost by the number of pounds to get the cost for one pound.

Q0399 (Test 11 · Matching table · moderate)

Match each rate with its unit rate (the amount for exactly 1 unit).

Answer: 16 miles in 4 hours → 4; 16 dollars for 2 pounds → 8; 72 pages in 6 minutes → 12

Divide the first quantity by the second in each rate.

Q0455 (Test 12 · Multiple choice · easy)

A bus travels 180 miles in 2 hours at a constant speed.

At this rate, how far does the bus travel in 1 hour?

Choices: 1/90 miles | 360 miles | 90 miles | 178 miles

Answer: 90 miles

A unit rate tells how much for exactly one unit, so divide miles by hours.

Q0486 (Test 13 · Multiselect · moderate)

A 8-pound box costs \$64.00.

Select **all** the expressions that give the cost per pound, in dollars.

Choices: $64 \div 8$ | $8 \div 64$ | $64 - 8$ | 64×8 | $64/8$ | $128 \div 16$

Answer: $64 \div 8$, $64/8$, $128 \div 16$

The unit rate is the total cost divided by the number of pounds.

Q0528 (Test 14 · Multiple choice · challenging)

A store sells 7 pounds of guavas for \$35.00.

What is the unit price, in dollars per pound?

Choices: \$35.00 | \$5.00 | \$1.40 | \$42.00

Answer: \$5.00

Divide the total cost by the number of pounds to get the cost for one pound.

MA.6.AR.3.3 — Extend previous understanding of fractions and numerical patterns to generate or complete a two- or three-column table to display equivalent part-to-part ratios and part-to-part-to-whole ratios.**Q0035** (Test 1 · Multiple choice · easy)

The table shows equivalent ratios of red beads to blue beads.

What number belongs in the box marked ?

Red	Blue
5	6
10	12
10	?

Equivalent ratios

Choices: 14 | 8 | 10 | 12

Answer: 12

Each row multiplies the first ratio by the same number, here 2.

Q0043 (Test 2 · Equation editor · moderate)

The table shows equivalent ratios of juice to water.

Enter the number that belongs in the box marked ?

Cups of juice	Cups of water
5	6
25	?

Answer: 30

Both quantities are multiplied by 5.

Q0092 (Test 3 · Multiple choice · challenging)

The table shows equivalent part-to-part-to-whole ratios.

What number belongs in the box marked ?

Boys	Girls	Total
5	6	11
15	18	?

Part-to-part-to-whole ratios

Choices: 21 | 36 | 33 | 11

Answer: 33

The total is the sum of the two parts in the same row.

Q0127 (Test 4 · Editing task choice · easy)

The table shows equivalent ratios of buses to vans.

Use the drop-down menus to complete the table: the first ? is [[1]] and the second ? is [[2]].

Buses	Vans
3	5
?	10
9	?

Blanks: 3 / 5 / 6 / 10 || 5 / 8 / 9 / 15

Answer: 6, 15

Multiply both numbers in a ratio by the same factor to keep it equivalent.

Q0166 (Test 5 · Multiple choice · moderate)

The table shows equivalent ratios of red beads to blue beads.

What number belongs in the box marked ?

Red	Blue
3	4
6	8
15	?

Equivalent ratios

Choices: 9 | 25 | 15 | 20

Answer: 20

Each row multiplies the first ratio by the same number, here 5.

Q0220 (Test 6 · Multiselect · moderate)Select **all** the ratios that belong in a table of ratios equivalent to 5:6.

Choices: 20:24 | 6:7 | 10:6 | 6:5 | 5:6 | 10:12

Answer: 5:6, 10:12, 20:24

Multiplying both parts of a ratio by the same number keeps it equivalent; adding the same number does not.

Q0246 (Test 7 · Multiple choice · easy)

The table shows equivalent part-to-part-to-whole ratios.

What number belongs in the box marked ?

Boys	Girls	Total
3	5	8
9	15	?

Part-to-part-to-whole ratios

Choices: 27 | 8 | 14 | 24

Answer: 24

The total is the sum of the two parts in the same row.

Q0272 (Test 8 · Equation editor · moderate)

The table shows equivalent ratios of juice to water.

Enter the number that belongs in the box marked ?

Cups of juice	Cups of water
4	7
16	?

Answer: 28

Both quantities are multiplied by 4.

Q0337 (Test 9 · Multiple choice · challenging)

The table shows equivalent ratios of red beads to blue beads.

What number belongs in the box marked ?

Red	Blue
4	7
8	14
8	?

Equivalent ratios

Choices: 9 | 8 | 16 | 14

Answer: 14

Each row multiplies the first ratio by the same number, here 2.

Q0352 (Test 10 · Editing task choice · moderate)

The table shows equivalent ratios of buses to vans.

Use the drop-down menus to complete the table: the first ? is [[1]] and the second ? is [[2]].

Buses	Vans
3	5
?	10
9	?

Blanks: 3 / 5 / 6 / 10 || 5 / 8 / 9 / 15

Answer: 6, 15

Multiply both numbers in a ratio by the same factor to keep it equivalent.

Q0426 (Test 12 · Multiple choice · easy)

The table shows equivalent part-to-part-to-whole ratios.

What number belongs in the box marked ?

Boys	Girls	Total
4	7	11
12	21	?

Part-to-part-to-whole ratios

Choices: 19 | 36 | 33 | 11

Answer: 33

The total is the sum of the two parts in the same row.

Q0463 (Test 13 · Multiselect · moderate)

Select **all** the ratios that belong in a table of ratios equivalent to 5:6.

Choices: 10:12 | 5:6 | 6:5 | 6:7 | 20:24 | 10:6

Answer: 5:6, 10:12, 20:24

Multiplying both parts of a ratio by the same number keeps it equivalent; adding the same number does not.

Q0517 (Test 14 · Multiple choice · challenging)

The table shows equivalent ratios of red beads to blue beads.

What number belongs in the box marked ?

Red	Blue
3	5
6	10
12	?

Equivalent ratios

Choices: 9 | 20 | 12 | 24

Answer: 20

Each row multiplies the first ratio by the same number, here 4.

MA.6.AR.3.4 — Apply ratio relationships to solve mathematical and real-world problems involving percentages using the relationship between two quantities.

Q0023 (Test 1 · Multiple choice · easy)

There are 200 students in the sixth grade at Ponce de Leon Middle. 60% of them ride the bus.

How many students ride the bus?

Choices: 140 | 120 | 1200 | 3 1/3

Answer: 120

60% means 60 out of every 100, so multiply the total by 60/100.

Q0056 (Test 2 · Equation editor · moderate)

A pair of headphones costs \$60.00 and is on sale for 50% off.

What is the sale price, in dollars? Enter your answer.

Answer: 30

Find the discount, then subtract it from the original price.

Q0111 (Test 3 · Multiple choice · challenging)

16 students is 40% of the students in a grade.

How many students are in the grade?

Choices: 40 | 1600 | 640 | 56

Answer: 40

Divide the part by the percent written as a decimal to find the whole.

Q0129 (Test 4 · Editing task choice · easy)

Use the drop-down menus to complete the statements about 240 students.

20% of 240 is students, and 50% of 240 is students.

Blanks: 20 / 48 / 96 / 220 || 60 / 120 / 240 / 480

Answer: 48, 120

Write each percent as a fraction of 100 and multiply by the total.

Q0178 (Test 5 · Multiple choice · moderate)

There are 120 students in the sixth grade at Palmetto Middle. 40% of them ride the bus.

How many students ride the bus?

Choices: 3 | 48 | 80 | 480

Answer: 48

40% means 40 out of every 100, so multiply the total by 40/100.

Q0227 (Test 6 · Multiselect · moderate)

Select **all** the expressions that give 60% of 240.

Choices: 60×240 | 240×0.6 | $(60/100) \times 240$ | $60\% \times 240$ | $240 \div 60$ | $240 - 60$

Answer: $60\% \times 240$, $(60/100) \times 240$, 240×0.6

A percent of a number is that fraction of 100 multiplied by the number.

Q0250 (Test 7 · Multiple choice · easy)

150 students is 75% of the students in a grade.

How many students are in the grade?

Choices: 225 | 15000 | 11250 | 200

Answer: 200

Divide the part by the percent written as a decimal to find the whole.

Q0285 (Test 8 · Equation editor · moderate)

A pair of headphones costs \$45.00 and is on sale for 50% off.

What is the sale price, in dollars? Enter your answer.

Answer: 22.5

Find the discount, then subtract it from the original price.

Q0317 (Test 9 · Multiple choice · challenging)

There are 200 students in the sixth grade at Riviera Middle. 40% of them ride the bus.

How many students ride the bus?

Choices: 160 | 80 | 5 | 800

Answer: 80

40% means 40 out of every 100, so multiply the total by 40/100.

Q0351 (Test 10 · Editing task choice · moderate)

Use the drop-down menus to complete the statements about 80 students.

80% of 80 is students, and 50% of 80 is students.

Blanks: 0 / 64 / 80 / 128 || 20 / 40 / 80 / 160

Answer: 64, 40

Write each percent as a fraction of 100 and multiply by the total.

Q0387 (Test 11 · Multiple choice · easy)

20 students is 50% of the students in a grade.

How many students are in the grade?

Choices: 70 | 2000 | 1000 | 40

Answer: 40

Divide the part by the percent written as a decimal to find the whole.

Q0488 (Test 13 · Multiselect · moderate)

Select **all** the expressions that give 15% of 80.

Choices: $(15/100) \times 80$ | 80×0.15 | $15\% \times 80$ | $80 - 15$ | $80 \div 15$ | 15×80

Answer: $15\% \times 80$, $(15/100) \times 80$, 80×0.15

A percent of a number is that fraction of 100 multiplied by the number.

Q0509 (Test 14 · Multiple choice · challenging)

There are 40 students in the sixth grade at Coral Way K-8. 40% of them ride the bus.

How many students ride the bus?

Choices: 16 | 1 | 0 | 160

Answer: 16

40% means 40 out of every 100, so multiply the total by 40/100.

MA.6.AR.3.5 — Solve mathematical and real-world problems involving ratios, rates and unit rates, including comparisons, mixtures, ratios of lengths and conversions within the same measurement system.

Q0026 (Test 1 · Multiple choice · easy)

A banner is 36 feet long.

How many inches long is the banner? (1 foot = 12 inches)

Choices: 432 inches | 108 inches | 48 inches | 3 inches

Answer: 432 inches

Each foot is 12 inches, so multiply the number of feet by 12.

Q0048 (Test 2 · Equation editor · moderate)

A fence is 6 yards long.

How many feet long is the fence? Enter your answer. (1 yard = 3 feet)

Answer: 18

Each yard is 3 feet, so multiply by 3.

Q0091 (Test 3 · Multiple choice · challenging)

Store A sells 8 bottles of water for \$4.00. Store B sells 16 bottles for \$9.00.

What is the unit price at Store A, in dollars per bottle?

Choices: 0.5 | 12 | 0.5625 | 2

Answer: 0.5

Divide the price by the number of bottles to compare unit prices.

Q0146 (Test 4 · Matching table · easy)

Match each measurement with its equivalent value. Use the Grade 6 reference sheet.

Answer: 2 pounds, in ounces → 32; 3 gallons, in quarts → 12; 5 meters, in centimeters → 500

Multiply by the number of smaller units in one larger unit.

Q0176 (Test 5 · Multiple choice · moderate)

A banner is 36 feet long.

How many inches long is the banner? (1 foot = 12 inches)

Choices: 432 inches | 48 inches | 3 inches | 108 inches

Answer: 432 inches

Each foot is 12 inches, so multiply the number of feet by 12.

Q0219 (Test 6 · Multiselect · moderate)

A pitcher holds 8 cups of juice.

Select **all** the expressions that give the number of fluid ounces the pitcher holds. (1 cup = 8 fluid ounces)

Choices: $8 \div 8$ | $8 + 8$ | 8×2 | 64×1 | 8×8

Answer: 8×8 , 64×1

Each cup is 8 fluid ounces, so multiply the number of cups by 8.

Q0258 (Test 7 · Multiple choice · easy)

Store A sells 6 bottles of water for \$3.00. Store B sells 12 bottles for \$8.00.

What is the unit price at Store A, in dollars per bottle?

Choices: 0.5 | 0.6667 | 9 | 2

Answer: 0.5

Divide the price by the number of bottles to compare unit prices.

Q0290 (Test 8 · Equation editor · moderate)

A fence is 6 yards long.

How many feet long is the fence? Enter your answer. (1 yard = 3 feet)

Answer: 18

Each yard is 3 feet, so multiply by 3.

Q0341 (Test 9 · Multiple choice · challenging)

A banner is 15 feet long.

How many inches long is the banner? (1 foot = 12 inches)

Choices: 45 inches | 27 inches | 180 inches | 1.25 inches

Answer: 180 inches

Each foot is 12 inches, so multiply the number of feet by 12.

Q0364 (Test 10 · Matching table · moderate)

Match each measurement with its equivalent value. Use the Grade 6 reference sheet.

Answer: 2 pounds, in ounces → 32; 3 gallons, in quarts → 12; 5 meters, in centimeters → 500

Multiply by the number of smaller units in one larger unit.

Q0391 (Test 11 · Multiple choice · easy)

Store A sells 6 bottles of water for \$3.00. Store B sells 12 bottles for \$8.00.

What is the unit price at Store A, in dollars per bottle?

Choices: 9 | 2 | 0.6667 | 0.5

Answer: 0.5

Divide the price by the number of bottles to compare unit prices.

Q0440 (Test 12 · Multiselect · moderate)

A pitcher holds 8 cups of juice.

Select **all** the expressions that give the number of fluid ounces the pitcher holds. (1 cup = 8 fluid ounces)

Choices: 8×2 | 8×8 | $8 \div 8$ | $8 + 8$ | 64×1

Answer: 8×8 , 64×1

Each cup is 8 fluid ounces, so multiply the number of cups by 8.

Q0516 (Test 14 · Multiple choice · challenging)

A banner is 36 feet long.

How many inches long is the banner? (1 foot = 12 inches)

Choices: 108 inches | 432 inches | 3 inches | 48 inches

Answer: 432 inches

Each foot is 12 inches, so multiply the number of feet by 12.

Geometric Reasoning, Data Analysis, and Probability

MA.6.DP.1.1 — Recognize and formulate a statistical question that would generate numerical data.

Q0008 (Test 1 · Multiple choice · easy)

Which question is a statistical question?

Choices: What time does the first bell ring at my school? | How many minutes long is my bus ride to school? | How old is my teacher? | How many text messages did each student send yesterday?

Answer: How many text messages did each student send yesterday?

A statistical question expects different answers from different people, so it produces a set of numerical data.

Q0049 (Test 2 · Multiselect · moderate)

Select **all** the questions that are statistical questions.

Choices: How many text messages did each student send yesterday? | How many minutes did each sixth grader spend on the bus this morning? | How old is my teacher? | What time does the first bell ring at my school? | How many books did I read last month?

Answer: How many minutes did each sixth grader spend on the bus this morning?, How many text messages did each student send yesterday?

Statistical questions anticipate variability in the answers.

Q0114 (Test 3 · Multiple choice · challenging)

Which question is **not** a statistical question?

Choices: How many books did each student in the class read last month? | How old is my teacher? | How many minutes did each sixth grader spend on the bus this morning? | What are the heights, in inches, of the students in the band?

Answer: How old is my teacher?

A question with only one possible answer does not produce a data set.

Q0148 (Test 4 · Selectable hot text · easy)

Select **all** the questions below that would generate numerical data with variability.

Choices: How many text messages did each student send yesterday? | What are the heights, in inches, of the students in the band? | How many books did I read last month? | How old is my teacher? | How many minutes long is my bus ride to school?

Answer: How many text messages did each student send yesterday?, What are the heights, in inches, of the students in the band?

Look for questions that many people would answer with different numbers.

Q0168 (Test 5 · Multiple choice · moderate)

Which question is a statistical question?

Choices: What time does the first bell ring at my school? | How old is my teacher? | How many books did I read last month? | What are the heights, in inches, of the students in the band?

Answer: What are the heights, in inches, of the students in the band?

A statistical question expects different answers from different people, so it produces a set of numerical data.

Q0206 (Test 6 · Editing task choice · moderate)

Use the drop-down menu to complete the statistical question.

How many minutes did [[1]] spend reading last night?

Blanks: each student in the class / I / my teacher

Answer: each student in the class

Asking about a whole group produces data that vary; asking about one person produces a single answer.

Q0286 (Test 8 · Multiple choice · easy)

Which question is **not** a statistical question?

Choices: How many books did each student in the class read last month? | How many text messages did each student send yesterday? | How old is my teacher? | What are the heights, in inches, of the students in the band?

Answer: How old is my teacher?

A question with only one possible answer does not produce a data set.

Q0321 (Test 9 · Multiselect · moderate)

Select **all** the questions that are statistical questions.

Choices: How tall is the tallest student in my class? | How many text messages did each student send yesterday? | What time does the first bell ring at my school? | How many minutes did each sixth grader spend on the bus this morning? | How old is my teacher?

Answer: How many text messages did each student send yesterday?, How many minutes did each sixth grader spend on the bus this morning?

Statistical questions anticipate variability in the answers.

Q0363 (Test 10 · Multiple choice · challenging)

Which question is a statistical question?

Choices: How many minutes long is my bus ride to school? | How many minutes did each sixth grader spend on the bus this morning? | How many books did I read last month? | How tall is the tallest student in my class?

Answer: How many minutes did each sixth grader spend on the bus this morning?

A statistical question expects different answers from different people, so it produces a set of numerical data.

Q0392 (Test 11 · Selectable hot text · moderate)

Select **all** the questions below that would generate numerical data with variability.

Choices: How many minutes did each sixth grader spend on the bus this morning? | How many books did each student in the class read last month? | How tall is the tallest student in my class? | What time does the first bell ring at my school? | How old is my teacher?

Answer: How many minutes did each sixth grader spend on the bus this morning?, How many books did each student in the class read last month?

Look for questions that many people would answer with different numbers.

Q0437 (Test 12 · Multiple choice · easy)

Which question is **not** a statistical question?

Choices: How many text messages did each student send yesterday? | How tall is the tallest student in my class? | How many hours did each student in the class sleep last night? | How many minutes did each sixth grader spend on the bus this morning?

Answer: How tall is the tallest student in my class?

A question with only one possible answer does not produce a data set.

Q0457 (Test 13 · Editing task choice · moderate)

Use the drop-down menu to complete the statistical question.

How many minutes did [[1]] spend reading last night?

Blanks: each student in the class / I / my teacher

Answer: each student in the class

Asking about a whole group produces data that vary; asking about one person produces a single answer.

Q0515 (Test 14 · Multiple choice · challenging)

Which question is a statistical question?

Choices: How many text messages did each student send yesterday? | How old is my teacher? | What time does the first bell ring at my school? | How many minutes long is my bus ride to school?

Answer: How many text messages did each student send yesterday?

A statistical question expects different answers from different people, so it produces a set of numerical data.

MA.6.DP.1.2 — Given a numerical data set within a real-world context, find and interpret mean, median, mode and range.**Q0031** (Test 1 · Multiple choice · easy)

The table shows the number of minutes six students spent on the bus.

What is the mean number of minutes?

Minutes	#1	#2	#3	#4	#5	#6
Minutes	6	13	14	17	25	27

Choices: 17 | 102 | 21 | 19

Answer: 17

Add all the values, then divide by how many values there are.

Q0045 (Test 2 · Equation editor · moderate)

The table shows the points scored in five games.

What is the mean number of points? Enter your answer.

Points	#1	#2	#3	#4	#5
Points	17	22	23	25	33

Answer: 24

Add the five values and divide the total by 5.

Q0081 (Test 3 · Multiple choice · challenging)

Ximena recorded these scores:

3, 4, 4, 5, 6, 10, 18

What is the median of the data set?

Choices: 5 | 7 | 15 | 4

Answer: 5

The median is the middle value when the data are in order.

Q0150 (Test 4 · Matching table · easy)

A data set is shown.

11, 15, 17, 20, 22, 23

Match each measure with its value.

Answer: Mean → 18; Median → 18 1/2; Range → 12

Find the mean by dividing the total, the median as the middle value, and the range as the greatest minus the least.

Q0174 (Test 5 · Multiple choice · moderate)

The table shows the number of minutes six students spent on the bus.

What is the mean number of minutes?

Minutes	#1	#2	#3	#4	#5	#6
Minutes	17	21	22	24	27	27

Choices: 24 | 10 | 138 | 23

Answer: 23

Add all the values, then divide by how many values there are.

Q0201 (Test 6 · Multiselect · moderate)

A data set is shown.

8, 11, 13, 15, 16, 21

Select **all** the statements that are true about the data set.

Choices: The range is 17. | The median is 16. | The mean is 14. | The median is 14. | The range is 13. | The mean is 17.

Answer: The mean is 14., The median is 14., The range is 13.

Check each statement by computing that measure from the data.

Q0263 (Test 7 · Multiple choice · easy)

Ximena recorded these scores:

3, 3, 5, 9, 10, 13, 14

What is the mode of the data set?

Choices: 11 | 8 | 3 | 9

Answer: 3

The mode is the value that appears most often.

Q0331 (Test 9 · Equation editor · moderate)

The table shows the points scored in five games.

What is the mean number of points? Enter your answer.

Points	#1	#2	#3	#4	#5
Points	11	11	16	17	35

Answer: 18

Add the five values and divide the total by 5.

Q0362 (Test 10 · Multiple choice · challenging)

The table shows the number of minutes six students spent on the bus.

What is the mean number of minutes?

Minutes	#1	#2	#3	#4	#5	#6
Minutes	13	15	25	26	29	30

Choices: 138 | 23 | 17 | 26

Answer: 23

Add all the values, then divide by how many values there are.

Q0397 (Test 11 · Matching table · moderate)

A data set is shown.

6, 10, 12, 17, 17, 22

Match each measure with its value.

Answer: Mean → 14; Median → 14 1/2; Range → 16

Find the mean by dividing the total, the median as the middle value, and the range as the greatest minus the least.

Q0449 (Test 12 · Multiple choice · easy)

Javier recorded these scores:

6, 7, 8, 8, 8, 14, 18

What is the median of the data set?

Choices: 12 | 8 | 9 | 10

Answer: 8

The median is the middle value when the data are in order.

Q0493 (Test 13 · Multiselect · moderate)

A data set is shown.

8, 12, 13, 17, 23, 23

Select **all** the statements that are true about the data set.

Choices: The median is 17. | The mean is 16. | The median is 15. | The mean is 19. | The range is 15. | The range is 19.

Answer: The mean is 16., The median is 15., The range is 15.

Check each statement by computing that measure from the data.

Q0495 (Test 14 · Multiple choice · challenging)

The table shows the number of minutes six students spent on the bus.

What is the mean number of minutes?

Minutes	#1	#2	#3	#4	#5	#6
Minutes	8	14	15	17	18	24

Choices: 16 | 96 | 18 | 17

Answer: 16

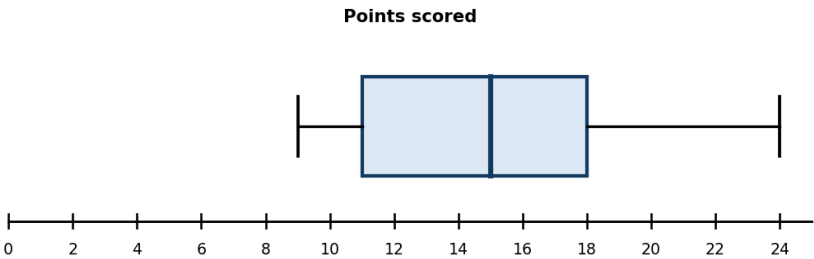
Add all the values, then divide by how many values there are.

MA.6.DP.1.3 — Given a box plot within a real-world context, determine the minimum, the lower quartile, the median, the upper quartile and the maximum. Use this summary of the data to describe the spread and distribution of the data.

Q0021 (Test 1 · Multiple choice · easy)

The box plot shows a set of data.

What is the median of the data?



Choices: 15 | 18 | 11 | 17

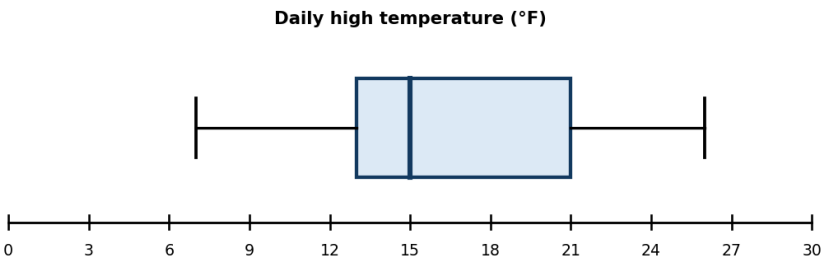
Answer: 15

The median is shown by the vertical line inside the box.

Q0053 (Test 2 · Equation editor · moderate)

The box plot shows a set of data.

What is the maximum? Enter your answer.



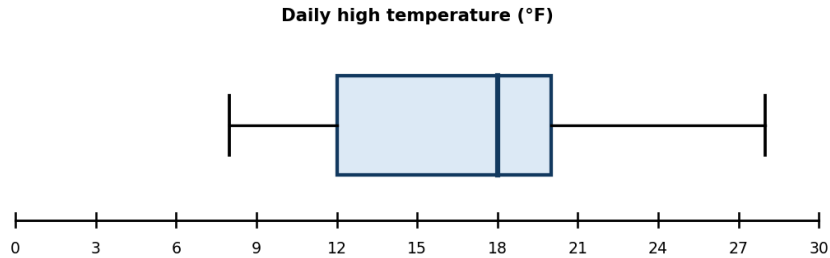
Answer: 26

Read the value at that mark on the box plot.

Q0094 (Test 3 · Multiple choice · challenging)

The box plot shows a set of data.

What is the interquartile range of the data?



Choices: 10 | 18 | 8 | 20

Answer: 8

The interquartile range is the upper quartile minus the lower quartile.

Q0122 (Test 4 · Matching table · easy)

A box plot of a data set has the five-number summary shown below the table.

Match each part of the summary with its value.

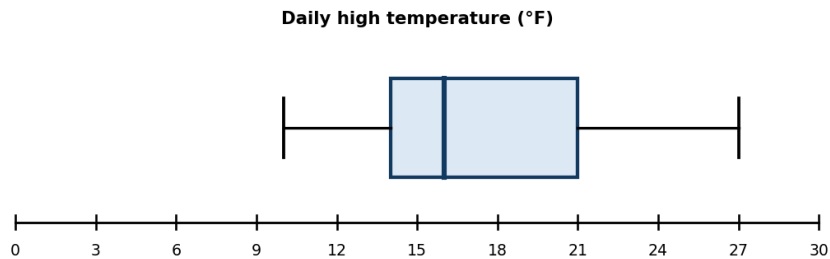
Answer: Minimum → 6; Median → 15; Upper quartile → 18

Read the five-number summary from left to right on the box plot.

Q0165 (Test 5 · Multiple choice · moderate)

The box plot shows a set of data.

What is the median of the data?



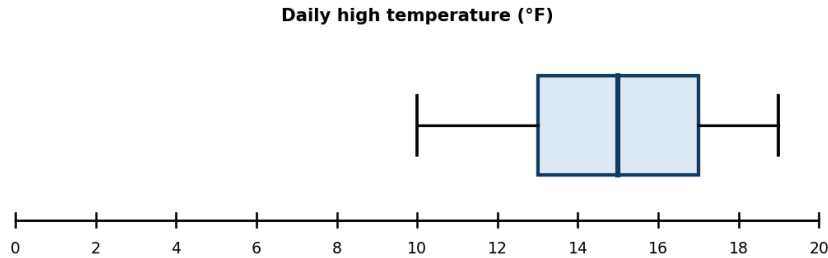
Choices: 21 | 17 | 16 | 14

Answer: 16

The median is shown by the vertical line inside the box.

Q0226 (Test 6 · Multiselect · moderate)

The box plot shows a set of data.

Select **all** the values that lie between the lower quartile and the upper quartile, inclusive.

Choices: 15 | 10 | 17 | 22 | 13 | 19 | 14

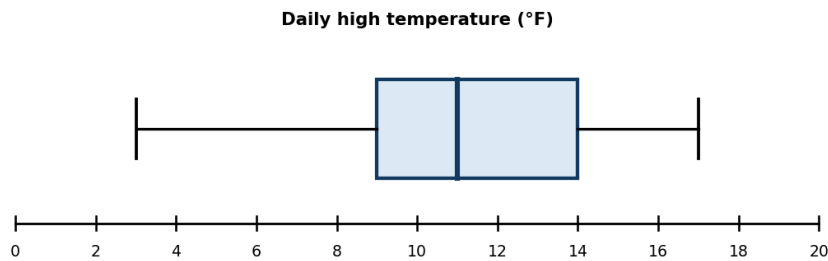
Answer: 13, 14, 15, 17

The box itself covers the values from the lower quartile to the upper quartile.

Q0245 (Test 7 · Multiple choice · easy)

The box plot shows a set of data.

What is the interquartile range of the data?



Choices: 7 | 11 | 14 | 5

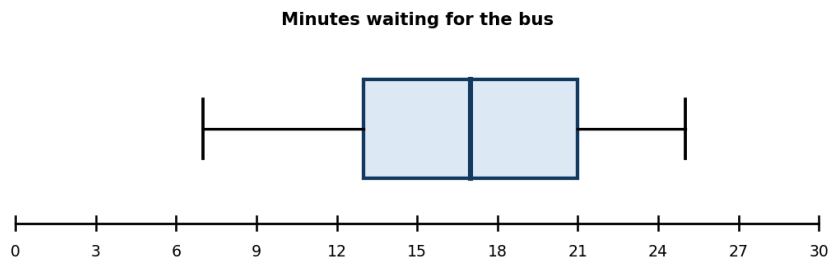
Answer: 5

The interquartile range is the upper quartile minus the lower quartile.

Q0269 (Test 8 · Equation editor · moderate)

The box plot shows a set of data.

What is the maximum? Enter your answer.

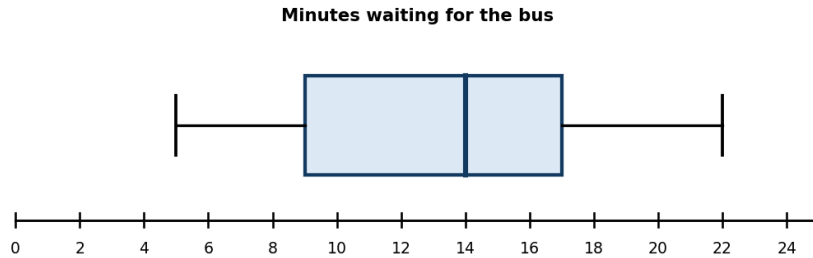
**Answer:** 25

Read the value at that mark on the box plot.

Q0349 (Test 10 · Multiple choice · challenging)

The box plot shows a set of data.

What is the median of the data?



Choices: 9 | 16 | 17 | 14

Answer: 14

The median is shown by the vertical line inside the box.

Q0407 (Test 11 · Matching table · moderate)

A box plot of a data set has the five-number summary shown below the table.

Match each part of the summary with its value.

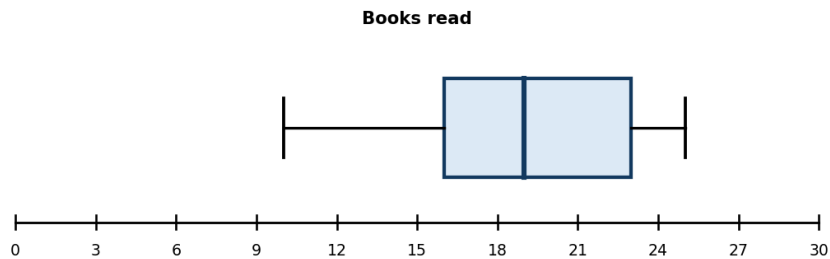
Answer: Minimum → 2; Median → 9; Upper quartile → 14

Read the five-number summary from left to right on the box plot.

Q0430 (Test 12 · Multiple choice · easy)

The box plot shows a set of data.

What is the range of the data?



Choices: 23 | 19 | 15 | 7

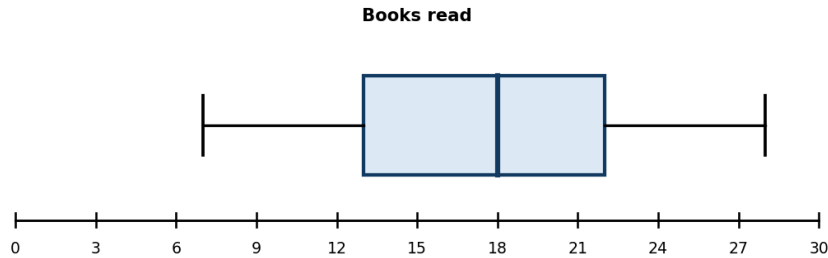
Answer: 15

The range is the maximum minus the minimum.

Q0460 (Test 13 · Multiselect · moderate)

The box plot shows a set of data.

Select **all** the values that lie between the lower quartile and the upper quartile, inclusive.



Choices: 31 | 13 | 22 | 28 | 7 | 18 | 15

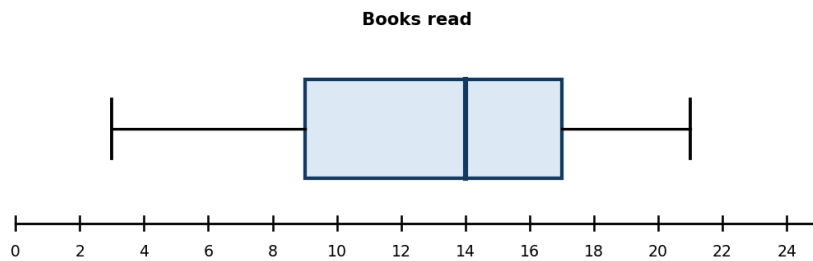
Answer: 13, 15, 18, 22

The box itself covers the values from the lower quartile to the upper quartile.

Q0503 (Test 14 · Multiple choice · challenging)

The box plot shows a set of data.

What is the median of the data?



Choices: 14 | 9 | 17 | 18

Answer: 14

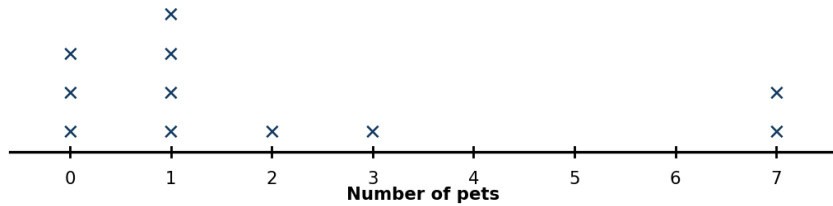
The median is shown by the vertical line inside the box.

MA.6.DP.1.4 — Given a histogram or line plot within a real-world context, qualitatively describe and interpret the spread and distribution of the data, including any symmetry, skewness, gaps, clusters, outliers and the range.

Q0001 (Test 1 · Multiple choice · easy)

The line plot shows number of pets reported by a group of students.

How many students reported 2 or more?



Choices: 7 | 6 | 5 | 4

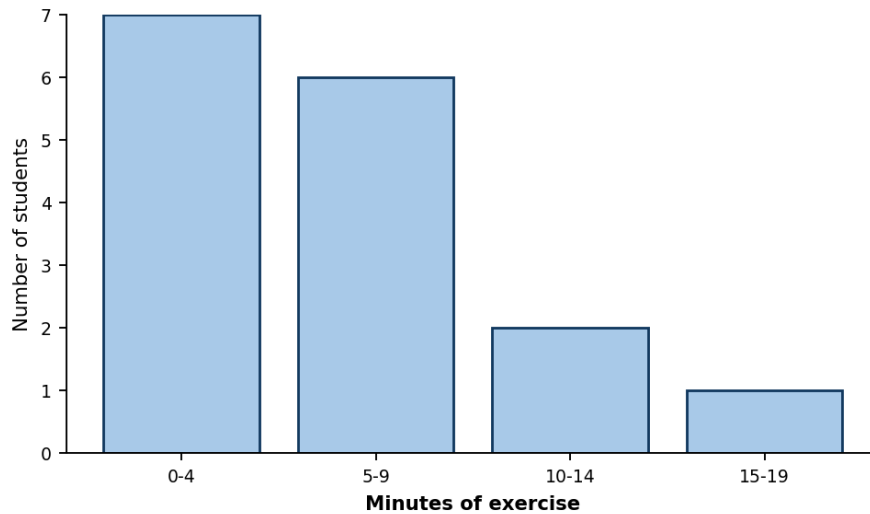
Answer: 4

Count one dot for each student at 2 and to the right of 2.

Q0040 (Test 2 · Multiple choice · moderate)

The histogram shows the data collected from a group of students.

Which statement best describes the shape of the data?



Choices: The data are symmetric, with about the same number of values on each side. | The data have no values in the lowest interval. | The data are skewed to the left, with most values in the higher intervals. | The data are skewed to the right, with most values in the lower intervals.

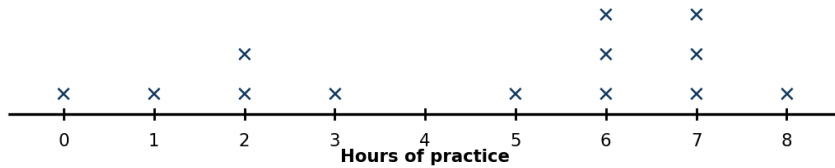
Answer: The data are skewed to the right, with most values in the lower intervals.

Most of the data sit in the low intervals with a tail stretching to the right, so the distribution is skewed right.

Q0096 (Test 3 · Multiple choice · challenging)

The line plot shows hours of practice.

What is the range of the data?



Choices: 10 | 9 | 13 | 8

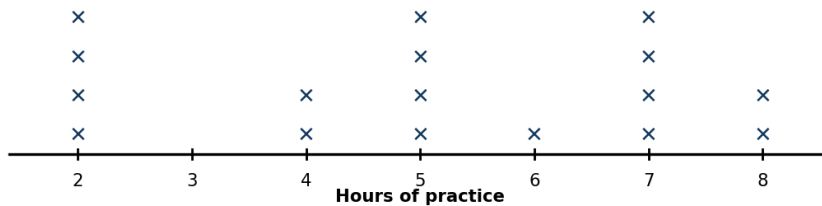
Answer: 8

The range is the greatest value minus the least value shown on the plot.

Q0137 (Test 4 · Equation editor · easy)

The line plot shows hours of practice.

How many students are represented on the line plot? Enter your answer.

**Answer: 17**

Count every dot on the plot, including repeats.

Q0173 (Test 5 · Multiple choice · moderate)

The line plot shows books read last month reported by a group of students.

How many students reported 4 or more?



Choices: 8 | 11 | 9 | 13

Answer: 11

Count one dot for each student at 4 and to the right of 4.

Q0192 (Test 6 · Multiselect · moderate)

The line plot shows books read last month.

Select **all** the values that were reported by exactly 2 students.



Choices: 5 | 4 | 3 | 7 | 2 | 1 | 8 | 0 | 6

Answer: 0, 3

Look for the columns that have exactly 2 dots.

Q0235 (Test 7 · Multiple choice · easy)

The line plot shows books read last month.

What is the range of the data?



Choices: 10 | 19 | 9 | 8

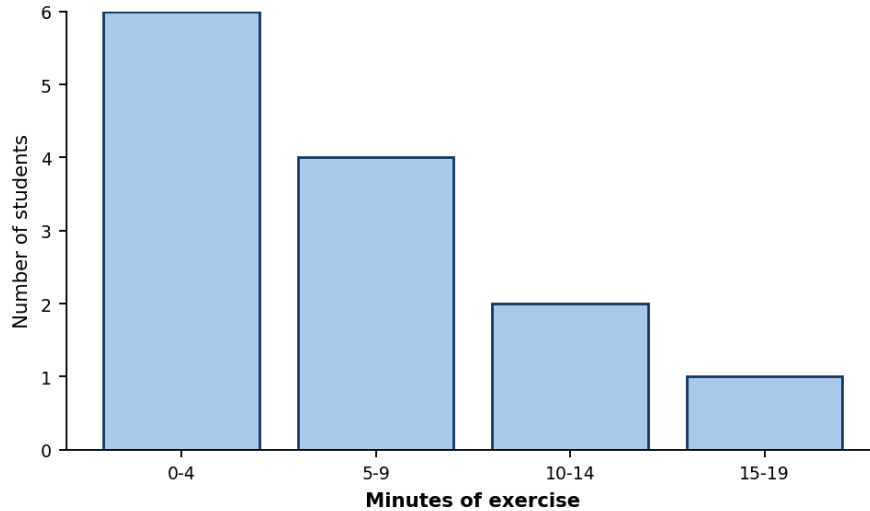
Answer: 8

The range is the greatest value minus the least value shown on the plot.

Q0304 (Test 8 · Multiple choice · moderate)

The histogram shows the data collected from a group of students.

Which statement best describes the shape of the data?



Choices: The data are skewed to the left, with most values in the higher intervals. | The data are skewed to the right, with most values in the lower intervals. | The data are symmetric, with about the same number of values on each side. | The data have no values in the lowest interval.

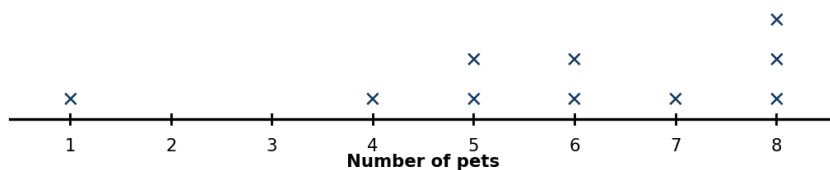
Answer: The data are skewed to the right, with most values in the lower intervals.

Most of the data sit in the low intervals with a tail stretching to the right, so the distribution is skewed right.

Q0315 (Test 9 · Multiple choice · challenging)

The line plot shows number of pets reported by a group of students.

How many students reported 6 or more?



Choices: 8 | 4 | 6 | 7

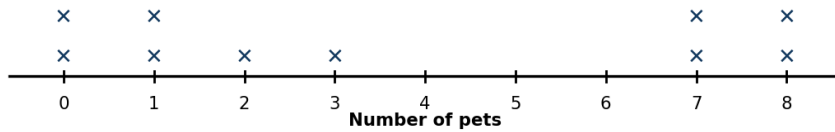
Answer: 6

Count one dot for each student at 6 and to the right of 6.

Q0398 (Test 11 · Equation editor · moderate)

The line plot shows number of pets.

How many students are represented on the line plot? Enter your answer.

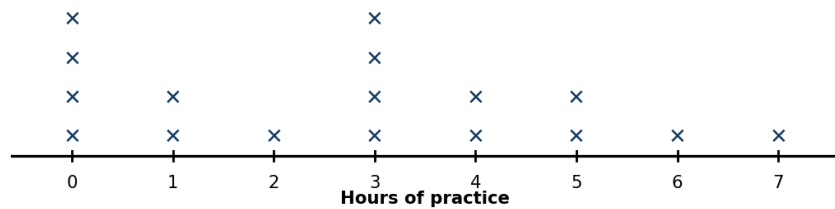
**Answer: 10**

Count every dot on the plot, including repeats.

Q0454 (Test 12 · Multiple choice · easy)

The line plot shows hours of practice.

What is the range of the data?



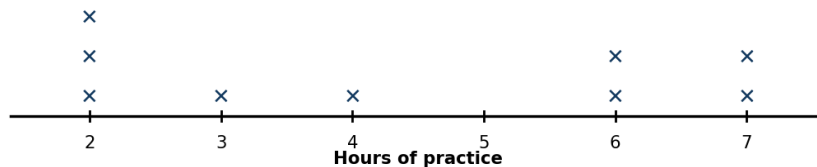
Choices: 9 | 17 | 7 | 8

Answer: 7

The range is the greatest value minus the least value shown on the plot.

Q0475 (Test 13 · Multiselect · moderate)

The line plot shows hours of practice.

Select **all** the values that were reported by exactly 2 students.

Choices: 7 | 2 | 4 | 3 | 6

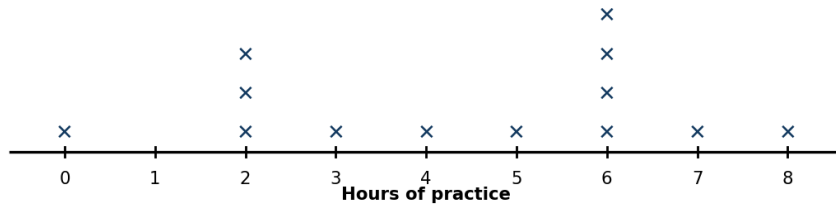
Answer: 6, 7

Look for the columns that have exactly 2 dots.

Q0529 (Test 14 · Multiple choice · challenging)

The line plot shows hours of practice reported by a group of students.

How many students reported 5 or more?



Choices: 9 | 7 | 6 | 8

Answer: 7

Count one dot for each student at 5 and to the right of 5.

MA.6.DP.1.5 — Create box plots and histograms to represent sets of numerical data within real-world contexts.

Q0003 (Test 1 · Multiple choice · easy)

A data set is shown.

2, 3, 14, 16, 24, 27, 33, 34, 37

Which list gives the five-number summary (minimum, lower quartile, median, upper quartile, maximum) of the data?

Choices: 2, 24, $17\frac{1}{2}$, $67\frac{1}{2}$, 37 | 2, $17\frac{1}{2}$, 16, $67\frac{1}{2}$, 37 | 2, 3, 24, 34, 37 | 2, $8\frac{1}{2}$, 24, $33\frac{1}{2}$, 37

Answer: 2, $8\frac{1}{2}$, 24, $33\frac{1}{2}$, 37

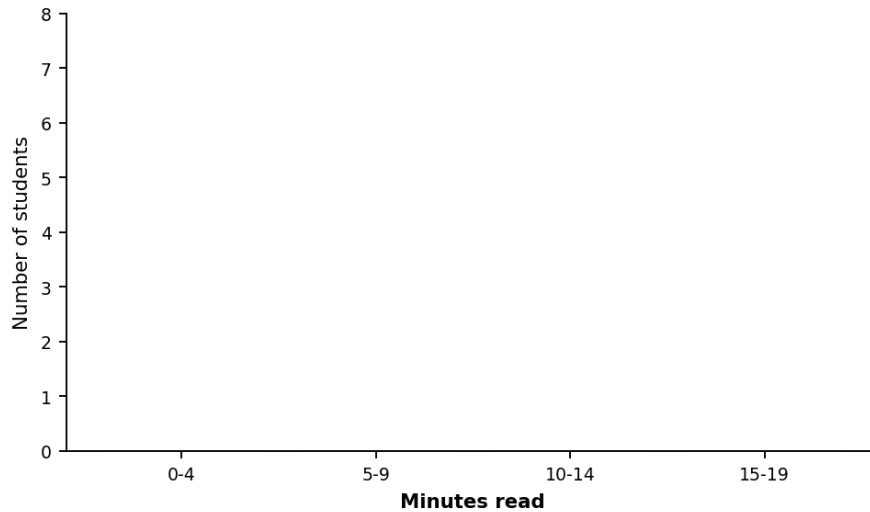
Order the data, find the median, then find the medians of the lower and upper halves.

Q0060 (Test 2 · Graphing (GRID) · moderate)

A data set is shown.

1, 2, 2, 4, 4, 4, 7, 10, 10, 11, 11, 17

Build the histogram by setting the height of each bar.

**Answer:** 6, 1, 4, 1

Count how many values fall in each interval, then set that bar height.

Q0077 (Test 3 · Multiple choice · challenging)

A data set is shown.

0, 2, 2, 3, 4, 4, 4, 6, 6, 8, 11, 17

A histogram is made with the intervals 0-4, 5-9, 10-14 and 15-19.

How tall is the bar for the interval 10-14?

Choices: 11 | 2 | 1 | 3

Answer: 1

Count how many values fall inside that interval, including both endpoints.

Q0141 (Test 4 · Equation editor · easy)

A data set is shown.

2, 3, 4, 10, 12, 13, 14, 16, 16, 18, 23

How many values belong in the interval 19-24 of a histogram? Enter your answer.

Answer: 1

Count the values from the list that fall inside that interval.

Q0190 (Test 5 · Multiple choice · moderate)

A data set is shown.

2, 7, 11, 17, 20, 23, 26, 34, 37

Which list gives the five-number summary (minimum, lower quartile, median, upper quartile, maximum) of the data?

Choices: 2, 20, 9, 30, 37 | 2, 9, 17, 30, 37 | 2, 7, 20, 34, 37 | 2, 9, 20, 30, 37

Answer: 2, 9, 20, 30, 37

Order the data, find the median, then find the medians of the lower and upper halves.

Q0195 (Test 6 · Multiselect · moderate)

A data set is shown.

0, 1, 1, 2, 3, 4, 5, 6, 9, 10, 11, 14, 14, 14, 26, 26, 26, 27

Select **all** the histogram intervals that would contain exactly 0 values.

Choices: 15-19 | 0-4 | 5-9 | 10-14 | 20-24 | 25-29

Answer: 15-19, 20-24

Count the values in each interval and compare.

Q0248 (Test 7 · Multiple choice · easy)

A data set is shown.

2, 4, 4, 5, 7, 9, 11, 13, 13, 14, 17, 18

A histogram is made with the intervals 0-4, 5-9, 10-14 and 15-19.

How tall is the bar for the interval 10-14?

Choices: 8 | 4 | 6 | 5

Answer: 4

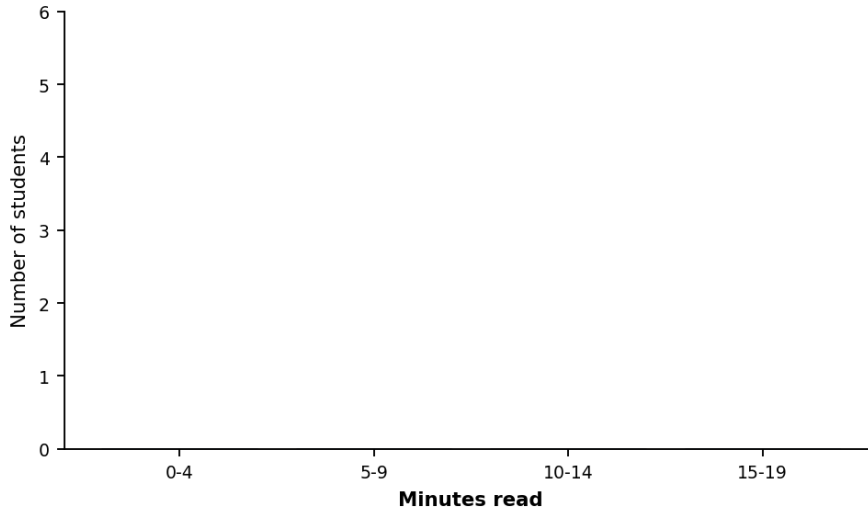
Count how many values fall inside that interval, including both endpoints.

Q0267 (Test 8 · Graphing (GRID) · moderate)

A data set is shown.

0, 2, 5, 6, 8, 9, 11, 12, 15, 17, 19, 19

Build the histogram by setting the height of each bar.

**Answer:** 2, 4, 2, 4

Count how many values fall in each interval, then set that bar height.

Q0325 (Test 9 · Multiple choice · challenging)

A data set is shown.

2, 5, 8, 17, 23, 26, 27, 28, 29

Which list gives the five-number summary (minimum, lower quartile, median, upper quartile, maximum) of the data?

Choices: 2, 5, 23, 28, 29 | 2, 23, $13\frac{1}{2}$, $55\frac{1}{2}$, 29 | 2, $6\frac{1}{2}$, 23, $27\frac{1}{2}$, 29 | 2, $13\frac{1}{2}$, 17, $55\frac{1}{2}$, 29**Answer:** 2, $6\frac{1}{2}$, 23, $27\frac{1}{2}$, 29

Order the data, find the median, then find the medians of the lower and upper halves.

Q0371 (Test 10 · Equation editor · moderate)

A data set is shown.

1, 1, 1, 1, 2, 4, 5, 6, 8, 19, 22

How many values belong in the interval 13-18 of a histogram? Enter your answer.

Answer: 0

Count the values from the list that fall inside that interval.

Q0425 (Test 12 · Multiple choice · easy)

A data set is shown.

1, 3, 4, 5, 5, 7, 9, 10, 12, 13, 13, 14

A histogram is made with the intervals 0-4, 5-9, 10-14 and 15-19.

How tall is the bar for the interval 15-19?

Choices: 12 | 0 | 1 | 2

Answer: 0

Count how many values fall inside that interval, including both endpoints.

Q0473 (Test 13 · Multiselect · moderate)

A data set is shown.

1, 3, 5, 5, 9, 9, 10, 13, 17, 19, 22, 23, 23, 23, 24, 25, 28, 29

Select **all** the histogram intervals that would contain exactly 2 values.

Choices: 10-14 | 25-29 | 15-19 | 0-4 | 20-24 | 5-9

Answer: 0-4, 10-14, 15-19

Count the values in each interval and compare.

Q0507 (Test 14 · Multiple choice · challenging)

A data set is shown.

2, 14, 17, 19, 21, 24, 27, 30, 37

Which list gives the five-number summary (minimum, lower quartile, median, upper quartile, maximum) of the data?

Choices: 2, 21, $31\frac{1}{2}$, $57\frac{1}{2}$, 37 | 2, 14, 21, 30, 37 | 2, $15\frac{1}{2}$, 21, $28\frac{1}{2}$, 37 | 2, $31\frac{1}{2}$, 19, $57\frac{1}{2}$, 37

Answer: 2, $15\frac{1}{2}$, 21, $28\frac{1}{2}$, 37

Order the data, find the median, then find the medians of the lower and upper halves.

MA.6.DP.1.6 — Given a real-world scenario, determine and describe how changes in data values impact measures of center and variation.

Q0010 (Test 1 · Multiple choice · easy)

A data set is shown.

6, 7, 7, 8, 22

The value 26 is added to the data set. What is the mean of the new data set?

Choices: 26 | $12\frac{2}{3}$ | 10 | $15\frac{1}{5}$

Answer: $12\frac{2}{3}$

Add the new value to the total and divide by the new number of values.

Q0076 (Test 2 · Equation editor · moderate)

A data set is shown.

6, 8, 12, 22, 25, 27, 28

The value 6 is removed. What is the median of the new data set? Enter your answer.

Answer: 23 $\frac{1}{2}$

Reorder the remaining values and find the middle; with six values, average the two in the middle.

Q0087 (Test 3 · Multiple choice · challenging)

A data set is shown.

10, 11, 11, 14, 19

The value 26 is added to the data set. What is the range of the new data set?

Choices: 9 | 19 | 26 | 16

Answer: 16

The new value becomes the greatest value, so the range grows.

Q0147 (Test 4 · Matching table · easy)

A data set is shown.

8, 9, 12, 17, 24

After 40 is added to the data set, match each measure with its new value.

Answer: Mean $\rightarrow$ 18 $\frac{1}{3}$; Median $\rightarrow$ 14 $\frac{1}{2}$; Range $\rightarrow$ 32

Recompute each measure using the six values of the new data set.

Q0182 (Test 5 · Multiple choice · moderate)

A data set is shown.

15, 20, 21, 21, 23

The value 28 is added to the data set. What is the mean of the new data set?

Choices: 25 $\frac{3}{5}$ | 21 $\frac{1}{3}$ | 20 | 28

Answer: 21 $\frac{1}{3}$

Add the new value to the total and divide by the new number of values.

Q0199 (Test 6 · Multiselect · moderate)

A data set is shown.

14, 14, 14, 17, 18, 19

The value 29 is added to the data set.

Select **all** the measures that change.

Choices: Maximum | Median | Mean | Range | Minimum

Answer: Mean, Range, Maximum, Median

Adding a large value shifts the total and the greatest value, but it may leave the least value unchanged.

Q0249 (Test 7 · Multiple choice · easy)

A data set is shown.

7, 8, 19, 20, 21

The value 35 is added to the data set. What is the range of the new data set?

Choices: 28 | 31 | 35 | 14

Answer: 28

The new value becomes the greatest value, so the range grows.

Q0291 (Test 8 · Equation editor · moderate)

A data set is shown.

8, 17, 21, 23, 24, 27, 29

The value 8 is removed. What is the median of the new data set? Enter your answer.

Answer: 23 $\frac{1}{2}$

Reorder the remaining values and find the middle; with six values, average the two in the middle.

Q0326 (Test 9 · Multiple choice · challenging)

A data set is shown.

11, 16, 18, 18, 22

The value 32 is added to the data set. What is the mean of the new data set?

Choices: 17 | 23 $\frac{2}{5}$ | 32 | 19 $\frac{1}{2}$

Answer: 19 $\frac{1}{2}$

Add the new value to the total and divide by the new number of values.

Q0369 (Test 10 · Matching table · moderate)

A data set is shown.

9, 12, 17, 19, 23

After 32 is added to the data set, match each measure with its new value.

Answer: Mean → $18 \frac{2}{3}$; Median → 18; Range → 23

Recompute each measure using the six values of the new data set.

Q0382 (Test 11 · Multiple choice · easy)

A data set is shown.

7, 9, 10, 11, 13

The value 37 is added to the data set. What is the range of the new data set?

Choices: 33 | 37 | 6 | 30

Answer: 30

The new value becomes the greatest value, so the range grows.

Q0467 (Test 13 · Multiselect · moderate)

A data set is shown.

12, 12, 13, 17, 17, 19

The value 29 is added to the data set.

Select **all** the measures that change.

Choices: Range | Maximum | Median | Mean | Minimum

Answer: Mean, Range, Maximum, Median

Adding a large value shifts the total and the greatest value, but it may leave the least value unchanged.

Q0511 (Test 14 · Multiple choice · challenging)

A data set is shown.

10, 13, 21, 22, 24

The value 25 is added to the data set. What is the mean of the new data set?

Choices: 25 | 23 | 18 | $19 \frac{1}{6}$

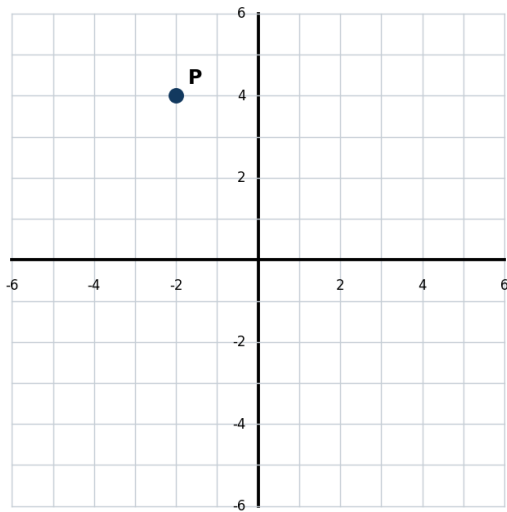
Answer: $19 \frac{1}{6}$

Add the new value to the total and divide by the new number of values.

MA.6.GR.1.1 — Extend previous understanding of the coordinate plane to plot rational number ordered pairs in all four quadrants and on both axes. Identify the x- or y-axis as the line of reflection when two ordered pairs have an opposite x- or y-coordinate.

Q0063 (Test 2 · Multiple choice · easy)

What are the coordinates of point P?



Choices: (2, 4) | (-2, -4) | (-2, 4) | (4, -2)

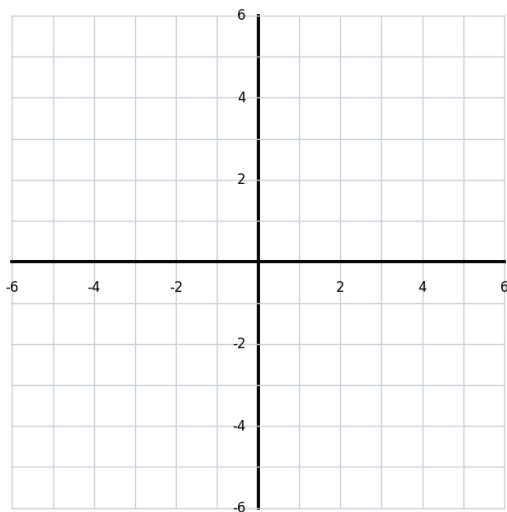
Answer: (-2, 4)

The first number tells how far to move right or left, and the second tells how far to move up or down.

Q0103 (Test 3 · Graphing (GRID) · moderate)

Plot these three points on the coordinate plane.

(-5, -5), (5, -5), (-5, 5)

**Answer:** (-5, -5), (5, -5), (-5, 5)

Start at the origin, move along the x-axis first, then up or down.

Q0132 (Test 4 · Multiple choice · challenging)

Point A is located at $(-3, -5)$. Point A is reflected across the y-axis.

What are the coordinates of the reflected point?

Choices: $(3, 5)$ | $(3, -5)$ | $(-3, -5)$ | $(-5, -3)$

Answer: $(3, -5)$

Reflecting across the y-axis changes the sign of the x-coordinate only.

Q0184 (Test 5 · Matching table · easy)

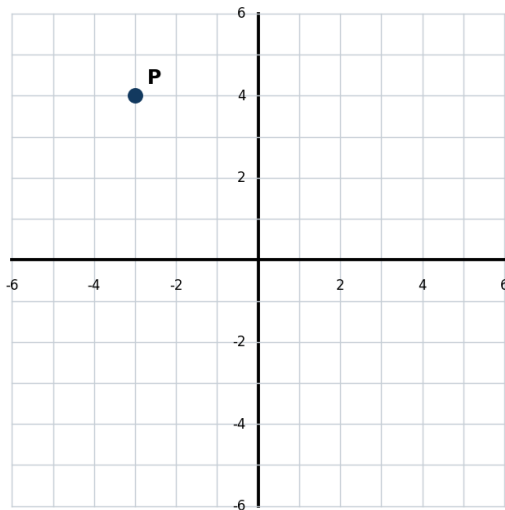
Match each point with the quadrant it is located in.

Answer: D $(5, -2) \rightarrow$ Quadrant IV; A $(3, 4) \rightarrow$ Quadrant I; C $(-4, -3) \rightarrow$ Quadrant III

The signs of the coordinates tell which quadrant a point is in.

Q0218 (Test 6 · Multiple choice · moderate)

What are the coordinates of point P?



Choices: $(4, -3)$ | $(3, 4)$ | $(-3, -4)$ | $(-3, 4)$

Answer: $(-3, 4)$

The first number tells how far to move right or left, and the second tells how far to move up or down.

Q0243 (Test 7 · Multiselect · moderate)

Select **all** the points that are located in Quadrant III.

Choices: $(2, 3)$ | $(-1, -5)$ | $(-2, 6)$ | $(-5, -4)$ | $(-3, -2)$ | $(4, -1)$

Answer: $(-3, -2)$, $(-5, -4)$, $(-1, -5)$

A point is in Quadrant III when both coordinates are negative.

Q0270 (Test 8 · Multiple choice · easy)

Point A is located at $(-2, -2)$. Point A is reflected across the y-axis.

What are the coordinates of the reflected point?

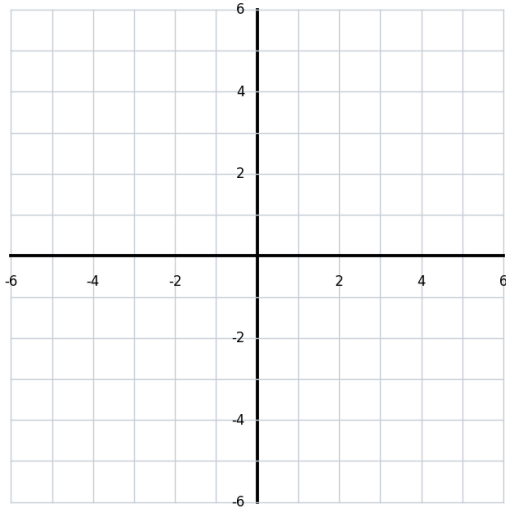
Choices: $(2, -2)$ | $(2, 2)$ | $(-2, -2)$ | $(4, -2)$

Answer: $(2, -2)$

Reflecting across the y-axis changes the sign of the x-coordinate only.

Q0342 (Test 9 · Graphing (GRID) · moderate)

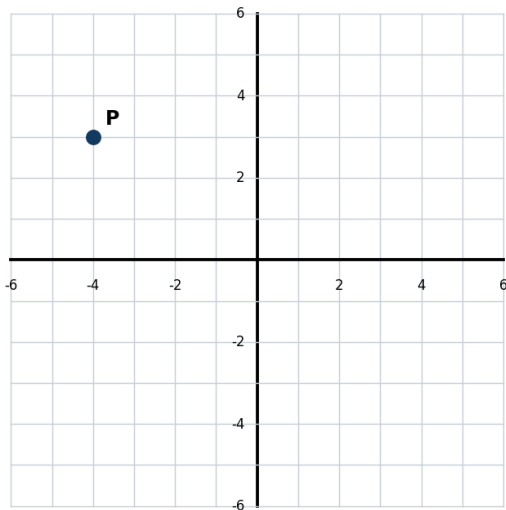
Plot these three points on the coordinate plane.

 $(-2, 5)$, $(2, 5)$, $(-2, -5)$ **Answer:** $(-2, 5)$, $(2, 5)$, $(-2, -5)$

Start at the origin, move along the x-axis first, then up or down.

Q0367 (Test 10 · Multiple choice · challenging)

What are the coordinates of point P?

Choices: $(3, -4)$ | $(4, 3)$ | $(-4, 3)$ | $(-4, -3)$ **Answer:** $(-4, 3)$

The first number tells how far to move right or left, and the second tells how far to move up or down.

Q0404 (Test 11 · Matching table · moderate)

Match each point with the quadrant it is located in.

Answer: A (3, 4) → Quadrant I; D (5, -2) → Quadrant IV; C (-4, -3) → Quadrant III

The signs of the coordinates tell which quadrant a point is in.

Q0420 (Test 12 · Multiple choice · easy)

Point A is located at (-5, -5). Point A is reflected across the x-axis.

What are the coordinates of the reflected point?

Choices: (-5, 5) | (5, 5) | (-3, 5) | (-5, -5)

Answer: (-5, 5)

Reflecting across the x-axis changes the sign of the y-coordinate only.

Q0525 (Test 14 · Multiselect · moderate)

Select **all** the points that are located in Quadrant III.

Choices: (-5, -4) | (-1, -5) | (-3, -2) | (4, -1) | (2, 3) | (-2, 6)

Answer: (-3, -2), (-5, -4), (-1, -5)

A point is in Quadrant III when both coordinates are negative.

MA.6.GR.1.3 — Solve mathematical and real-world problems by plotting points on a coordinate plane, including finding the perimeter or area of a rectangle. (Also assesses MA.6.GR.1.2: find distances between ordered pairs with the same x-coordinate or the same y-coordinate.)

Q0007 (Test 1 · Multiple choice · easy)

Point M is at (1, 1) and point N is at (5, 1).

What is the distance, in units, between point M and point N?

Choices: 2 | 4 | 6 | 8

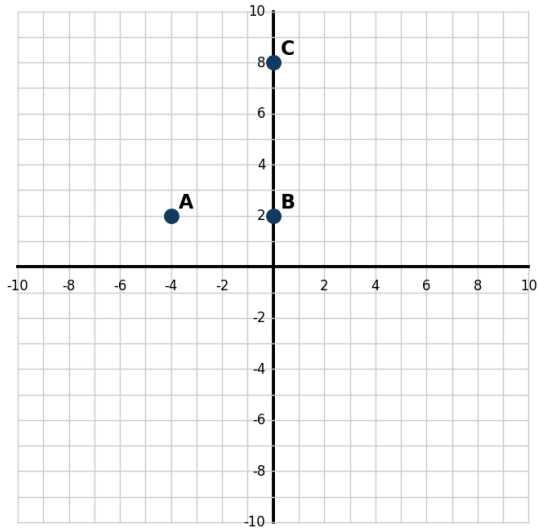
Answer: 4

Points with the same y-coordinate lie on a horizontal line, so subtract the x-coordinates.

Q0108 (Test 3 · Graphing (GRID) · moderate)

Points A, B and C are three vertices of a rectangle.

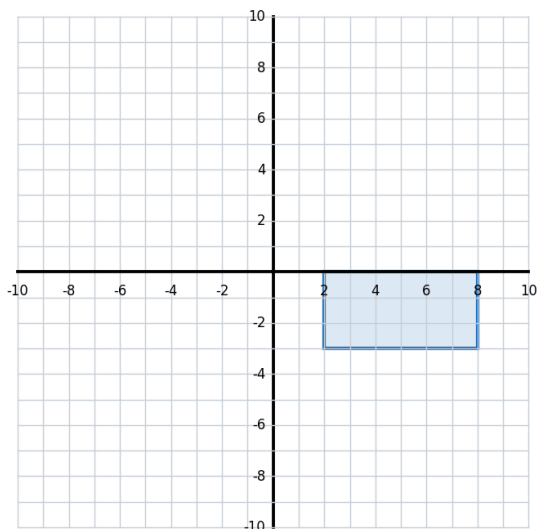
Plot the fourth vertex on the coordinate plane.

**Answer:** $(-4, 8)$

The fourth vertex lines up with A vertically and with C horizontally.

Q0118 (Test 4 · Multiple choice · challenging)A rectangle has vertices at $(2, -3)$, $(8, -3)$, $(8, 0)$ and $(2, 0)$.

What is the area, in square units, of the rectangle?



Choices: 18 | 9 | 21 | 36

Answer: 18

Find the side lengths by subtracting coordinates, then apply the formula.

Q0159 (Test 5 · Equation editor · easy)

A rectangular garden is drawn on a coordinate plane with vertices $(-4, -5)$, $(1, -5)$, $(1, 1)$ and $(-4, 1)$. Each unit is 1 meter.

What is the perimeter of the garden, in meters? Enter your answer.

Answer: 22

Add the four side lengths, or use 2 times the sum of the length and width.

Q0212 (Test 6 · Multiple choice · moderate)

Point M is at $(2, -4)$ and point N is at $(6, -4)$.

What is the distance, in units, between point M and point N?

Choices: 4 | 8 | 6 | 2

Answer: 4

Points with the same y-coordinate lie on a horizontal line, so subtract the x-coordinates.

Q0231 (Test 7 · Multiselect · moderate)

Select **all** the points that lie on the vertical line through $(5, 0)$.

Choices: $(7, 1)$ | $(5, -3)$ | $(5, 6)$ | $(2, 4)$ | $(5, 2)$ | $(7, -5)$

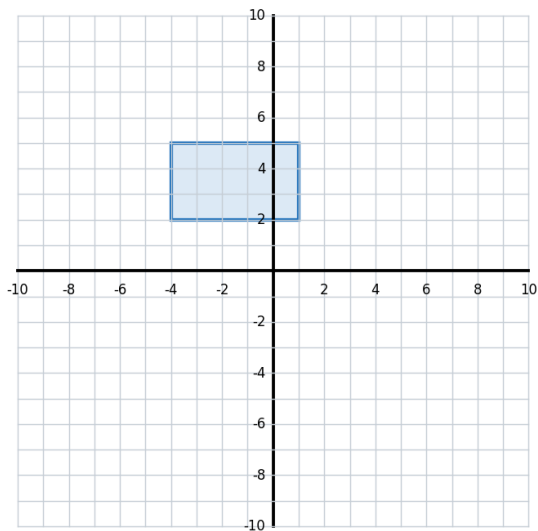
Answer: $(5, -3)$, $(5, 2)$, $(5, 6)$

Every point on a vertical line has the same x-coordinate.

Q0301 (Test 8 · Multiple choice · easy)

A rectangle has vertices at $(-4, 2)$, $(1, 2)$, $(1, 5)$ and $(-4, 5)$.

What is the area, in square units, of the rectangle?



Choices: 8 | 16 | 15 | 30

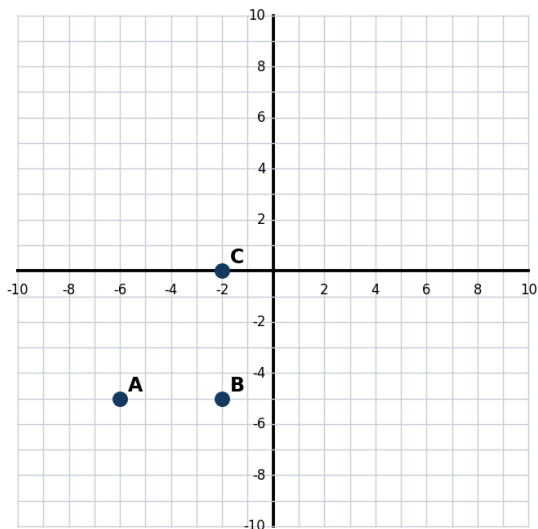
Answer: 15

Find the side lengths by subtracting coordinates, then apply the formula.

Q0311 (Test 9 · Graphing (GRID) · moderate)

Points A, B and C are three vertices of a rectangle.

Plot the fourth vertex on the coordinate plane.

**Answer:** $(-6, 0)$

The fourth vertex lines up with A vertically and with C horizontally.

Q0357 (Test 10 · Multiple choice · challenging)Point M is at $(-4, -5)$ and point N is at $(1, -5)$.

What is the distance, in units, between point M and point N?

Choices: 3 | 9 | 5 | -3

Answer: 5

Points with the same y-coordinate lie on a horizontal line, so subtract the x-coordinates.

Q0417 (Test 11 · Equation editor · moderate)A rectangular garden is drawn on a coordinate plane with vertices $(1, -4)$, $(9, -4)$, $(9, 2)$ and $(1, 2)$. Each unit is 1 meter.

What is the perimeter of the garden, in meters? Enter your answer.

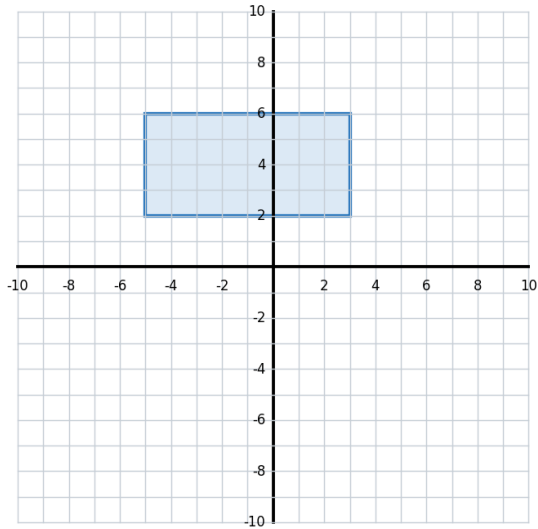
Answer: 28

Add the four side lengths, or use 2 times the sum of the length and width.

Q0450 (Test 12 · Multiple choice · easy)

A rectangle has vertices at $(-5, 2)$, $(3, 2)$, $(3, 6)$ and $(-5, 6)$.

What is the perimeter, in units, of the rectangle?



Choices: 32 | 24 | 12 | 64

Answer: 24

Find the side lengths by subtracting coordinates, then apply the formula.

Q0466 (Test 13 · Multiselect · moderate)

Select **all** the points that lie on the vertical line through $(3, 0)$.

Choices: $(3, 6)$ | $(5, -5)$ | $(3, 2)$ | $(5, 1)$ | $(0, 4)$ | $(3, -3)$

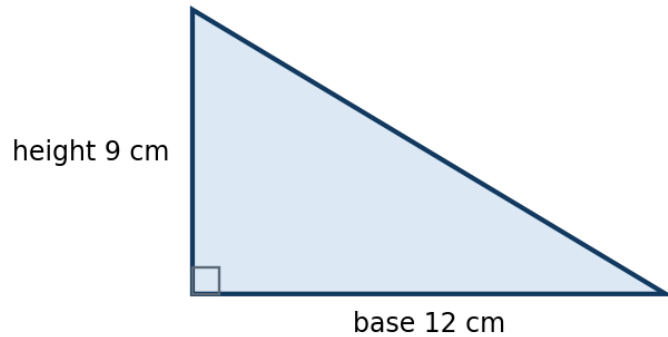
Answer: $(3, -3)$, $(3, 2)$, $(3, 6)$

Every point on a vertical line has the same x-coordinate.

MA.6.GR.2.1 — Derive a formula for the area of a right triangle using a rectangle. Apply a formula to find the area of a triangle.

Q0034 (Test 1 · Multiple choice · easy)

What is the area of the right triangle shown, in square centimeters?

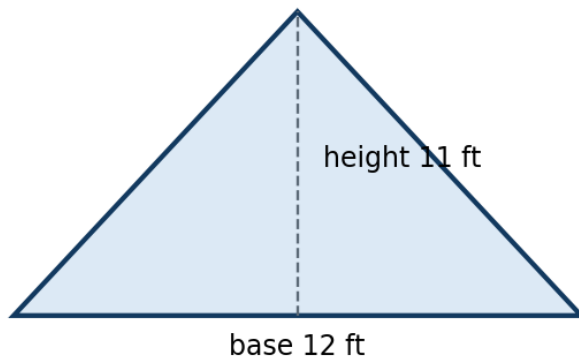
Choices: 108 | $10\frac{1}{2}$ | 54 | 21**Answer: 54**

A triangle is half of a rectangle with the same base and height, so use one half times base times height.

Q0061 (Test 2 · Equation editor · moderate)

A triangular sail has the base and height shown.

What is the area of the sail, in square feet? Enter your answer.

**Answer: 66**

Multiply the base by the height, then take half.

Q0117 (Test 4 · Multiple choice · challenging)

A triangle has an area of 33 square feet and a height of 11 feet.

What is the length of its base, in feet?

Choices: 8 | 3 | 66 | 6

Answer: 6

Work backwards: double the area, then divide by the height.

Q0157 (Test 5 · Matching table · easy)

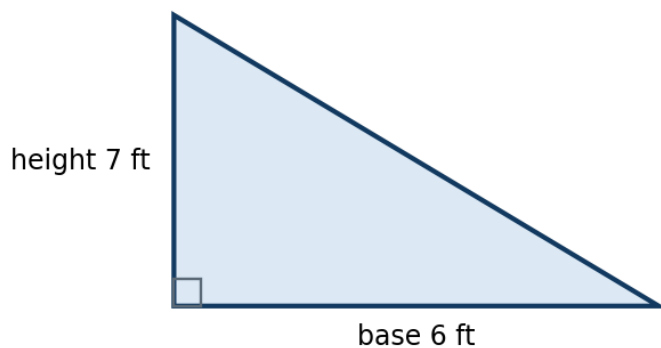
Match each triangle with its area, in square centimeters.

Answer: base 7 centimeters, height 4 centimeters → 14; base 9 centimeters, height 6 centimeters → 27; base 6 centimeters, height 10 centimeters → 30

Use one half times base times height for each triangle.

Q0222 (Test 6 · Multiple choice · moderate)

What is the area of the right triangle shown, in square feet?



Choices: $6\frac{1}{2}$ | 42 | 13 | 21

Answer: 21

A triangle is half of a rectangle with the same base and height, so use one half times base times height.

Q0261 (Test 7 · Multiselect · moderate)

A triangle has a base of 14 inches and a height of 9 inches.

Select **all** the expressions that give its area in square inches.

Choices: $(14 + 9) \div 2$ | 14×9 | $2 \times 14 \times 9$ | $0.5 \times 14 \times 9$ | $(14 \times 9) \div 2$ | $14 \times 9 \times 0.5$

Answer: $(14 \times 9) \div 2$, $0.5 \times 14 \times 9$, $14 \times 9 \times 0.5$

The area of a triangle is half the product of the base and the height.

Q0297 (Test 8 · Multiple choice · easy)

A triangle has an area of 15 square feet and a height of 3 feet.

What is the length of its base, in feet?

Choices: 12 | 5 | 10 | 30

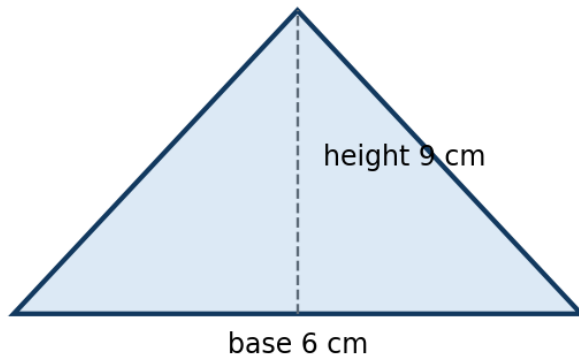
Answer: 10

Work backwards: double the area, then divide by the height.

Q0316 (Test 9 · Equation editor · moderate)

A triangular sail has the base and height shown.

What is the area of the sail, in square centimeters? Enter your answer.

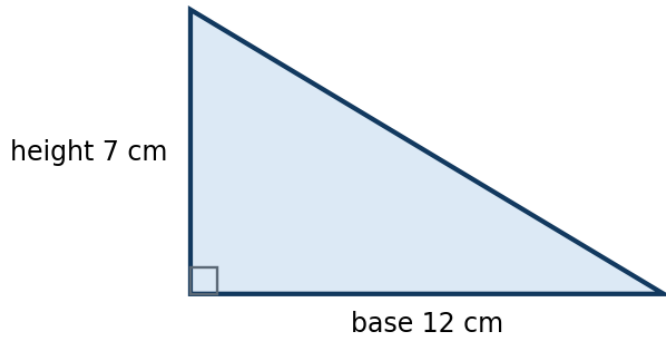


Answer: 27

Multiply the base by the height, then take half.

Q0379 (Test 10 · Multiple choice · challenging)

What is the area of the right triangle shown, in square centimeters?



Choices: 42 | 84 | 19 | $9\frac{1}{2}$

Answer: 42

A triangle is half of a rectangle with the same base and height, so use one half times base times height.

Q0405 (Test 11 · Matching table · moderate)

Match each triangle with its area, in square feet.

Answer: base 5 feet, height 8 feet → 20; base 9 feet, height 6 feet → 27; base 7 feet, height 4 feet → 14

Use one half times base times height for each triangle.

Q0422 (Test 12 · Multiple choice · easy)

A triangle has an area of 25 square feet and a height of 5 feet.

What is the length of its base, in feet?

Choices: 12 | 10 | 5 | 50

Answer: 10

Work backwards: double the area, then divide by the height.

Q0469 (Test 13 · Multiselect · moderate)

A triangle has a base of 8 centimeters and a height of 9 centimeters.

Select **all** the expressions that give its area in square centimeters.

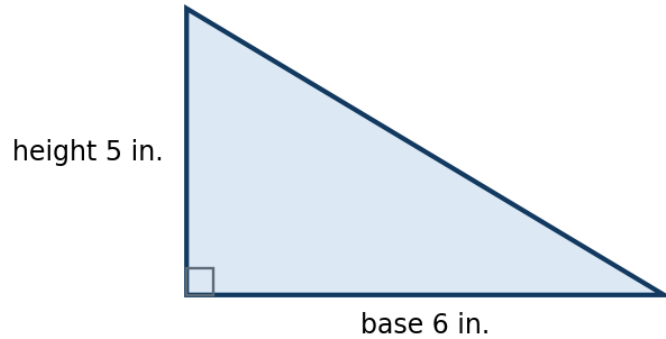
Choices: $0.5 \times 8 \times 9$ | $(8 \times 9) \div 2$ | 8×9 | $8 \times 9 \times 0.5$ | $(8 + 9) \div 2$ | $2 \times 8 \times 9$

Answer: $(8 \times 9) \div 2$, $0.5 \times 8 \times 9$, $8 \times 9 \times 0.5$

The area of a triangle is half the product of the base and the height.

Q0508 (Test 14 · Multiple choice · challenging)

What is the area of the right triangle shown, in square inches?



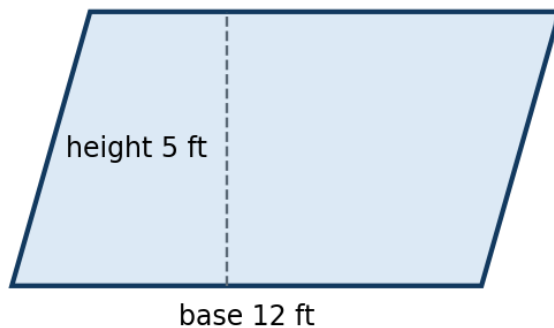
Choices: 15 | 30 | 11 | 5 1/2

Answer: 15

A triangle is half of a rectangle with the same base and height, so use one half times base times height.

MA.6.GR.2.2 — Solve mathematical and real-world problems involving the area of quadrilaterals and composite figures by decomposing them into triangles or rectangles.**Q0011** (Test 1 · Multiple choice · easy)

What is the area of the parallelogram shown, in square feet?



Choices: 17 | 34 | 30 | 60

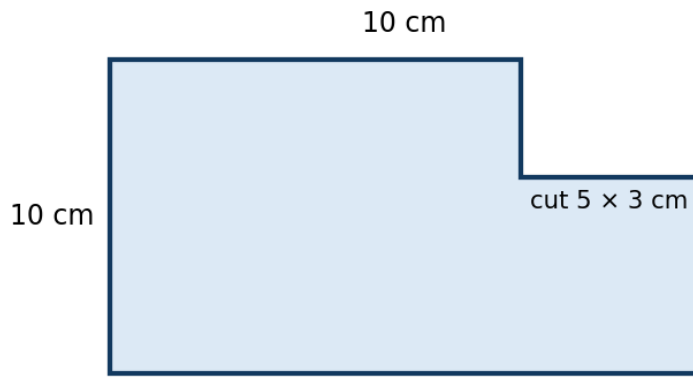
Answer: 60

The area of a parallelogram is base times height, using the height that is perpendicular to the base.

Q0062 (Test 2 · Equation editor · moderate)

The figure shows the floor of a room. All corners are right angles.

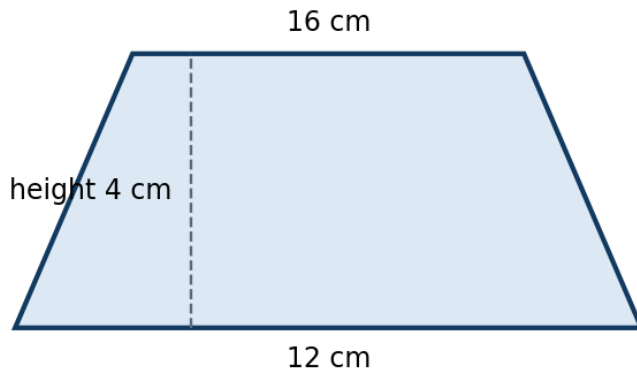
What is the area of the floor, in square centimeters? Enter your answer.

**Answer: 85**

Find the area of the whole rectangle, then subtract the corner that is cut out.

Q0112 (Test 3 · Multiple choice · challenging)

What is the area of the trapezoid shown, in square centimeters?



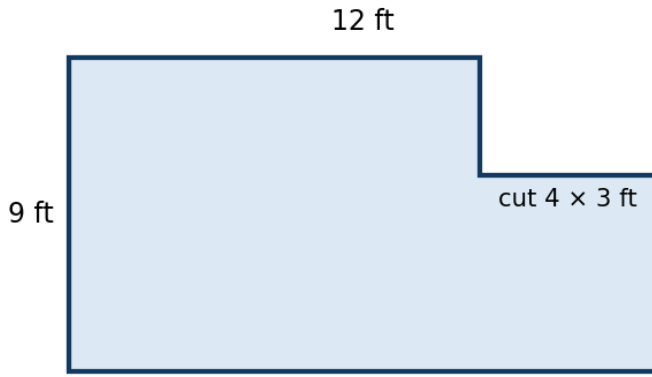
Choices: 56 | 112 | 48 | 16

Answer: 56

Decompose the trapezoid into a rectangle and triangles, or average the two bases and multiply by the height.

Q0181 (Test 5 · Editing task choice · easy)

Use the drop-down menus to find the area of the figure, in square feet.

The area of the whole rectangle is square feet, and the area of the cut-out corner is square feet.

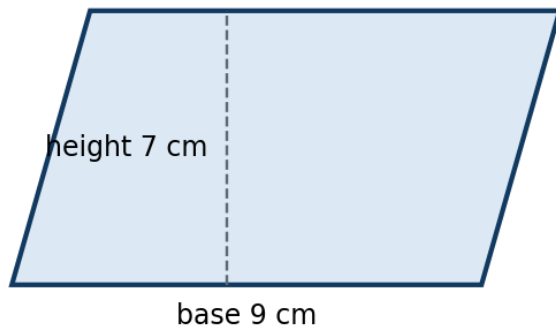
Blanks: 21 / 42 / 96 / 108 || 7 / 12 / 14 / 36

Answer: 108, 12

Decompose the figure: whole rectangle minus the missing corner.

Q0196 (Test 6 · Multiple choice · moderate)

What is the area of the parallelogram shown, in square centimeters?



Choices: 31 1/2 | 32 | 63 | 16

Answer: 63

The area of a parallelogram is base times height, using the height that is perpendicular to the base.

Q0236 (Test 7 · Multiselect · moderate)

A parallelogram has a base of 15 centimeters and a height of 5 centimeters.

Select **all** the expressions that give its area in square centimeters.

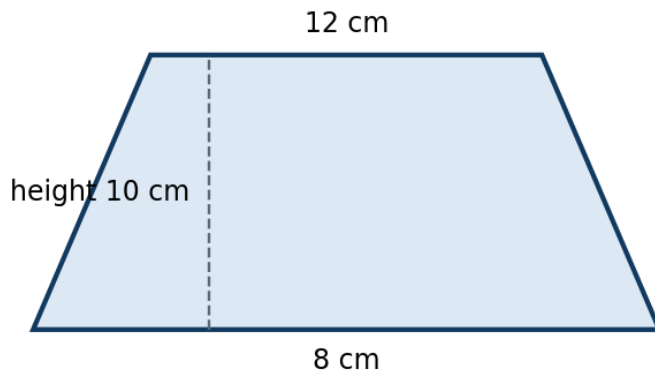
Choices: 5×15 | 15×5 | $2 \times (15 + 5)$ | 75×1 | $(15 \times 5) \div 2$ | $15 + 5$

Answer: 15×5 , 5×15 , 75×1

The area of a parallelogram is the base times the perpendicular height.

Q0303 (Test 8 · Multiple choice · easy)

What is the area of the trapezoid shown, in square centimeters?



Choices: 80 | 200 | 100 | 15

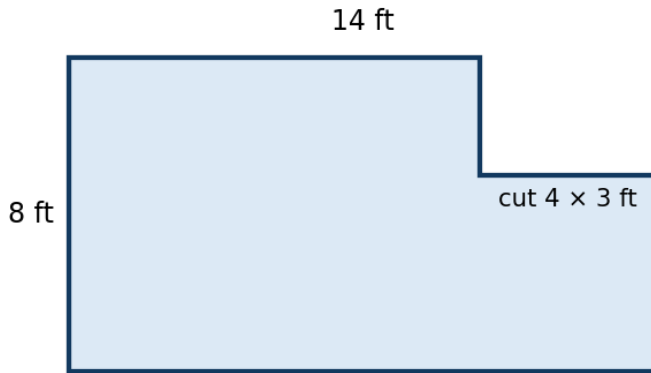
Answer: 100

Decompose the trapezoid into a rectangle and triangles, or average the two bases and multiply by the height.

Q0333 (Test 9 · Equation editor · moderate)

The figure shows the floor of a room. All corners are right angles.

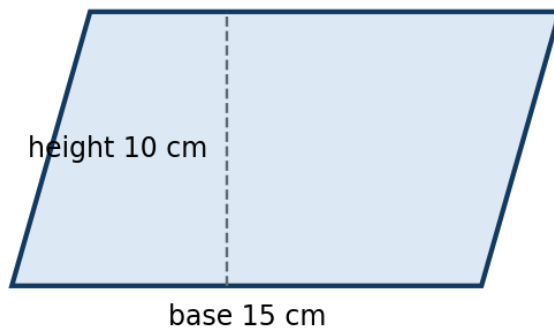
What is the area of the floor, in square feet? Enter your answer.

**Answer: 100**

Find the area of the whole rectangle, then subtract the corner that is cut out.

Q0376 (Test 10 · Multiple choice · challenging)

What is the area of the parallelogram shown, in square centimeters?



Choices: 150 | 25 | 75 | 50

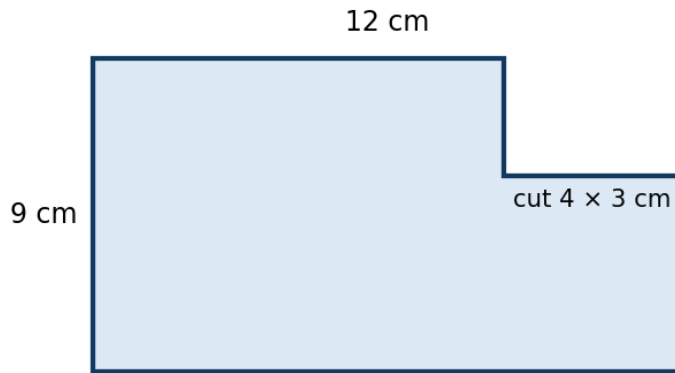
Answer: 150

The area of a parallelogram is base times height, using the height that is perpendicular to the base.

Q0400 (Test 11 · Editing task choice · moderate)

Use the drop-down menus to find the area of the figure, in square centimeters.

The area of the whole rectangle is square centimeters, and the area of the cut-out corner is square centimeters.



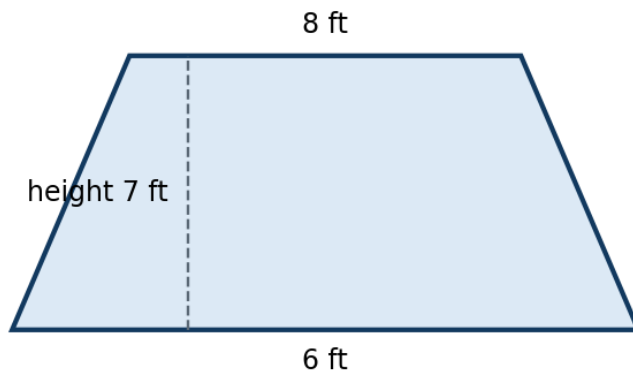
Blanks: 21 / 42 / 96 / 108 || 7 / 12 / 14 / 36

Answer: 108, 12

Decompose the figure: whole rectangle minus the missing corner.

Q0434 (Test 12 · Multiple choice · easy)

What is the area of the trapezoid shown, in square feet?



Choices: 42 | 98 | $10\frac{1}{2}$ | 49

Answer: 49

Decompose the trapezoid into a rectangle and triangles, or average the two bases and multiply by the height.

Q0482 (Test 13 · Multiselect · moderate)

A parallelogram has a base of 8 centimeters and a height of 4 centimeters.

Select **all** the expressions that give its area in square centimeters.

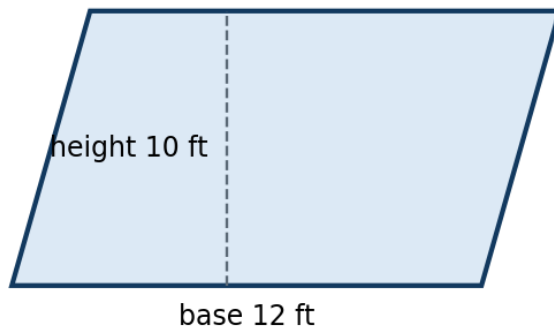
Choices: $2 \times (8 + 4)$ | 32×1 | 4×8 | $8 + 4$ | 8×4 | $(8 \times 4) \div 2$

Answer: 8×4 , 4×8 , 32×1

The area of a parallelogram is the base times the perpendicular height.

Q0496 (Test 14 · Multiple choice · challenging)

What is the area of the parallelogram shown, in square feet?



Choices: 60 | 44 | 120 | 22

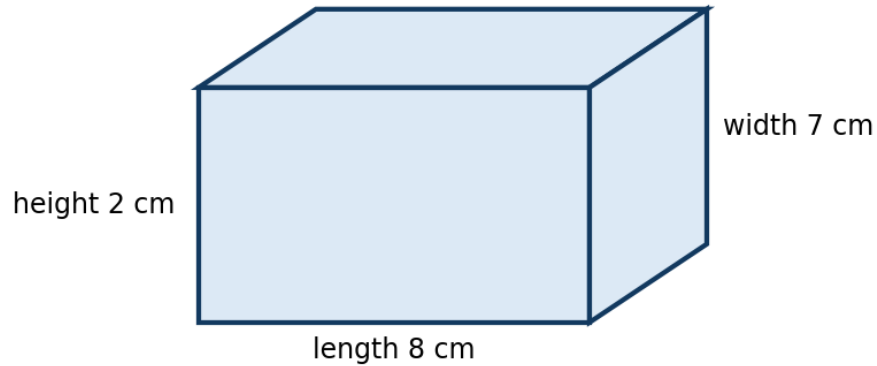
Answer: 120

The area of a parallelogram is base times height, using the height that is perpendicular to the base.

MA.6.GR.2.3 — Solve mathematical and real-world problems involving the volume of right rectangular prisms with positive rational number edge lengths using a visual model and a formula.

Q0033 (Test 1 · Multiple choice · easy)

What is the volume of the right rectangular prism shown, in cubic centimeters?



Choices: 112 | 56 | 172 | 17

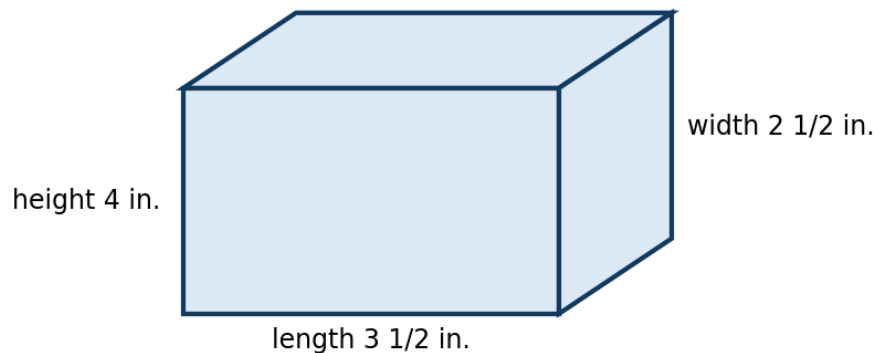
Answer: 112

Volume is length times width times height, or the area of the base times the height.

Q0072 (Test 2 · Equation editor · moderate)

A storage bin is a right rectangular prism with the dimensions shown.

What is its volume, in cubic inches? Enter your answer as a fraction, a mixed number or a decimal.

**Answer: 35**

Multiply the three edge lengths together.

Q0080 (Test 3 · Multiple choice · challenging)

A box shaped like a right rectangular prism has a volume of 96 cubic centimeters. Its base has an area of 32 square centimeters.

What is the height of the box, in centimeters?

Choices: 3 | 6 | 32 | 64

Answer: 3

Divide the volume by the area of the base to find the height.

Q0124 (Test 4 · Matching table · easy)

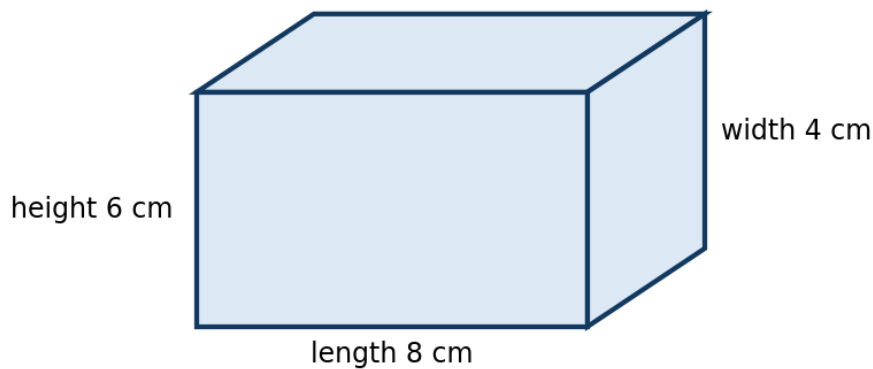
Match each right rectangular prism with its volume, in cubic inches.

Answer: 2 by 3 by 4 inches → 24; 2 by 5 by 6 inches → 60; 3 by 3 by 5 inches → 45

Multiply the three dimensions of each prism.

Q0197 (Test 6 · Multiple choice · moderate)

What is the volume of the right rectangular prism shown, in cubic centimeters?



Choices: 32 | 192 | 18 | 208

Answer: 192

Volume is length times width times height, or the area of the base times the height.

Q0241 (Test 7 · Multiselect · moderate)

A right rectangular prism measures 8 by 7 by 2 inches.

Select **all** the expressions that give its volume in cubic inches.

Choices: 8×7 | $2 \times 7 \times 8$ | $2 \times (8 + 7 + 2)$ | $8 + 7 + 2$ | 56×2 | $8 \times 7 \times 2$

Answer: $8 \times 7 \times 2$, 56×2 , $2 \times 7 \times 8$

Volume is the product of the three edge lengths, which is also the area of the base times the height.

Q0288 (Test 8 · Multiple choice · easy)

A box shaped like a right rectangular prism has a volume of 150 cubic inches. Its base has an area of 25 square inches.

What is the height of the box, in inches?

Choices: 25 | 125 | 6 | 12

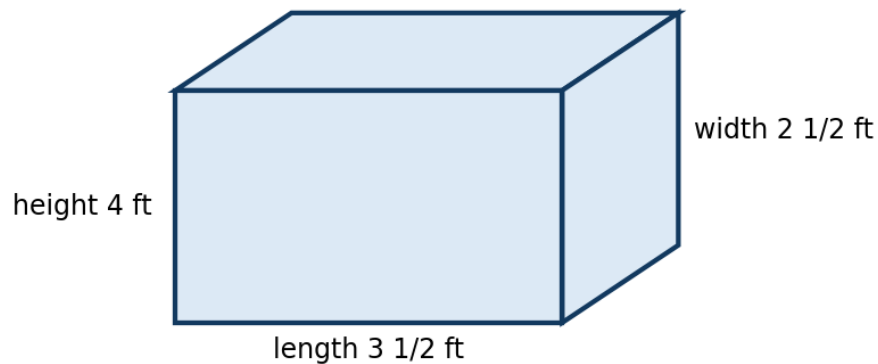
Answer: 6

Divide the volume by the area of the base to find the height.

Q0308 (Test 9 · Equation editor · moderate)

A storage bin is a right rectangular prism with the dimensions shown.

What is its volume, in cubic feet? Enter your answer as a fraction, a mixed number or a decimal.

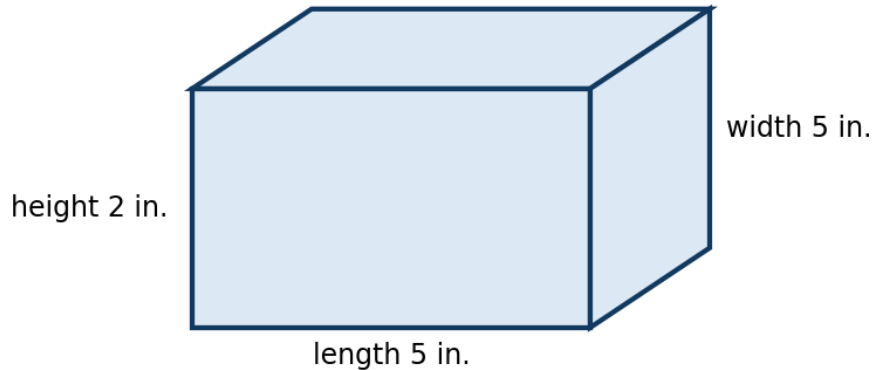


Answer: 35

Multiply the three edge lengths together.

Q0348 (Test 10 · Multiple choice · challenging)

What is the volume of the right rectangular prism shown, in cubic inches?



Choices: 90 | 25 | 12 | 50

Answer: 50

Volume is length times width times height, or the area of the base times the height.

Q0418 (Test 11 · Matching table · moderate)

Match each right rectangular prism with its volume, in cubic inches.

Answer: 2 by 5 by 6 inches → 60; 2 by 2 by 9 inches → 36; 4 by 4 by 3 inches → 48

Multiply the three dimensions of each prism.

Q0439 (Test 12 · Multiple choice · easy)

A box shaped like a right rectangular prism has a volume of 140 cubic inches. Its base has an area of 70 square inches.

What is the height of the box, in inches?

Choices: 2 | 4 | 70 | 0

Answer: 2

Divide the volume by the area of the base to find the height.

Q0465 (Test 13 · Multiselect · moderate)

A right rectangular prism measures 4 by 5 by 2 inches.

Select **all** the expressions that give its volume in cubic inches.

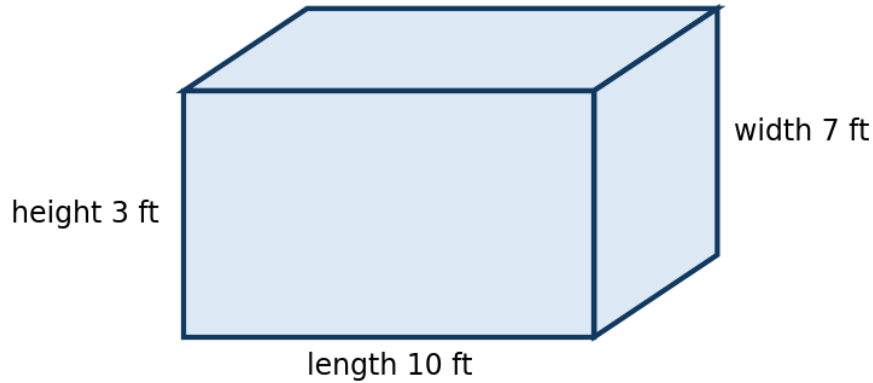
Choices: $4 \times 5 \times 2$ | $2 \times 5 \times 4$ | 20×2 | $2 \times (4 + 5 + 2)$ | 4×5 | $4 + 5 + 2$

Answer: $4 \times 5 \times 2$, 20×2 , $2 \times 5 \times 4$

Volume is the product of the three edge lengths, which is also the area of the base times the height.

Q0512 (Test 14 · Multiple choice · challenging)

What is the volume of the right rectangular prism shown, in cubic feet?



Choices: 20 | 70 | 242 | 210

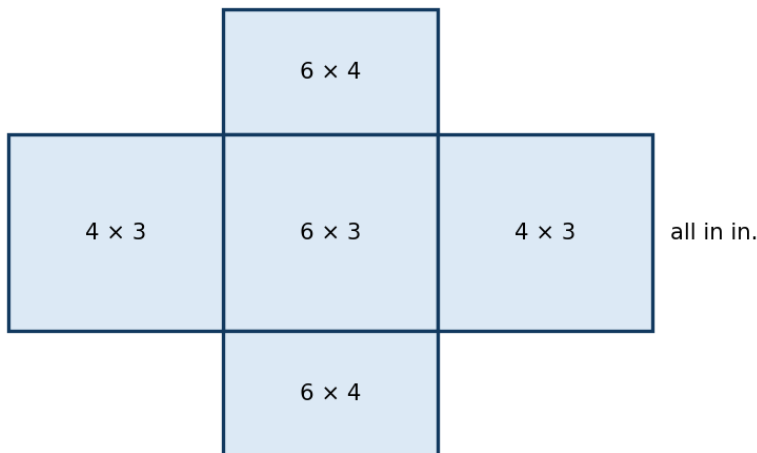
Answer: 210

Volume is length times width times height, or the area of the base times the height.

MA.6.GR.2.4 — Given a mathematical or real-world context, find the surface area of right rectangular prisms and right rectangular pyramids using the figure's net.**Q0028** (Test 1 · Multiple choice · easy)

The net of a right rectangular prism is shown.

What is the surface area of the prism, in square inches?



Choices: 54 | 72 | 108 | 48

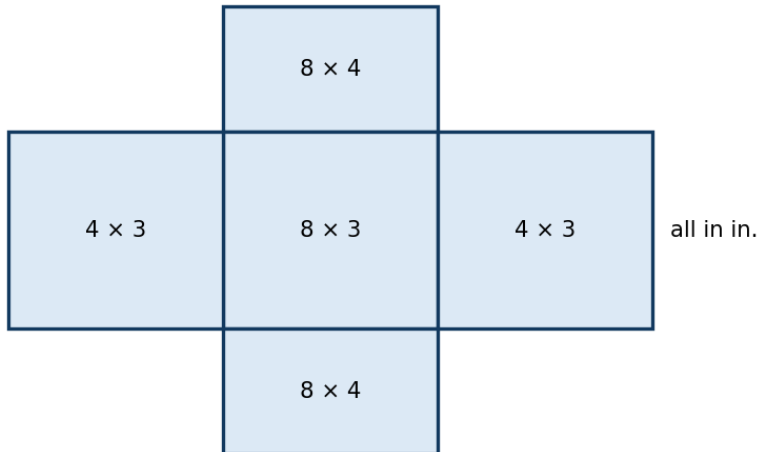
Answer: 108

Add the areas of all six rectangles in the net; opposite faces match.

Q0075 (Test 2 · Equation editor · moderate)

A gift box is a right rectangular prism measuring 8 by 4 by 3 inches. Its net is shown.

How many square inches of paper are needed to cover the box exactly? Enter your answer.

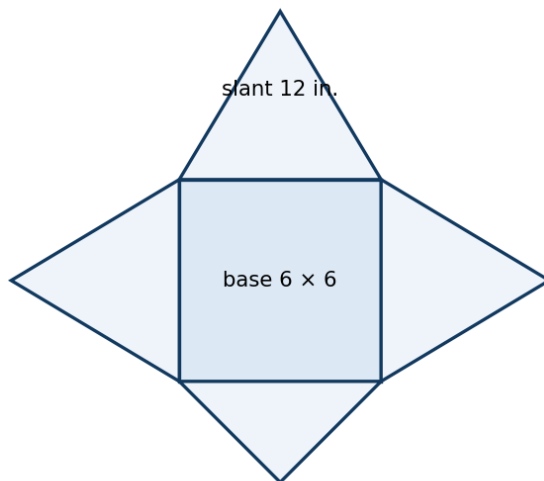
**Answer: 136**

Find the area of each face from the net and add all six.

Q0086 (Test 3 · Multiple choice · challenging)

The net of a right rectangular pyramid with a square base is shown.

What is the surface area of the pyramid, in square inches?



Choices: 36 | 180 | 288 | 324

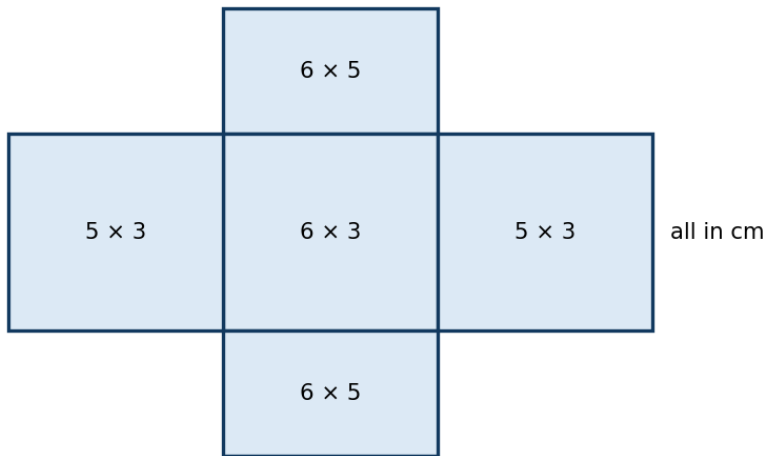
Answer: 180

Add the square base to the four triangular faces, each with area one half times base times slant height.

Q0126 (Test 4 · Editing task choice · easy)

Use the drop-down menus to find the surface area of the prism from its net.

The two 6 by 5 faces have a total area of square centimeters, and the two 6 by 3 faces have a total area of square centimeters.



Blanks: 11 / 30 / 60 / 120 || 9 / 18 / 36 / 72

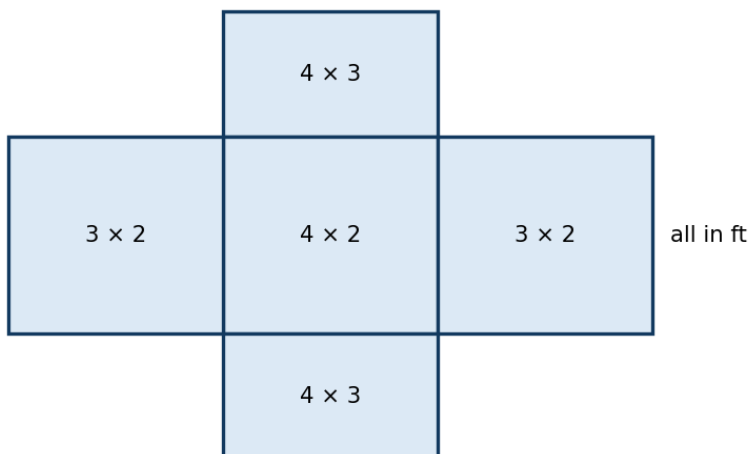
Answer: 60, 36

Each pair of opposite faces has the same area, so double the area of one.

Q0183 (Test 5 · Multiple choice · moderate)

The net of a right rectangular prism is shown.

What is the surface area of the prism, in square feet?



Choices: 24 | 26 | 58 | 52

Answer: 52

Add the areas of all six rectangles in the net; opposite faces match.

Q0242 (Test 7 · Multiselect · moderate)

A right rectangular prism measures 5 by 5 by 6 inches.

Select **all** the expressions that give its surface area in square inches.

Choices: $5 \times 5 + 5 \times 6 + 5 \times 6$ | $2(5 \times 5 + 5 \times 6 + 5 \times 6)$ | $6 \times 5 \times 5$ | 170×1 | $5 \times 5 \times 6$ | $2(5 \times 5) + 2(5 \times 6) + 2(5 \times 6)$

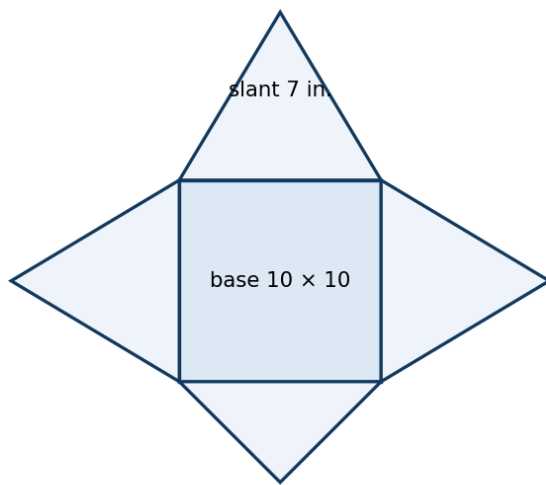
Answer: $2(5 \times 5) + 2(5 \times 6) + 2(5 \times 6)$, $2(5 \times 5 + 5 \times 6 + 5 \times 6)$, 170×1

Surface area adds the areas of all six faces, and opposite faces are equal.

Q0296 (Test 8 · Multiple choice · easy)

The net of a right rectangular pyramid with a square base is shown.

What is the surface area of the pyramid, in square inches?



Choices: 380 | 240 | 100 | 280

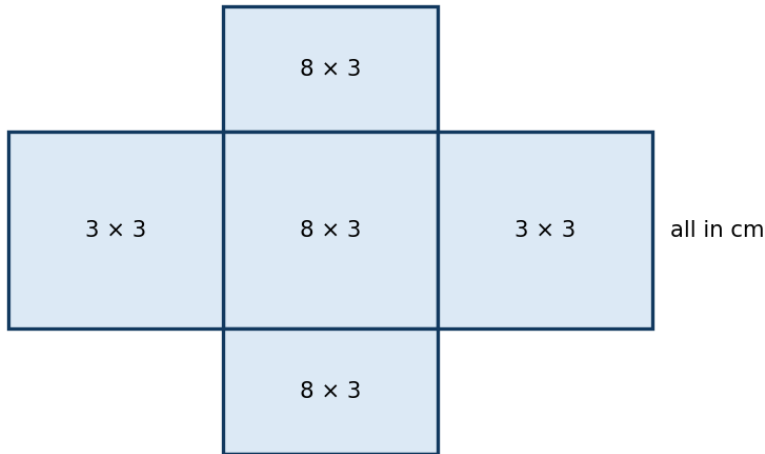
Answer: 240

Add the square base to the four triangular faces, each with area one half times base times slant height.

Q0334 (Test 9 · Equation editor · moderate)

A gift box is a right rectangular prism measuring 8 by 3 by 3 centimeters. Its net is shown.

How many square centimeters of paper are needed to cover the box exactly? Enter your answer.

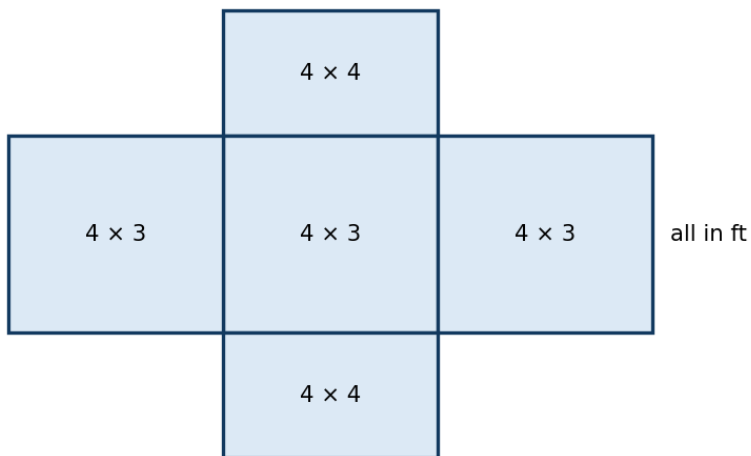
**Answer: 114**

Find the area of each face from the net and add all six.

Q0344 (Test 10 · Multiple choice · challenging)

The net of a right rectangular prism is shown.

What is the surface area of the prism, in square feet?



Choices: 32 | 48 | 80 | 40

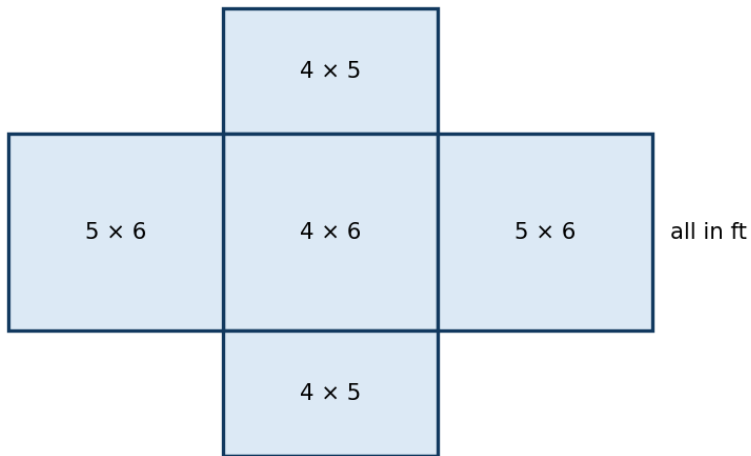
Answer: 80

Add the areas of all six rectangles in the net; opposite faces match.

Q0416 (Test 11 · Editing task choice · moderate)

Use the drop-down menus to find the surface area of the prism from its net.

The two 4 by 5 faces have a total area of square feet, and the two 4 by 6 faces have a total area of square feet.



Blanks: 9 / 20 / 40 / 80 || 10 / 24 / 48 / 96

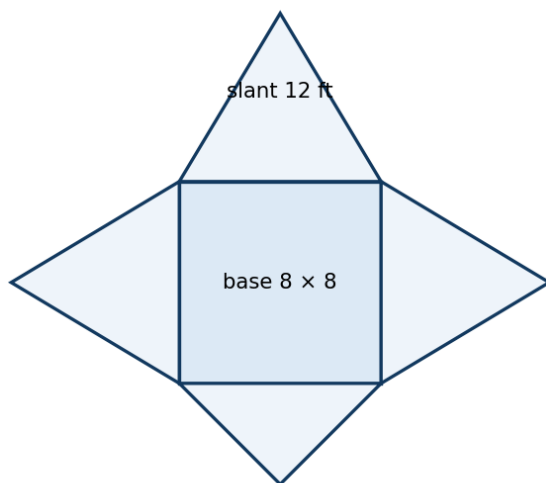
Answer: 40, 48

Each pair of opposite faces has the same area, so double the area of one.

Q0432 (Test 12 · Multiple choice · easy)

The net of a right rectangular pyramid with a square base is shown.

What is the surface area of the pyramid, in square feet?



Choices: 64 | 384 | 448 | 256

Answer: 256

Add the square base to the four triangular faces, each with area one half times base times slant height.

Q0481 (Test 13 · Multiselect · moderate)

A right rectangular prism measures 5 by 5 by 2 centimeters.

Select **all** the expressions that give its surface area in square centimeters.

Choices: 90×1 | $2(5 \times 5 + 5 \times 2 + 5 \times 2)$ | $2(5 \times 5) + 2(5 \times 2) + 2(5 \times 2)$ | $6 \times 5 \times 5$ | $5 \times 5 + 5 \times 2 + 5 \times 2$ | $5 \times 5 \times 2$

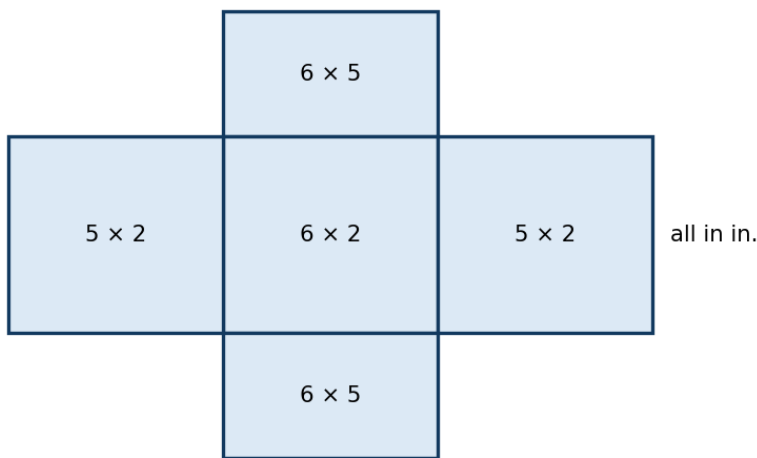
Answer: $2(5 \times 5) + 2(5 \times 2) + 2(5 \times 2)$, $2(5 \times 5 + 5 \times 2 + 5 \times 2)$, 90×1

Surface area adds the areas of all six faces, and opposite faces are equal.

Q0500 (Test 14 · Multiple choice · challenging)

The net of a right rectangular prism is shown.

What is the surface area of the prism, in square inches?



Choices: 60 | 52 | 110 | 104

Answer: 104

Add the areas of all six rectangles in the net; opposite faces match.